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Sweden

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OECD ECONOMIC SURVEYS

2004

Sweden

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Table of contents

Assessment and recommendations 9

1. Key challenges for living standards and the welfare state 21

Sweden's long-term growth path 22
The macroeconomic framework and the current economic situation 32
Medium and longer-term growth prospects 38

Notes 54

Bibliography 55

Annexes

1.1. Progress in structural reforms 56
1.2. Macroeconomic stability 62
1.3. Outlook to 2005 67

2. Product market competition and economic performance 71

Product market competition and macroeconomic performance 72
Competition legislation and enforcement 75
Regulatory policy and competition issues in specific markets/sectors 82
Competition and the public sector 93
Concluding remarks and priorities for policies 103

Notes 107

Bibliography 108

Annexes

2.1. Telecommunications competition in practice 110
2.2. Proposals by housing organisations 111
2.3. Alcohol retailing 113
2.4. Public procurement 116

3. Harnessing resources more effectively 119

Labour supply and use 119
Improving the mobility and optimal allocation of capital 127
Entrepreneurial activity 130
Innovation 137
Concluding remarks and priorities for policies 143

Notes 146

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Bibliography 148

Annexes

3.1. Income and other direct taxes in Sweden 150
3.2. Broad policy guidelines for fostering entrepreneurship 152
3.3. Turning science into business 154
3.4. Promoting innovation in the British National Health Service 156

4. More sustainable public finances 157

The fiscal position has weakened in recent years 158
Fiscal rules may have curbed spending pressures 163
Expenditure ceilings have come under strain 165
Local governments struggle to balance budgets 168
Concluding remarks and priorities for policies 174

Notes 176

Bibliography 178

Annex

4.1. Local government equalisation schemes 179

5. Some aspects of sustainable development 181

Air pollution 181
Water pollution 189
Waste management 193

Notes 198

Bibliography 200

Boxes

1.1. Developments in the telecommunications equipment sector 23
1.2. Sweden's monetary policy framework and the euro referendum 35
1.3. Population projections 38
1.4. The public pension system 49
2.1. The stance of competition in Sweden 74
2.2. Experiences with procurement at the municipal level 100
4.1. Spending reforms and tax expenditure in the 121-point plan 162
4.2. Measures to battle rising sickness expenditures 167
5.1. The integration of policies across sustainable development areas 182

Tables

1.1. General government financial balance 37
1.2. Population projections 39
1.3. Scenarios for total hours worked 44
2.1. Consumption exposed to competition 73
2.2. Relative price levels 77
2.3. Output, employment and productivity 78
2.4. Major competition cases ruled upon by the Market Court 79
2.5. Fines claimed and imposed in major competition cases 80
2.6. Electricity prices 83
2.7. Total consumption of alcohol 93

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2.8. State-owned enterprise sector 96
3.1. Net replacement rates 122
3.2. Agreement models for wage bargaining 126
3.3. METRs for domestic investment 128
3.4. Effective marginal tax rates for various combinations of ownership and financing, 2001 129
3.5. Intellectual property in public research organisations 141
4.1. Central and local government finances 159
4.2. Central government expenditure 164
4.3. Central government expenditure ceilings 165
4.4. State grants to local governments 170
5.1. Main indicators: air pollution 183
5.2. Performance indicators: water pollution 190
5.3. Performance indicators: waste management 194
5.4. Household waste 195
5.5. Recycling rates and targets 196

Annexes

1.A3.1. Summary of short-term projections 68
4.A1.1. An example of income equalisation in 2004 180

Figures

1.1. Real GDP per person 24
1.2. Driving forces of GDP per capita growth 24
1.3. Labour resources that could be mobilised 25
1.4. Labour force participation 25
1.5. Average hours worked 26
1.6. Composition of employment and labour productivity 28
1.7. Business investment rate 29
1.8. ICT investments 30
1.9. Population with at least an upper-secondary qualification 31
1.10. Dependency ratios 31
1.11. Forces shaping output 33
1.12. Output gap 34
1.13. Interest rates and exchange rate developments 36
1.14. Dependency ratios 40
1.15. Labour market performance, 2002 41
1.16. Demographic impact on employment 42
1.17. Trends in labour supply 43
1.18. Public finance position 48
1.19. Projected change in age-related government spending 50
1.20. Public finances in the longer term 51
2.1. Indicators on competition 76
2.2. Market concentration in electricity 83
2.3. Telephone charges 85
2.4. Completed rental dwellings 88
2.5. Relative pharmaceutical prices 91
2.6. General government consumption and employment 94
2.7. Public enterprise sector in EU countries 95
2.8. Openly advertised public procurement 99
2.9. State aid to enterprises in the EU 103

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3.1. Employment gaps for immigrants 121
3.2. Average job tenure by age 125
3.3. Strictness of employment protection 126
3.4. Starting or running a business 131
3.5. Entry, exit and survival of firms 132
3.6. Swedish private equity 133
3.7. Venture capital investment in OECD countries 134
3.8. Attitudes towards business failure 136
3.9. Resources used in R&D 138
3.10. Internationalisation of manufacturing R&D 139
3.11. Tax subsidies for R&D 140
4.1. Government net lending and debt 158
4.2. Cyclically-adjusted general government net lending 161
4.3. Local government tax rates 169
4.4. Local government income 169
5.1. Air pollution 185
5.2. Particulate matter emission 187
5.3. Nitrate concentrations 191
5.4. Nitrogen supply to the Baltic Sea proper and Öresund 191

Annexes

1.A2.1. Factors affecting macroeconomic stability 63
1.A2.2. Consumer price inflation 65
1.A3.1. Retail sales and confidence indicators 67

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BASIC STATISTICS OF SWEDEN

THE LAND

Land area (1 000 sq. km)	411
Lakes (1 000 sq. km)	39
Arable area (1 000 sq. km) (2002)	27
Woodland (1 000 sq. km) (1997-2001)	226

Inhabitants in major cities, including suburbs (31 December 2002) thousands	
Stockholm	1 684
Göteborg	810
Malmö	533

THE PEOPLE

Population (31 December 2002), thousands	8 941
Number of inhabitants per sq. km	22
Net natural increase (average 1998-2002) thousands	–3.1

Net natural increase per 1 000 inhabitants (average 1998-2002)	–0.4
Net migration (average 1998-2002), thousands	21.9

THE PRODUCTION

Gross domestic product in 2002 (Kr billion)	2 340.0
GDP per head, USD	27 000

Gross fixed capital formation in 2002	
Per cent of GDP	17.1
Per head, USD	4 612
Employment, 2002:	
Total civilian, thousands	4 241
Per cent of total	
Agriculture, forestry, fishing	2.1
Industry	23.1
Other	74.6

THE GOVERNMENT

Per cent of GDP in 2001:	
Public consumption	26.7
General government current revenue	57.3
Public gross fixed capital formation	2.3

Composition of Parliament	Number of seats
Social democrats	144
Moderates	55
Liberals	48
Christian democrats	33
Left	30
Centre	22
Greens	17
Total	349

Last general election: September 2002
Next general election: September 2006

THE FOREIGN TRADE

Exports of goods and services, 2002 (per cent of GDP)	45.4
Main merchandise exports (per cent of total)	
Forestry products	14.0
Mineral products	9.6
Chemical products	11.3
Energy products	3.0
Engineering products	50.8
Other products	11.3

Imports of goods and services, 2002 (per cent of GDP)	34.9
Main merchandise imports (per cent of total)	
Forestry products	3.5
Mineral products	8.3
Chemical products	10.8
Energy products	8.9
Engineering products	47.8
Other products	20.7

THE CURRENCY

Monetary unit: Krona

Currency units per USD, average of daily figures:	
Year 2003	8.08
January 2004	7.25

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

•

The economic situation and policies of Sweden were reviewed by the Committee on 12 January 2004. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 27 January 2004.

•

The Secretariat's draft report was prepared for the Committee by Deborah Roseveare, Martin Jørgensen and Douglas Sutherland under the supervision of Peter Jarrett.

•

The previous Survey of Sweden was issued in August 2002.

Assessment and recommendations

Sweden's recent good macroeconomic performance allows policy-makers to focus on longer-term goals…

The performance of the Swedish economy during the recent international slowdown has demonstrated the value of a sound macroeconomic policy approach. After Swedish voters decided not to adopt the euro for the time being, the monetary framework continues as a well-designed inflation targeting regime. The budgetary framework has delivered solid public finances, providing room for automatic stabilisers to work during the recent global slowdown, while monetary easing helped to stabilise activity without compromising price stability. In addition, tax cuts and other discretionary policy measures have helped to underpin output. As a result, growth in both 2002 and 2003 surpassed the European average by about one percentage point, and the OECD has projected economic growth of around 2¼ per cent in 2004 and 2¾ per cent in 2005. Despite current softness in some economic indicators, once activity has clearly picked up pace and spare capacity shrinks, monetary policy will need to be tightened in line with the inflation target. Overall, Sweden is in the enviable position of being able to focus its economic strategy towards longer-term requirements, which is also the main theme of this Survey. GDP per capita has grown faster than the OECD average in recent years after a long period of slower than average growth. But maintaining Sweden's per capita growth rate will become harder in the longer term as the dependency ratio starts to rise. Thus, the government's renewed emphasis on policies to boost labour supply and potential growth is welcome.

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… of raising living standards and maintaining the welfare state as the population ages

As in many other OECD countries, the working-age population will start to shrink towards the end of this decade unless it can be supplemented by net immigration, while the old-age dependency ratio will start climbing quite sharply. In this context, the two broad economic policy challenges facing Sweden are to:

- Continue raising living standards through faster productivity growth and greater utilisation of resources available in the economy.
- Maintain the solidarity principles of the welfare state, despite a sharp increase in the dependency ratio.

More product market competition could boost productivity growth…

Establishing the conditions for greater product market competition is part of the answer, given its central role in bolstering productivity growth. Over the past 10 to 15 years, Sweden has strengthened competition legislation and deregulated a number of sectors, including electricity, telecommunications and parts of transport. The scope for competition from non-agricultural imports has also been augmented by increasing economic integration within the European Union. However, Sweden may not obtain the full benefits from these reforms until all its product markets are sufficiently exposed to competition, and labour and capital are able to move quickly to their most productive uses.

… but the institutional framework needs strengthening

Vigorous competition needs to be underpinned by both a strong institutional framework and vigilance in identifying and eliminating anti-competitive behaviour. Although Sweden's arrangements are already broadly in line with international best practice, there are several ways in which the approach could be rendered more effective:

- Strengthening the Competition Authority's capacity to combat hard-core cartels and other serious breaches of competition law so that fear of detection, prosecution and damaged public image provide an effective and strong deterrent to collusive behaviour.
- Providing for substantial sanctions on individuals actively and knowingly engaged in anti-competitive activities in a way that reinforces the effectiveness of the newly introduced leniency system.
- Streamlining the process and reducing the time taken for competition cases to be dealt with.

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Specific measures could increase competition in several sectors…

More competition could be injected into several sectors, including some areas where extensive improvements have already been made. This could involve the following:

- Making further investments in the Nordic electricity infrastructure including the Swedish system to reduce the scope for any producer to exercise market power at times when transmission bottlenecks prevent the regional Nordpool electricity market from functioning efficiently.
- Exposing all the activities of the state-owned passenger rail company to competition, including inter-regional routes, and requiring it to operate under a hard budget constraint, with no prospects of further government capital injections to bail it out.
- Eliminating the tendency towards collusive behaviour in building and construction by tougher enforcement of competition law.
- Examining carefully the equity and efficiency implications of a more market-oriented allocation of rental housing, instead of the present controls on rent-setting.
- Modifying the planning process to require that the benefits of increased competition be explicitly taken into account when local authorities are deciding on applications for supermarket sites.
- Considering alternative regulatory mechanisms that would allow the introduction of competition for retailing of alcohol without sacrificing control over access to liquor.

… while the public sector should operate on a level playing field

There is also room for greater efforts to boost competition in Sweden's large and decentralised public sector and in its interactions with the private sector. Over the longer term the Swedish authorities should consider whether the private sector is better placed to provide some of the goods and services that the public sector currently provides. Although the government has taken a number of steps already to increase exposure to competition and to promote a more level playing field, some further changes would help, including:

- Strengthening the legal framework governing the market activities of municipalities and other public institutions, bringing them clearly within the scope of the competition

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legislation and providing greater opportunity for competitors to seek redress.

- Rationalising the present supervisory structure for public procurement into a single agency covering all levels of government and with scope to impose sanctions, and modifying legislation to require that tenders from internal units and external suppliers be treated equally.
- Increasing outsourcing of public services, which would not only boost competition and lead to higher quality and/or lower costs, but would also provide more opportunities for nascent entrepreneurs to emerge in currently under-developed parts of the personal and social services sector.
- Monitoring more effectively the activities of state agencies and their effects on competition, which would also demonstrate the government's commitment to promoting competition in the public sector.

More generally, efforts need to be made to ensure that all rules affecting private and public suppliers are designed so that providers can compete on fair and equal terms in every respect.

Young people could complete education more quickly…

Expanding the country's output depends on making effective use of its available resources through overall labour utilisation and by ensuring that labour productivity is maximised. With already high employment rates overall, policies are now being reoriented to target those groups where participation is especially low and where there is scope for increases. One such group is young people, who could complete their education and enter the workforce at a younger average age if the rules for admission to tertiary education were rationalised to facilitate the transition from secondary to higher education and to encourage improved performance in upper secondary school. The incentives for more rapid completion of post-secondary education would be strengthened, for example, by further limiting the availability of study grants to the normal duration of study programmes. The merits of shifting to a more direct "voucher" type of arrangement for funding of tertiary education could also be considered as a mechanism for more efficiently matching supply and demand for places within the tertiary system.

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… and immigrants need jobs to integrate faster into the economy

Immigrants are also under-represented in the workforce, with a significantly lower participation rate than native Swedes. The reasons for this are complex, not least because immigrants are a heterogeneous group and may face a range of obstacles to economic integration, depending on their circumstances. Improving their labour market prospects will require a combination of greater efforts to improve their language and job-related skills and appropriate activation strategies, as well as vigilance against discrimination. Low-skilled foreigners face an extra hurdle because the economy has relatively few jobs at their level, given the wage compression that is a hallmark of the Swedish model of the labour market. If the social partners concentrate wage increases on raising the lowest pay rates in forthcoming agreements, this will make it harder for the low-skilled, whether immigrants or native Swedes, to find jobs commensurate with their current skills and productivity levels.

Sick leave needs to be tightly administered

Effective labour supply is also restricted by people absent from the workplace because of sickness, while long-term sick leave and disability pensions have provided a permanent exit route from the workforce for some older workers. Indeed, the growing recourse to sick leave in recent years has made it one of the most urgent labour supply issues to address. The government has made strenuous efforts in the last year to curb the upward trend by tightening administrative controls and giving employers a stronger incentive to reduce its incidence. It also intends to convert disability pensions into time-limited, though renewable, sickness compensation. To be effective, these measures need to be enacted as intended in every local social insurance office; strengthening the powers of the national agency over local insurance boards would help to ensure this. If the measures already taken do not yield strong and durable reductions in days lost, further control measures to address moral hazard will need to be considered: this would avoid the less palatable alternative of scaling back benefit rates significantly.

Income tax cuts should focus on reducing high marginal tax rates

Boosting average working hours is another way of raising labour utilisation. While trimming back working hours is a natural response to rising living standards, in Sweden's case, hours worked are also heavily influenced by high taxes on

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labour income. But fewer working hours diminish the output produced, shrink the tax base and make it harder to finance the welfare state. At the same time, negotiated cutbacks in working hours may also push up the overall cost of labour-intensive public services, where offsetting productivity gains are difficult to achieve. Taken altogether, the long-term implications for public finances of reduced hours would be significant, and it is therefore crucial to moderate the tax burden on earned income. Once the last step of the current programme of cuts is implemented, future tax reforms could concentrate on producing stronger labour supply benefits by further lowering high effective marginal tax rates on earned income. In addition, reducing the state income tax would be justified in order to increase returns to education and promote growth in labour productivity. The revenue foregone could be financed by trimming public spending, reducing tax expenditures, raising taxes on property and removing preferential VAT rates and/or preferential treatment of pension savings.

Obstacles to job mobility could be reduced

The productivity of labour resources depends partly on whether they are able to move easily and quickly in response to changes arising from vigorous product market competition. Swedes stay longer in the same job than do employees in other countries, which can have advantages for developing firm-specific human capital. But it may also mean that some workers remain in jobs where their output is smaller than it would be if they moved to another position where their productivity could also grow more rapidly over time. Sweden's relatively strict employment protection legislation may persuade workers to stay where they are, rather than risk moving to a different job – even if it would suit them better than their current one – because of the associated loss of job protection. The bias against hiring older workers, in part reflecting employment protection rights, also discourages job mobility. Furthermore, with wages still relatively compressed, it can be more difficult for workers to identify opportunities for increasing their long-term earnings by moving to a higher productivity position. These aspects of the labour market could be addressed by:

- Amending employment protection legislation so that a better balance is struck between providing job security and facilitating the movement of people to jobs where

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they are most productive, where their earnings potential is greatest, and where they would prefer to be.

- Allowing relative wages to play a greater role in providing information about the demand for workers and skills between firms, sectors and professions. As wages are negotiated between the social partners, this may involve bargaining outcomes that reflect a greater tolerance for wage dispersion.

Tax rules distort investment decisions…

Raising inter-sectoral capital mobility could also contribute to higher productivity. Current tax rules inhibit the process of shifting capital rapidly to its most productive uses. Although Sweden's corporate tax regime is internationally competitive, more uniform tax treatment of different forms of corporate finance and different types of investment would improve efficiency. A cut of the tax rate financed by base-broadening measures could contribute to this. Wealth taxation discourages households from saving outside of institutional vehicles such as occupational pension schemes and may make it more attractive to hold assets offshore for tax reasons.

… and also discourage entrepreneurial activity

Entrepreneurial activity is one of the most important, though elusive, elements of a dynamic economy. It is still weak in Sweden, although the government has given it more emphasis and public attitudes have become more favourable in recent years. Several factors work against entrepreneurial efforts. The more favourable corporate tax treatment of debt is disadvantageous, especially for small companies and start-ups, for which equity financing is often more appropriate. The income splitting (so-called "3:12") rules applying to closely held companies provide a neutral treatment of different organisational business forms and prevent large-scale income shifting in the dual income tax system. Although the imputed capital income formula has been chosen to reflect an average *ex ante* risk premium, it nevertheless limits the *ex post* rewards for risk-taking and remains relatively onerous for highly profitable businesses, even after the recently announced changes. The risks associated with shifting from employment to self-employment may also be magnified by the loss of accumulated employment protection and the high weight of previous earnings in the amount of unemployment insurance benefit. Overall, it may

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be unrealistic to expect entrepreneurs to flourish in such a climate, which almost certainly could be improved with very little overall impact on public finances.

Joint rewards to institutions and inventors would stimulate commercialisation of R&D

Although innovation is clearly important for technical progress, Sweden's high rate of investment in R&D does not appear to have yielded commensurate increases in aggregate productivity that would contribute to long-term economic growth. This suggests that it would be appropriate to maintain a relatively cautious approach to policies for R&D *per se*, and the optimal strategy would ensure that economic conditions overall provide market-based financial rewards to firms that innovate. However, in the public sector, the incentives to commercialise research could be strengthened by sharing the intellectual property rights between the researcher and the employing institution. Setting up structures that help institutions develop effective strategies for bringing research to market could also facilitate technology transfer. A similar approach could also help diffuse innovative practices within the public sector and stimulate employers and employees alike to search more actively for such improvements.

Ageing will put public finances under pressure…

While boosting labour utilisation and productivity will help maintain GDP growth at a higher rate, Sweden's rising dependency ratio over coming decades implies substantially greater redistribution *via* public finances. The reforms designed to put public pensions onto a robust financial basis have eased future budgetary pressures significantly. Nevertheless, health and elderly care expenditures are still expected to absorb a steadily rising share of GDP over the next 40 to 50 years. These pressures will be attenuated by greater utilisation of available labour, especially where it involves people shifting from public benefits into employment, but it will also require vigilance in the management of public finances.

… but strong financial surpluses for the next 15 years would help

The government, well aware of the importance of good fiscal management, has set itself the ongoing target of a fiscal surplus of 2 per cent of GDP over the cycle, in order to provide for long-term sustainability of public finances. Achieving this target for the next 15 years or so would allow

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net assets to be accumulated that can later be drawn down to finance the additional spending pressures arising as demands on health and elderly care services climb. With a prudent fiscal framework in place, Sweden's public finances are already in much better shape than in most other OECD countries and the country has virtually no net public debt. According to OECD estimates, structural budget surpluses have averaged around 1¾ per cent of GDP since 1997. But tax cuts and expenditure increases, including for sickness and disability benefits, have produced an estimated structural surplus of only ¾ per cent of GDP in 2003. Hence, strict fiscal discipline will be necessary to achieve the government's objective of a 2 per cent surplus over the cycle. The costs of slippage in meeting this target could be significant and might jeopardise long-term fiscal sustainability.

Central government budget margins need to be preserved…

The government's present projections show a return to a 2 per cent general government surplus by 2006, but only on the assumption that outlays remain well under the expenditure ceilings. If additional spending were to eat up these margins, as has tended to happen in the past, then the return to a sustainable longer-term path would be delayed, unless taxes were raised. This alternative is unpalatable, given the already very high tax wedges. The risk could be reduced if the expenditure ceilings were redesigned to ensure that budget margins are not used for discretionary spending, but reserved for unexpected macroeconomic fluctuations. For the same reason, implementation of the 121-point plan agreed between the government and its supporting parties would compromise the medium-term path if it involved additional spending: adhering to the political commitment to finance any new measures is thus crucial. More generally, ongoing reassessment of expenditure priorities and pruning low-priority programmes would help to make room for new initiatives and/or further tax cuts.

… and upward trends in local government taxes and expenditures curbed

Local authorities are also facing growing expenditure pressures, which translate into higher tax rates because they are required to balance their budgets. Large local tax hikes have partly undermined the labour supply benefits of the state income tax cuts. At the same time, the equalisation

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scheme may to some extent discourage local authorities from expanding their tax base. One option might be to shift part of local governments' revenue base to property, which would also make local tax revenues less susceptible to fluctuations in economic activity, while another alternative would be to make counter-cyclical adjustments to state grants. Both of these would reduce the tendency for the present balanced-budget rule to ratchet up local expenditures during economic upswings, and then increase tax rates when activity slows.

Refining the fiscal framework would help long-term policy appraisal

For the longer term, it would be useful to refine the current framework so that all the instruments and targets are brought together in a more integrated and transparent way. This would involve interlinking the expenditure ceilings, the balanced budget requirement on local authorities and the employment rate target of 80 per cent with the government's objective for the national accounts-based general government surplus. It would also make it easier to assess the additional policy efforts required, indicate how improvements on one element would flow through to the others, and provide a consistent benchmark for measuring progress against long-term objectives.

Environmental policies have been successful, but the cost-effectiveness of policy can be improved

Another element of long-term sustainability concerns the environment. Considerable progress has been made in reducing air and water pollution and increasing recycling rates. Further policy developments in these areas would benefit from greater use of cost-benefit analysis.

- For air pollution, further progress requires action reducing emissions of particles from mobile and smaller sources. Aligning vehicle taxes with emission characteristics and encouraging the purchase of lower-emission domestic boilers would be effective instruments.

- With respect to water pollution, measures aimed at reducing nitrogen balances of farms and trimming back agricultural support could help in meeting nitrogen load targets. However, due to the high costs of meeting pollution targets for the Baltic Sea in Sweden alone, an agreement with other states on introducing an area-wide

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abatement trading scheme would yield more efficient results.

- Waste management and recovery policies could be based more clearly on cost-effectiveness, in order to meet policy objectives at least cost, and avoid using policies that exceed the external costs at the margin. For example, the prohibitions on landfilling could be relaxed in some regions to allow greater flexibility in implementing waste management policy using the most cost-effective option.

Summing up

With sound monetary and fiscal frameworks and a good macroeconomic position, Sweden is better placed to face the longer-term challenges of an ageing population than many other OECD countries. As elsewhere, these demographic trends will put downward pressure on growth in potential output and increase burdens on public expenditure over time. Managing this situation effectively within the welfare state approach while enjoying the highest possible living standards will be easier if the country can raise productivity growth, expand the effective labour supply, and get public finances quickly back onto their intended medium-term path. Productivity growth could be boosted further by increasing the pressures from competition in all parts of the economy, building on earlier liberalisation efforts. Effectively addressing cartels and other anti-competitive behaviour, removing regulatory obstacles and further exposing the large public sector to competition would all help. Productivity growth could also be raised if workers and capital moved more quickly and easily in response to shifts in costs and demand for different outputs. However, taking the rewards for productivity improvements in the form of lower working hours diminishes the tax base; further reductions in high marginal tax rates would reduce the incentive to do so. Although employment rates are already high overall, more could be done to get young people qualified and into the workforce earlier and to improve the integration of immigrants. Reducing the numbers of people drawing sickness and disability benefits will not only boost effective labour supply but also help ease expenditure pressures. More generally, tight

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expenditure restraint will be necessary to attain the government's surplus target. The budget framework could be refined in ways that would help to ensure that public finances are sustainable in the long term, while also providing room to moderate marginal tax rates in order to boost incentives to work, save and invest. Further progress towards environmental sustainability could focus on improving the cost-effectiveness of policies. Ultimately, creating faster growth and maintaining the welfare state depend on continuation of far-sighted economic management.

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1. Key challenges for living standards and the welfare state

Sweden's economic performance has remained robust despite the recent global slowdown. The macroeconomic framework put in place in the 1990s has worked well and places the country in the enviable position of being able to focus its attention primarily on addressing longer-term challenges. A large number of other OECD countries face similar challenges, often to a greater degree and from a less favourable initial position. This might tempt some in Sweden to believe that taking steps to adjust policies can be postponed to some later date, when problems have become evident and pressing. But taking steps in the right direction sooner rather than later leaves more options and alternatives available and allows more scope for phasing in policies that allow people and businesses a reasonable time to adjust. Sweden has already made considerable progress in implementing structural reforms (Annex 1.1). It could find itself much better off in the longer run by taking the opportunity now to address those remaining areas where policies could contribute more positively to meet the twin challenges of raising average living standards and preserving the solidarity principles at the heart of the welfare state.

Demographic developments are probably the single most significant and largely predictable factor determining Sweden's economic and social future. As with most OECD countries, Sweden faces a significant rise in the old-age dependency ratio in coming years: this will both slow the rate of growth in GDP per person and put public finances under pressure. Of course, GDP per person is an imperfect measure of living standards, because it does not take into account a range of factors that contribute to the quality of life, such as environmental sustainability and social structure and networks. It also takes no account of the value of leisure, which generally rises with incomes. These missing factors can reduce the value of cross-country comparisons at any point in time, but probably matter less for the evolution through time, especially among OECD countries that share many similar values and aspirations. In any case, examining developments in GDP per person does provide a useful framework for assessing the factors that contribute to a country's economic well-being.

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Clearly the most favourable outcome for Sweden in the future would involve a strong increase in living standards and a well-functioning welfare state, while the worst would involve both slipping living standards and an inability to maintain the present approach to social protection. On present policies, such a worst-case scenario looks most unlikely. However, policy efforts and adjustments made in the near future, using the current window of opportunity before the demographically driven costs start to rise sharply, will be critical in determining how close Sweden comes to achieving the most favourable outcome possible. This Chapter starts by reviewing the economy's growth performance to date before analysing medium and longer-term GDP growth prospects, given present policies and demographic projections. The pressures on public finances and the requirements for longer-term fiscal sustainability are then discussed.

Sweden's long-term growth path

The Swedish economy experienced exceptionally strong growth in the second half of the 1990s, and such rates of expansion are unlikely to be seen again in the medium term. This partly reflected a cyclical rebound from the very deep recession in the early 1990s. It was also undoubtedly partly attributable to significant one-off gains from the new macroeconomic framework, which delivered both a dramatic consolidation of public finances and brought inflation down to stable, low rates. Growth was also boosted by the improvement in export competitiveness flowing from the sharp depreciation of the krona in 1992. A further contributing factor was the extraordinary experience of the telecommunications equipment sector where exports expanded by around 50 per cent over the three years to 2000, when they accounted for around 20 per cent of the total (albeit at the same time pushing down the commodity terms of trade) (Box 1.1). The international ICT bubble has since burst, and such spectacular growth rates are unlikely to be repeated.

Over the long term, Sweden has experienced an almost continuous rise in living standards but at a slower rate than for the European Union and the OECD as a whole (Figure 1.1). Even with the deep economic crisis of the early 1990s, Swedes were on average almost 20 per cent better off in 2002 than they were in 1990. However, trend growth in GDP per capita was slower than in most OECD countries for the 1990s as a whole (Figure 1.2). This is the combined effect of a high rate of increase in GDP per employed person and a major shedding of labour from the workforce. Understanding the driving forces behind these developments depends on looking more closely at the factors determining Sweden's GDP per capita.

Generating output growth

Economic output grows with increases in the total number of hours worked in the economy as a whole and improvements in labour and multi-factor productivity. In Sweden's case, trends in labour utilisation have tended to reduce

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Box 1.1. Developments in the telecommunications equipment sector

The telecommunications equipment sector expanded rapidly in Sweden during the 1990s. In the period 1994-2000, value added in the sector provided on average around ¾ percentage point to the annual average growth in GDP (as estimated from the production side, basic prices) of 3¾ per cent. However, with the collapse of the global market for telecommunication products, the value added of the sector fell markedly in 2001 and 2002.

The estimated volume growth of value added in the telecommunications equipment sector is highly dependent on the method used for deflation. In line with Eurostat recommendations, Statistics Sweden follows the principle of double deflation in the national accounts: the production value and the value of intermediate consumption in production are deflated separately, and value added is then obtained by the difference of the two volume series (Statistiska centralbyrån, 2003). Since the price index for production value has fallen around 9 per cent annually on average, while the price index for intermediate consumption has been stagnant or slightly increasing, this method gives rise to substantial recorded annual growth in value added (volume) of the sector, averaging nearly 70 per cent per year over 1994-2000.

Since overall GDP is primarily estimated from the expenditure side (*i.e.* consumption, investment and net export) in the Swedish national accounts, the high volume growth in value added of the telecommunications equipment sector affects recorded GDP volume growth to the extent that it is reflected in expenditure components. For instance, the falling product-price index of telecommunications equipment also affects the deflators for various expenditures, and thus ultimately the GDP-deflator. As most of the production of telecommunications equipment is exported, this is especially evident in the development of export prices. While Sweden's terms of trade excluding telecommunications equipment fell by nearly 1¾ per cent between 1993 and 2002, the total terms of trade drop was actually 10 per cent (Konjunkturinstitutet, 2003a).

the potential growth rate. Total numbers employed shrank by 10 per cent between 1990 and 1997, and only around half of that loss has been restored by employment growth since then, leaving the employment rate for 20-64 year-olds still around three percentage points below the government's 80 per cent target. As Sweden already has relatively high employment rates, the pool of additional labour that could be mobilised is smaller than in most countries (Figure 1.3). However, there is a larger gap between those employed and those actually at work in Sweden than for most countries, with the difference mainly accounted for by those on sickness and maternity leave (OECD, 2003a). Furthermore, a closer look at Sweden's evolution indicates that part of the rise in the employment rate from 1998 onwards is accounted for by more people being absent from work[1] (Figure 1.4).

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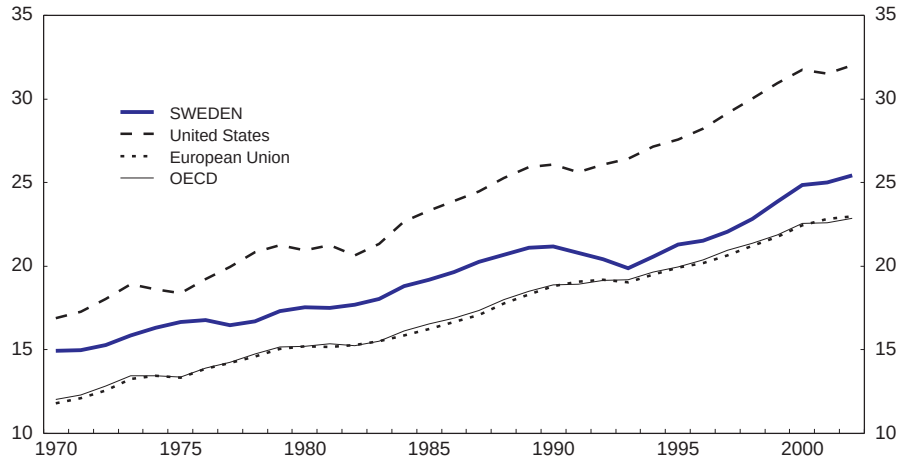
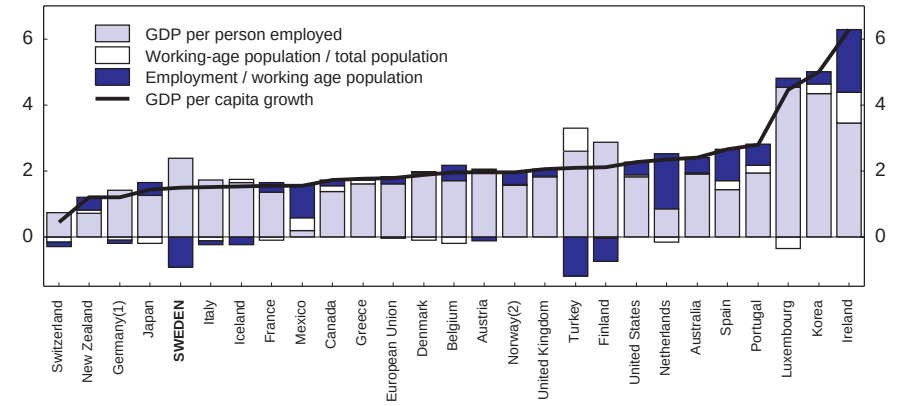
Figure 1.1. **Real GDP per person**
Thousand US dollars, 1995 PPPs



Source: OECD.

Figure 1.2. **Driving forces of GDP per capita growth**
Trend series, average annual percentage change, 1990-2000



1. 1991-2000.
2. Mainland only.

Source: OECD.

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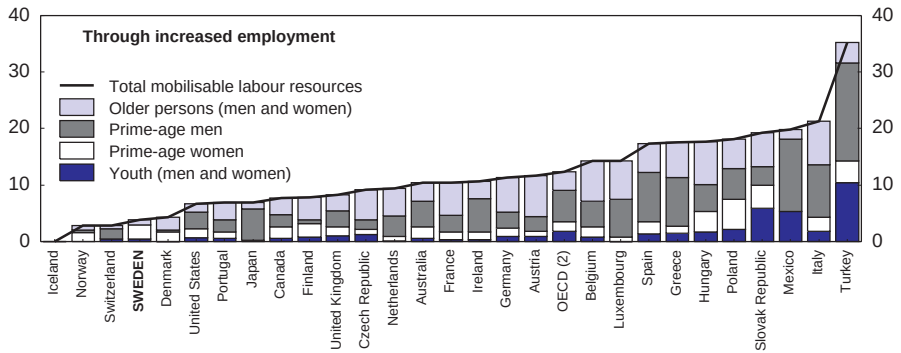
Figure 1.3. **Labour resources that could be mobilised**

Per cent of working-age population[1]

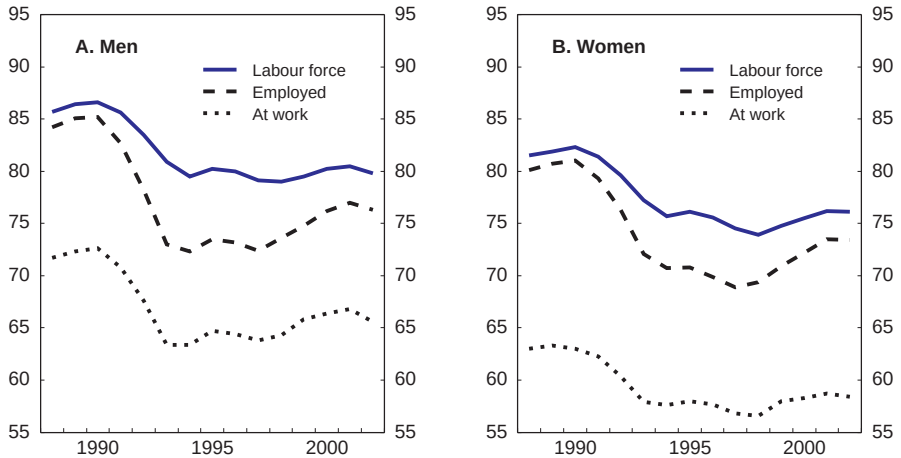


1. Mobilisable labour resources are shown as the vertical sum of excess unemployment and excess inactivity. Excess unemployment is defined as unemployment above 5 per cent of the labour force (if any). Excess inactivity is based on comparisons between actual participation rates for cells defined by age and gender, and international benchmark rates.
2. OECD: Population-weighted average for the 28 countries shown.

Source: OECD (2003a).

Figure 1.4. **Labour force participation**

Per cent of population



Source: Statistics Sweden; OECD.

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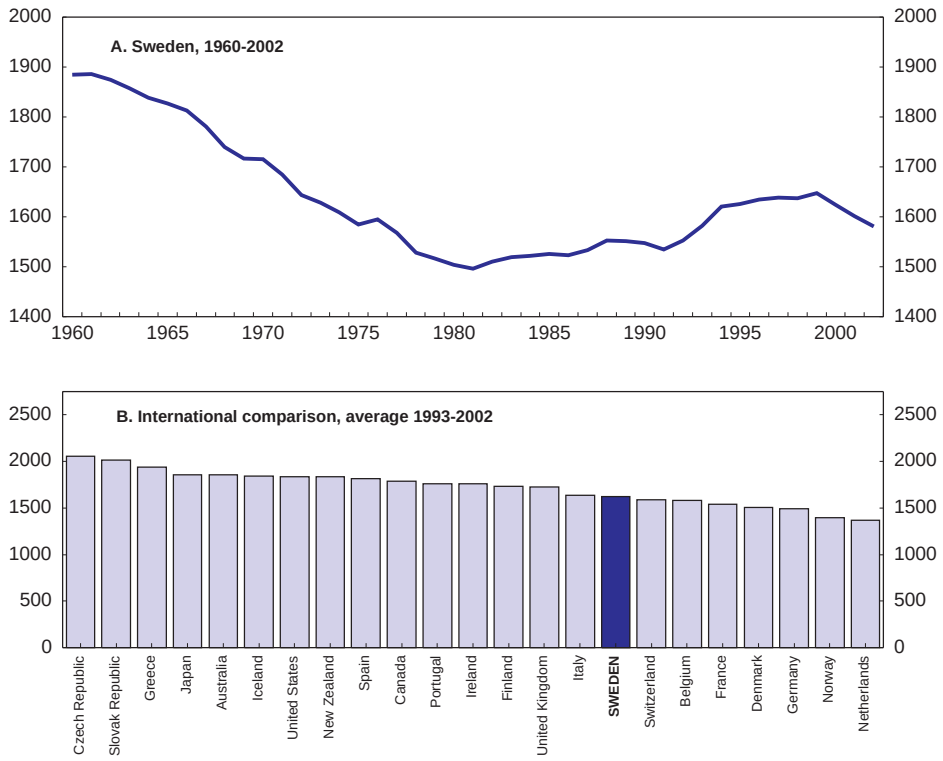
Overall, there might actually be more scope for increasing total hours worked than the already high employment rate might suggest.

Average hours worked per person is the other component of labour utilisation. Annual average hours worked in Sweden have fallen by around 15 per cent since 1960 (Figure 1.5), despite the rising proportion of women working full-time. Swedes also work fewer hours per year on average than workers do in most other OECD countries. While a higher proportion of women are employed than in other countries, they are significantly more likely to work part-time than Swedish men or than women in many other countries. This may be because more part-time jobs are available in Sweden. In the OECD overall, only a small proportion of part-time work is involuntary; this suggests that many women prefer to work part-time

Figure 1.5. **Average hours worked**

Annual average per worker



Source: OECD.

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(OECD, 2003a). To some extent, this downward trend in working hours reflects an increasing preference for leisure as incomes rise. Nevertheless, GDP per person today would have been noticeably greater if average working hours had not declined as sharply.

A striking feature of the Swedish story is the long-term shift in the composition of employment, which stands in sharp contrast to the OECD as a whole (Figure 1.6). This also helps clarify why Sweden's income per capita has slipped relative to the OECD. OECD-wide business sector employment is about 40 per cent larger now than it was in 1970, whereas in Sweden the business sector actually employs around 4 per cent fewer workers today than in 1970. Over the same period the share of government employment in the OECD workforce has remained fairly steady at around 13 per cent, but has climbed in Sweden from 19 per cent to 29 per cent. This evolution follows on from an 85 per cent increase in Swedish public sector employees between 1960 and 1970 and reflects the expansion in the provision of public services associated with the build-up of the Swedish welfare state. In turn, this has a significant impact on the evolution of overall productivity growth.

Overall productivity growth reflects the combined impact of productivity gains in the business sector and in the general government sector. Labour productivity growth in the Swedish business sector has been reasonably creditable over the past 40 years: it has outstripped the United States and is catching up to the European Union average. But it is more difficult to make labour productivity gains in the public sector because most public services are by nature labour intensive, and the absence of competition means less pressure to seek productivity improvements. Gains that are achieved are often very difficult to measure as there is no "price" for the outputs produced. In any case, productivity growth for the general government sector is set to zero by convention in the Swedish National Accounts. The increasing weight of this sector in the economy is a major factor in the slower overall labour productivity growth the country has delivered.

Sweden's business-sector productivity performance reflects three main factors. F*irst*, a high rate of investment in physical capital has provided greater opportunities for lifting labour productivity per hour worked, in part by diffusing new technology, as embodied in more recent vintages. Sweden's business investment rate has broadly followed developments in the OECD as a whole over a long period (Figure 1.7). S*econd*, and more recently, it has had one of the highest economy-wide rates of investment in ICT, especially software, which is increasingly recognised as a key element of firm-level productivity growth (Figure 1.8). T*hird*, Sweden has an increasingly skilled workforce, with a high proportion of people under 55 years old with at least an upper-secondary education (Figure 1.9).

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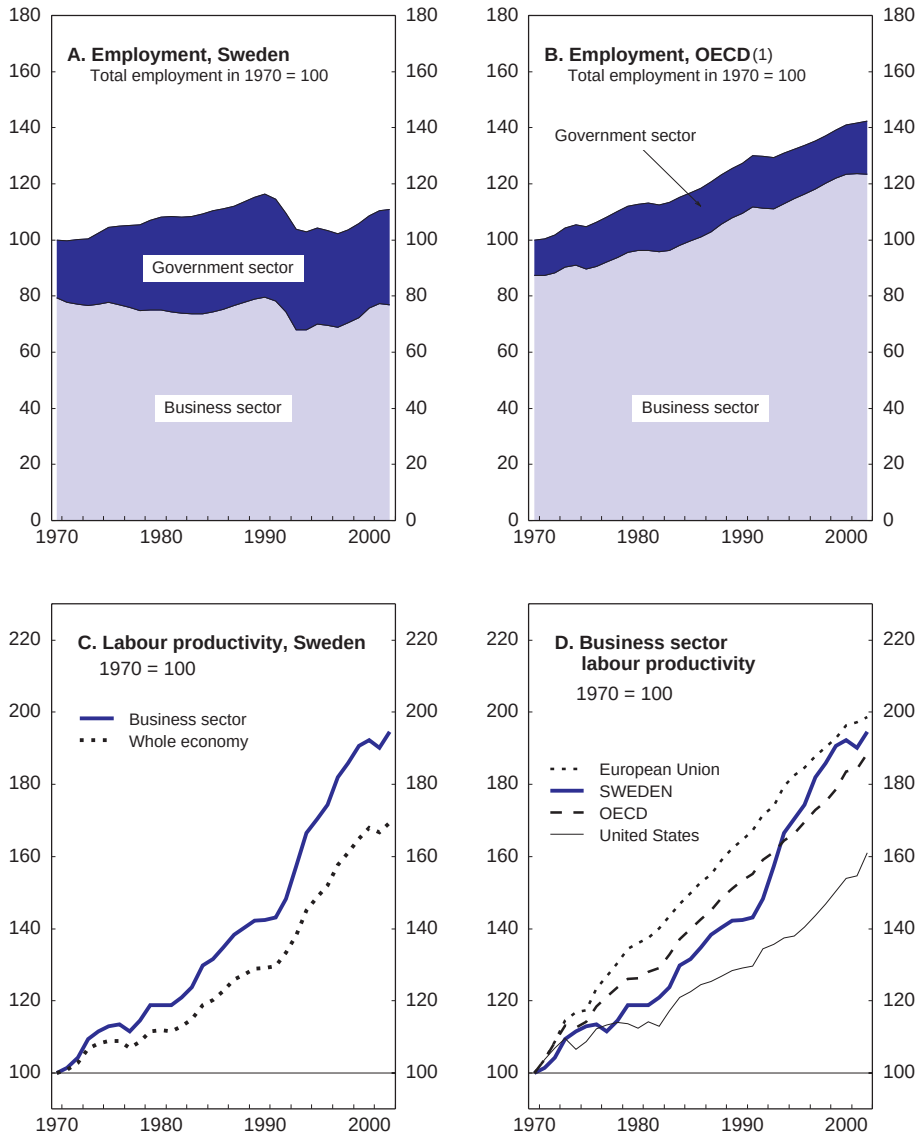
Figure 1.6. **Composition of employment and labour productivity**



1. OECD excluding Korea, Luxembourg, Czech Republic, Hungary, Poland, Slovak Republic, Mexico and Switzerland.
Source: OECD.

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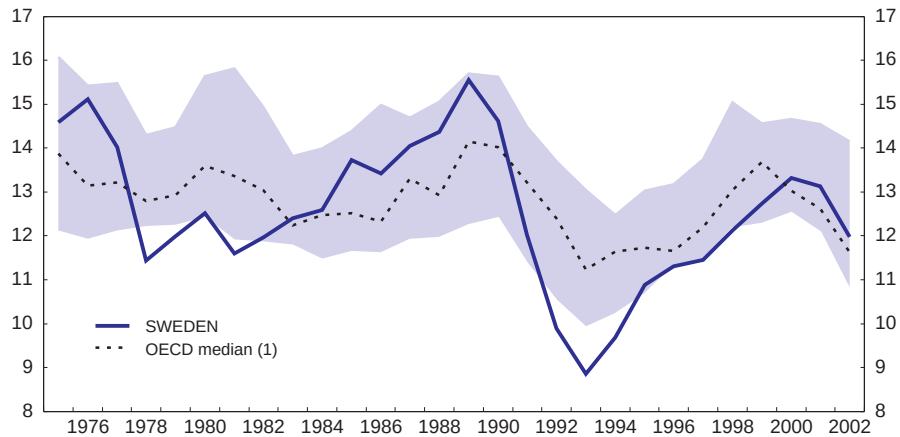
Figure 1.7. **Business investment rate**
Percentage of GDP



1. 23 countries. Shaded area shows the middle two quartiles (*i.e.* half the countries fall in this range).
Source: Statistics Sweden; OECD estimates and OECD.

The redistribution of output

GDP per person depends not only on the output produced, but also on how many people each worker is supporting. The dependency ratio (based on the working-age population) has played no significant part in the evolution of GDP per person in Sweden over recent history because it has remained broadly constant. But the *effective* dependency ratio, the ratio of all people not at work (including children and the elderly) to those actually at work on any given day has increased as employment has fallen and absences have risen (Figure 1.10). This greater redistribution element is reflected in a slower growth of GDP per person. And as almost every adult who is not working is receiving some sort of public income support in keeping with the Swedish welfare state approach, reducing the *effective* dependency ratio would also improve public finances by expanding the tax base and reducing public expenditures. Reducing absence due to sick leave would be a particular case in point.

Overall, these features of Sweden's long-term economic evolution also indicate the key factors that will determine its long-term outlook, namely labour supply, productivity and dependency. Sweden's current macroeconomic situation and short-term prospects indicate that the country has weathered the international slowdown relatively well, and as the next section illustrates, there are no indications of any underlying macroeconomic problems to be addressed. Furthermore, the recent slowdown illustrates the value of having put in place a well-

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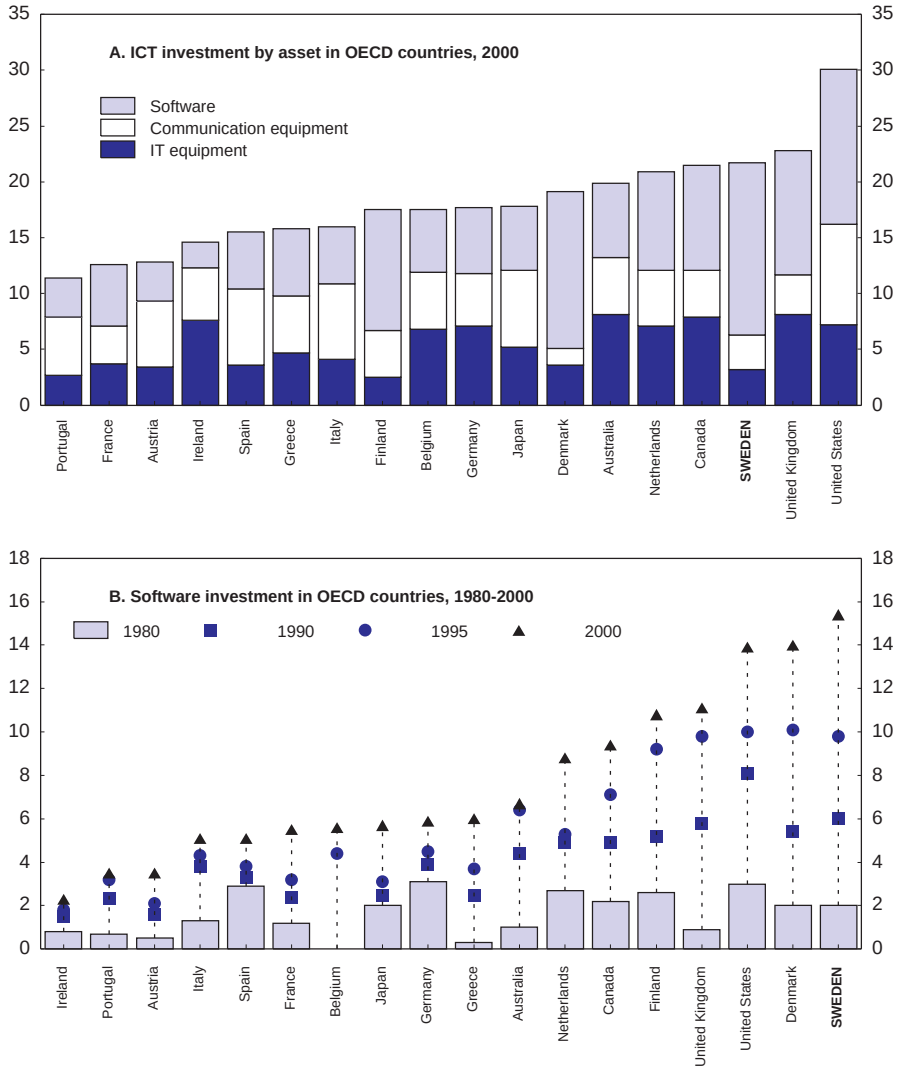
Figure 1.8. **ICT investments**[1]

Percentage of non-residential gross fixed capital formation, total economy



1. ICT equipment is defined as computer and office equipment and communication equipment; software includes both purchased and own account software. Software investment in Japan is likely to be underestimated, owing to methodological differences.

Source: OECD Science, Technology and Industry Scoreboard.

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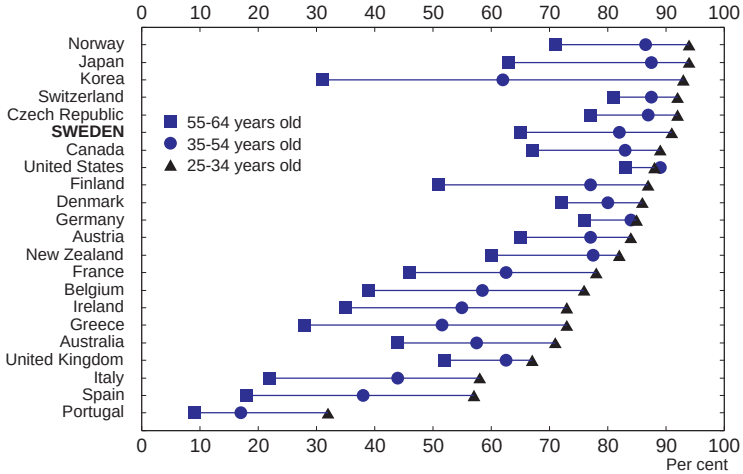
Figure 1.9. **Population with at least an upper-secondary qualification**

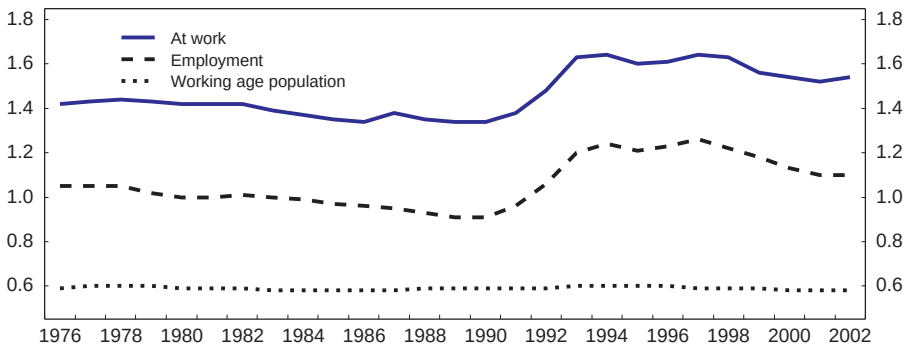
2001, per cent



Source: OECD.

Figure 1.10. **Dependency ratios**

Per cent[1]



1. The dependency ratio for working age is calculated as the proportion of those aged 0-15 or 65 and over relative to 16-64 year olds. The employment dependency ratio indicates the proportion of the non-employed (including children and elderly) relative to the employed. The at work dependency ratio indicates the number of people supported by each person actually at the workplace.

Source: OECD.

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designed macroeconomic framework. More importantly, recent macroeconomic developments should not be used as an excuse for inaction: a cyclical upswing will not resolve the longer-term challenges and putting off new measures until the economic situation is more buoyant will simply shorten the time remaining before the demographic window of opportunity closes.

The macroeconomic framework and the current economic situation

Sweden has done relatively well during the recent international downturn and enters the projected upswing without any underlying macroeconomic imbalances (Annex 1.2). The main forces shaping recent developments are the following:

- The strong automatic stabilisers have been free to operate, as public finances were brought into a substantial surplus position during the previous upturn (Figure 1.11) and public debt was significantly reduced. The timing of large tax cuts and other discretionary fiscal easing in 2001 and 2002 proved to be fortunate.
- With the tax cuts, household disposable income growth has remained buoyant, despite stagnating overall employment since early 2001. However, against the background of a collapsing stock market and rising unemployment, households reacted by raising their savings ratio further in 2001 and 2002. Developments on the housing market, on the other hand, have continued to nourish household spending during the economic slowdown. The real price of owner-occupied housing has risen further – although at a slower pace in recent years – providing security for additional take-up of loans.
- The collapse of international markets for telecommunication products hit this sector hard and led to substantial restructuring efforts, although other parts of the business sector have fared better. The business sector reacted to a sharp fall in capacity utilisation in manufacturing during 2001 by gradually reducing its investment ratio. Adjustments of capacity have also taken place *via* labour input: total hours worked in the business sector fell by 2 per cent in 2002 and a further fall took place in 2003.
- Increased hiring in the public sector has sustained total employment, while unemployment started rising from mid-2002, especially for young people, as net inflows to the labour force increased. Part of the increase in unemployment is related to the completion of the Adult Education Initiative (AEI), which had previously taken up to 4 per cent of the unemployed out of the labour force while they acquired educational qualifications.
- With the recent slowdown in GDP growth, the OECD's estimate of the output gap turned negative in 2003, indicating that some spare capacity

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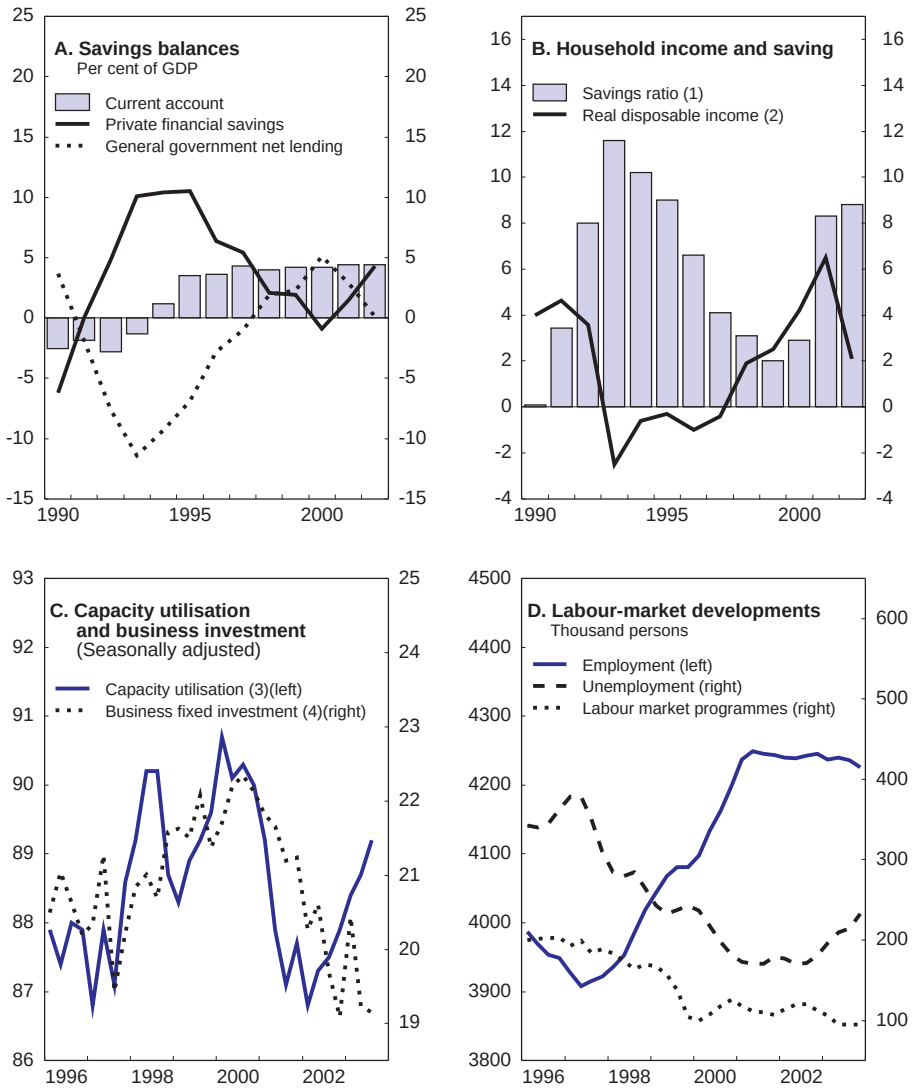
Figure 1.11. **Forces shaping output**



1. Per cent of disposable income. Including household saving in pension fund reserves.
2. Annual percentage change
3. In manufacturing.
4. Per cent of business sector value added.

Source: Statistics Sweden; OECD.

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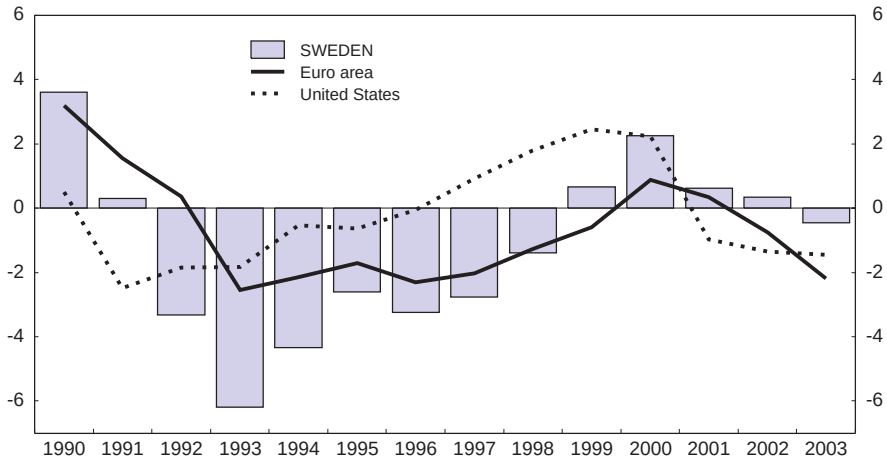
Figure 1.12. **Output gap**
Per cent of potential GDP



Source: OECD.

is available in the economy (Figure 1.12). Inflation pressures have also eased, as reflected in underlying inflation developments and lower wage increases.

Macroeconomic policies

Monetary policy-making remains unchanged after a solid majority of Swedes voted "no" to joining the euro at the referendum in September 2003 (Box 1.2), with the Riksbank setting its policy rates to steer inflation towards the target rate of 2 per cent on a one to two year horizon. The Riksbank has eased monetary policy since November 2002, and the repo rate is now at an unprecedented low level of 2¾ per cent (Figure 1.13). Consumer price inflation has fluctuated somewhat over the same period, but these fluctuations have been well understood as temporary and have not required a shift in policy stance. An appreciation of the Swedish krona by around 14 per cent in the last 2½ years – following the reversal of interest spreads *vis-à-vis* the euro area in 2001 – has helped to dampen price increases, and inflation expectations one to two years ahead stand at 2.2 to 2.4 per cent (according to Prospera Research), not far from the Riksbank's target.

Economic activity received a significant boost from fiscal policy in 2001 and 2002, with large income tax cuts (primarily the second and third steps in the

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Box 1.2. Sweden's monetary policy framework and the euro referendum

Sweden's present monetary policy framework emerged from the decision to allow the krona to float in late 1992. According to the Riksbank Act, the monetary policy target is to maintain price stability: since January 1993 the Riksbank has defined this as holding the rate of CPI inflation at 2 per cent ±1 percentage point. Broadly speaking, if the Riksbank's forecasts for one to two years ahead indicate that inflation will deviate from the target then the Bank will adjust the repo rate.

When Sweden negotiated its membership of the European Union from 1995, it was announced that consideration of the introduction of the common currency would be put to the Swedish Parliament as a separate issue. In view of weak public support for the euro in Sweden, Parliament decided that the country would not participate from its outset in 1999. At the same time it requested that work on euro preparations be carried out, in order to facilitate a future introduction of the euro. In its assessment in May 1998, the Ecofin Council stated that Sweden did not fulfill the necessary conditions for adopting the euro.

In March 2003, Parliament decided to hold a consultative referendum on the introduction of the euro on 14 September 2003, where the question "Do you think Sweden should introduce the euro as its official currency?" would be put to the people. Some 56 per cent voted "no" and 42 per cent voted "yes", with a voter turnout of just over 81 per cent. As a result, Sweden will remain outside the euro area for the time being and the monetary policy framework will continue to apply unchanged.

In any event, Sweden is now well placed to fulfill the economic convergence criteria set out for euro membership. This reflects the country's pursuit of prudent macroeconomic policies within the current framework.

government's multi-year tax-cut programme) and other discretionary measures together amounting to around 1¾ per cent of GDP each year. General government net lending consequently plunged from 4.6 per cent of GDP in 2000 to 0.1 per cent of GDP in 2002, when adjustments are made for one-off effects (Table 1.1), with the working of automatic stabilisers over the course of the economic downturn adding to the deterioration. The government's objective is to maintain a 2 per cent surplus for general government finances over the cycle. But tax cuts and expenditure increases, including for sickness and disability benefits, have pushed the OECD estimate of the structural budget surplus somewhat below 1 per cent of GDP in recent years, as the economy has not deviated very far from potential and currently operates with a relatively small negative output gap. In 2003, the structural position of public finances improved, even though the output gap widened. But

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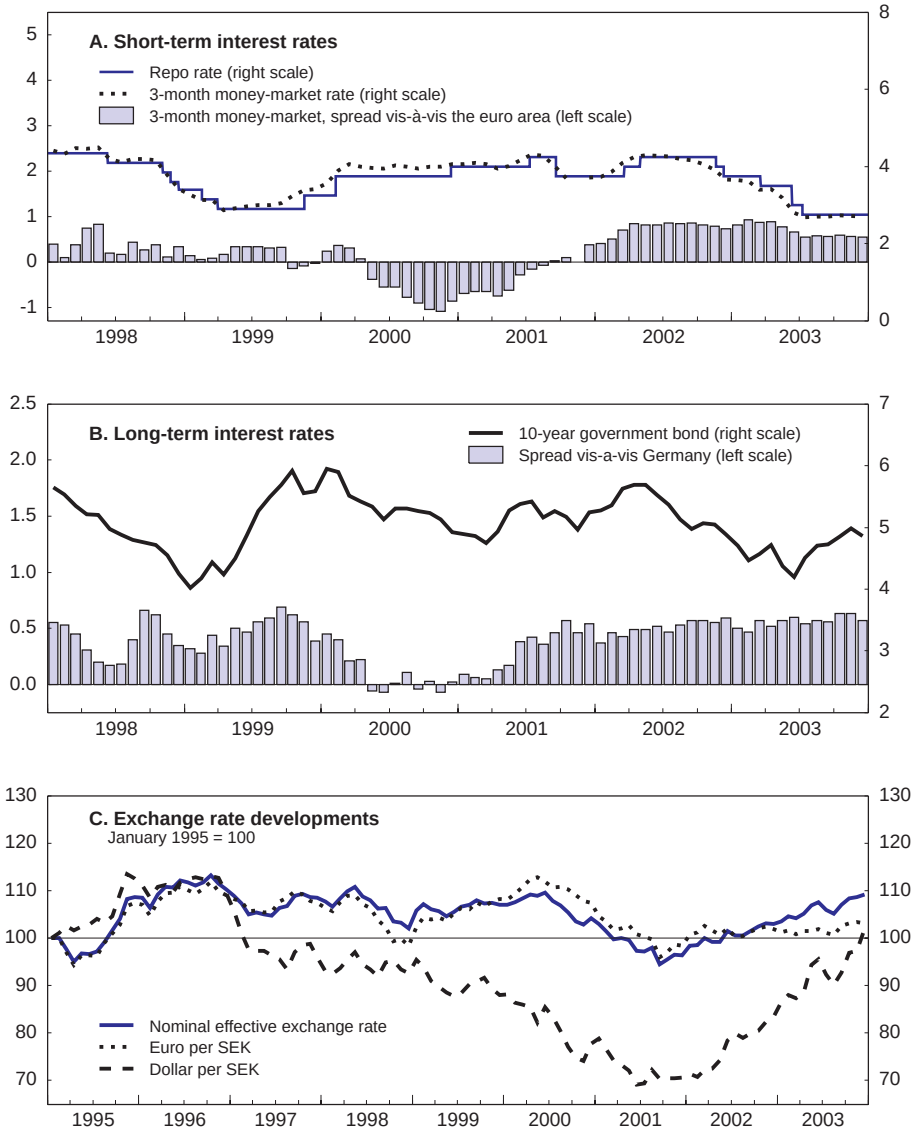
Figure 1.13. **Interest rates and exchange rate developments**

Per cent



1. Tor the euro area German government interest rates are used before 1999.

Source: Statistics Sweden; Sveriges Riksbank; OECD.

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Table 1.1. **General government financial balance**

	1998	1999	2000	2001	2002	2003[2]
Net lending	1.9	2.3	5.1	2.9	0.1	0.4
One-off effects[1]	–0.9	0.0	–0.5	0.0	0.0	0.0
Adjusted net lending	1.0	2.3	4.6	2.9	0.1	0.4
Discretionary central government fiscal policy[1]	0.2	–0.9	–1.1	–1.7	–1.8	–0.6
Cyclically-adjusted net lending[2]	2.9	1.9	3.6	2.4	–0.1	0.8

1. Contribution to change in net lending; Ministry of Finance estimates.
2. OECD estimates. Numbers differ from *Economic Outlook* 74, as account is taken of the new accrual-based national accounts. Projected net lending in 2003 is 0.2 percentage points higher, corresponding to the Ministry of Finance's estimate of the effect from accrued taxes.

Source: Statistics Sweden; Ministry of Finance; OECD.

developments at both the central and local levels of government indicate that strains may be building up in the fiscal framework, and strict fiscal discipline will be necessary in order to reach the surplus target in the medium term (see Chapter 4).

With economic growth projected to rise to 2¼ per cent in 2004 and 2¾ per cent in 2005, the output gap could soon be closed again (Annex 1.3). Given the relative position of output gaps in Sweden and the euro area, the Riksbank will probably need to react at an earlier stage of the projected upswing than the European Central Bank. These prospects also suggest that additional fiscal stimulus through discretionary easing at this point in the cycle would be unhelpful, and it would in any case make the government's task of restoring public finances to the 2 per cent surplus target more difficult. Specifically, if the fourth (and final) step of income tax reductions is carried out when the economy improves, offsetting measures on the expenditure side would be appropriate. The same holds for promises of restoring benefit levels. Even so, general government net lending is projected to rise from almost ½ per cent of GDP in 2003 to 1¼ per cent of GDP in 2005 (with taxes recorded on an accruals basis[2]), thus remaining somewhat below the 2 per cent target.

Overall, the macroeconomic framework continues to function well in stabilising the economy in the short term, allowing policy makers to concentrate their attention on medium and longer-term structural issues. But one area where short-term considerations might have very long-term consequences is in wage negotiations in the private and public sectors.[3] Following the rise in open unemployment in the last 1½ years, the forthcoming agreements are likely to involve lower increases in financial compensation than in recent years. However, some part of the overall envelope might materialise as a further reduction of working hours, fol-

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lowing the example of collective agreements in the private sector in 2001.[4] The apparent endorsement of shorter working hours in the 2004 budget could encourage such deals. This would further limit Sweden's growth potential in the medium term and slow the rate of increase in living standards, which already faces major demographic challenges.

Medium and longer-term growth prospects

Demographic and labour supply prospects

The demographic transition to a more elderly-dominated population is probably the single most important challenge to the Swedish economy in the medium and longer term. Although Sweden already has a relatively old population, it is projected to experience a rapid rise in the number of people older than 65 years (like most other OECD countries), and especially in the number of very old people, aged 80 years or more. The number of people of working age (20-64 years) will also increase in absolute numbers according to the latest population projections from Statistics Sweden (Box 1.3), although declining as a share of the total population (Table 1.2). Net immigration alone accounts for this increase in the working-age population: without this contribution, the number of people in this age group would decrease substantially in the next 30 years.

Box 1.3. **Population projections**

The analysis in this section is based on the latest official population projections by Statistics Sweden for the period 2002-50, complemented with projections of immigrant patterns by the Ministry of Finance presented in the 2004 budget. The main assumptions of the projections are:

- Average lifetime increases from 82.1 years in 2002 to 86.2 years in 2050 for women and from 77.7 years to 83.6 years for men.
- Average fertility increases from 1.65 children per woman in 2002 to 1.85 in 2010, remaining constant thereafter.
- After a small up-tick in the next couple of years, annual net immigration falls gradually from around 33 000 persons in 2004 to 23 000 persons in 2050.

These assumptions generate an increase in total population from 8.9 million in 2002 to 10.6 million in 2050. The share of the population born outside Sweden increases from 12 to 18 per cent in the same period.

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Table 1.2. **Population projections**

	2002	2010	2020	2030	2040	2050
	1 000 persons					
By age group						
0-19 years	2 144	2 140	2 232	2 339	2 348	2 411
20-64 years	5 263	5 407	5 445	5 485	5 572	5 771
65 years and above	1 534	1 719	2 042	2 280	2 431	2 448
Total	8 941	9 265	9 719	10 103	10 351	10 630
	Index 2002 = 100					
0-19 years	100.0	99.8	104.1	109.1	109.5	112.4
20-64 years	100.0	102.7	103.5	104.2	105.9	109.7
65 years and above	100.0	112.0	133.1	148.6	158.5	159.6
Total	100.0	103.6	108.7	113.0	115.8	118.9
	Proportion of total population, per cent					
0-19 years	24.0	23.1	23.0	23.1	22.7	22.7
20-64 years	58.9	58.4	56.0	54.3	53.8	54.3
65 years and above	17.2	18.6	21.0	22.6	23.5	23.0
	1 000 persons					
By origin						
Native Swedes	7 887	7 995	8 228	8 438	8 544	8 712
Immigrants from EU and the Nordic countries	380	404	403	393	384	389
Immigrants from other parts of the world	673	867	1 088	1 273	1 423	1 529
	Index 2002 = 100					
Native Swedes	100.0	101.4	104.3	107.0	108.3	110.5
Immigrants from EU and the Nordic countries	100.0	106.1	106.0	103.2	101.1	102.2
Immigrants from other parts of the world	100.0	128.8	161.6	189.1	211.4	227.1
	Proportion of total population, per cent					
Native Swedes	88.2	86.3	84.7	83.5	82.5	82.0
Immigrants from EU and the Nordic countries	4.3	4.4	4.1	3.9	3.7	3.7
Immigrants from other parts of the world	7.5	9.4	11.2	12.6	13.7	14.4

Source: Statistics Sweden; Ministry of Finance; OECD calculations.

In total, the population is projected to increase by 1.7 million between 2002 and 2050, but more than half of the increase comprises people aged 65 or above. Thus, the old-age dependency ratio (the number of people aged 65 years or more to the number of people of working age) is set to begin a marked rise after 2005 and peak around 2040 at a level 50 per cent higher than its current level (Figure 1.14). Until 2014, a falling number of under-20 year-olds relative to the working-age population will ease the effect on the total dependency ratio. However,

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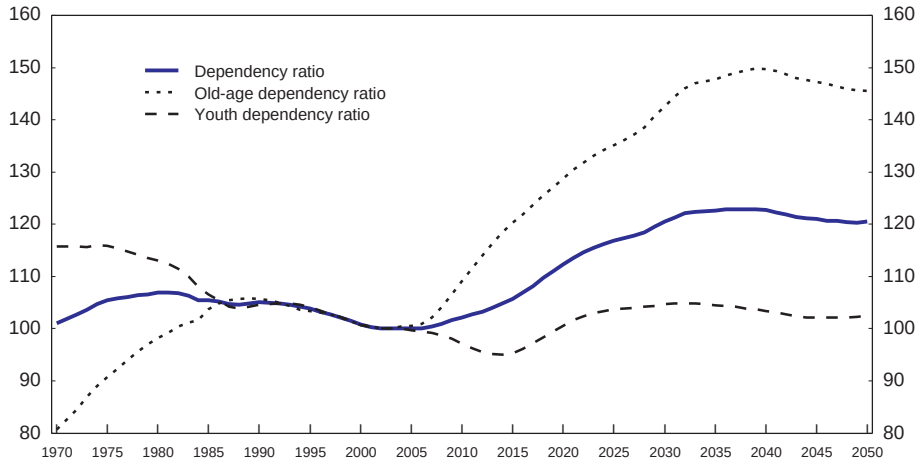
Figure 1.14. **Dependency ratios**
Index, 2002 = 100



Source: Statistics Sweden; OECD calculations.

for the rest of the period until 2050, a rising number of children will further add to an increase in the total dependency ratio, leaving it 20 per cent higher in 2050 than in 2002. It should be noted that while the ageing of the population can be predicted with a relatively high degree of certainty, the opposite is true for migration patterns. In the last decade, annual net immigration has fluctuated between 5 000 and 50 000 persons, and even small deviations from the assumptions in the projection would give rise to large cumulative effects by 2050.

As older generations within the working-age population and immigrants in general have lower labour force participation rates than native Swedes in the prime working-age group (30-55 years), a general shift towards the two former groups, both in absolute numbers and as a share of the total working-age population, will almost inevitably put a strain on labour supply. Although a high share of people in their later ages are active in the labour market when compared with other OECD countries, labour force participation rates drop around 25 percentage points for both men and women between 55-59 years and 60-64 years (Figure 1.15). A similar pattern can be seen in working hours, which to some extent results from absenteeism due to long-term sickness and unemployment being used as alternative routes of early retirement. For immigrants, the age profiles for labour force participation mirror those of native Swedes, but levels are significantly lower at all ages. This may reflect various barriers to entry into the labour

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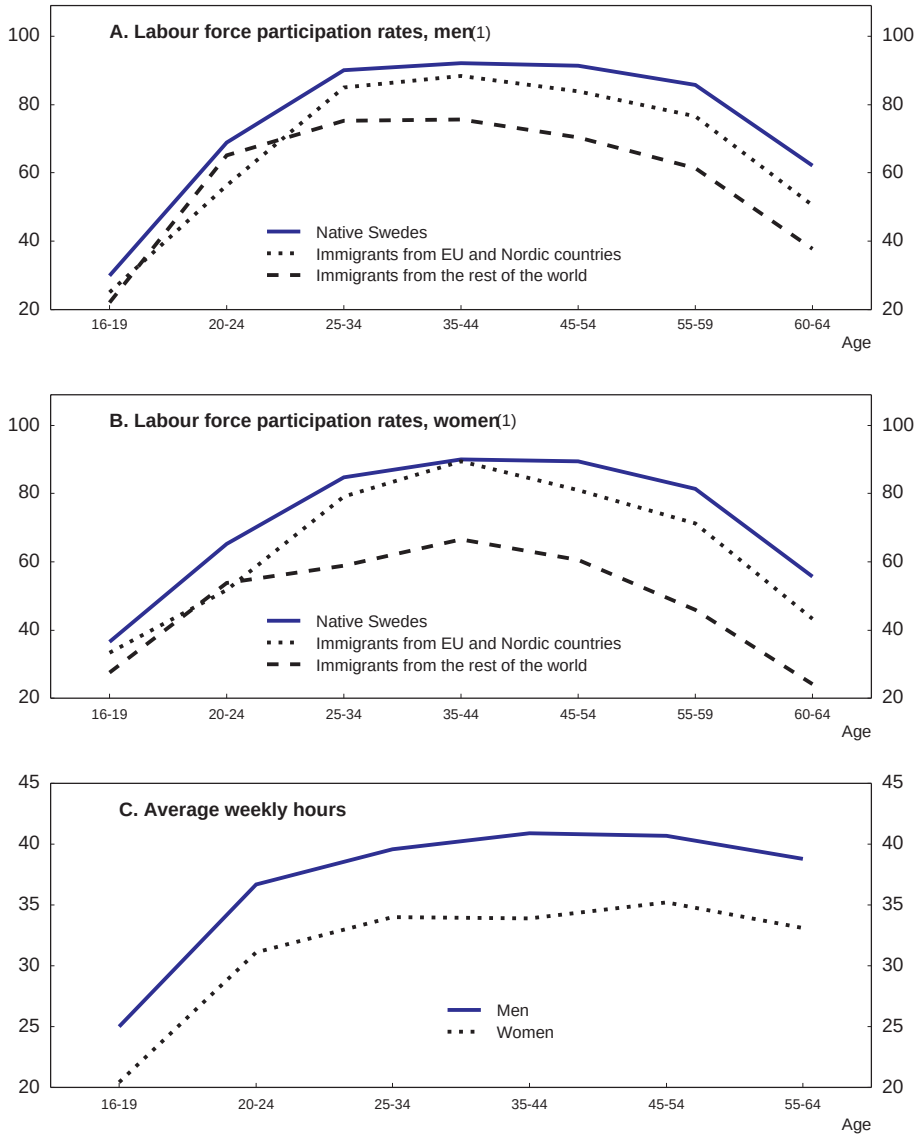
Figure 1.15. **Labour market performance, 2002**



1. Per cent of labour force.
Source: Statistics Sweden; Ministry of Finance; OECD calculations.

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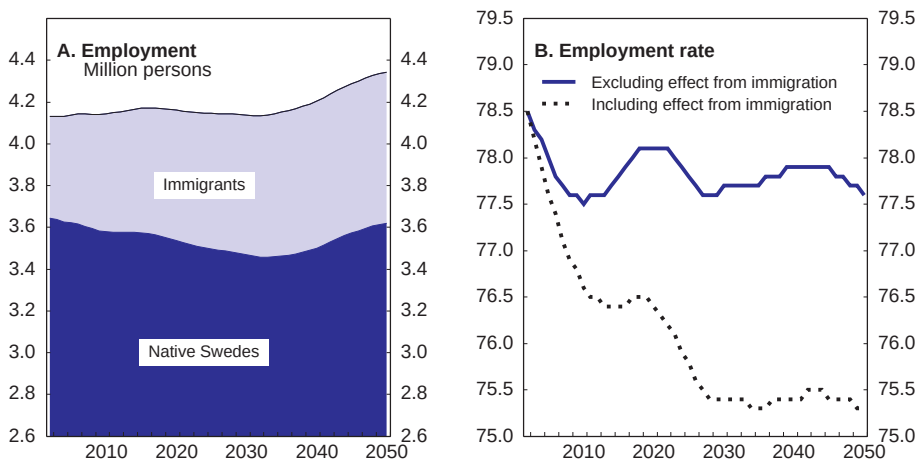
market, ranging from high replacement rates, qualification levels below those corresponding to minimum wages, and possibly some reluctance of employers to hire people with a different ethnic background (see Chapter 3).

The number of native Swedes in employment would fall substantially in the next 30 years, given present age- and gender-specific employment rates, while the assumed net inflow of immigrants could ensure a broadly unchanged total number of people in employment until 2035 (Figure 1.16). Further net immigration and increasing employment of native Swedes may raise total employment during the rest of the period until 2050, leaving it 5 per cent higher than in 2002. However, while the ageing of the population in itself would reduce the employment rate for 20-64 year olds by 1 percentage point by 2050, the assumed net inflow of immigrants would add another 2 percentage points to the fall – leaving it at nearly 75 per cent in 2050, well below the government's target of 80 per cent, and adding to, instead of reducing, the *effective* dependency ratio.

The possible effects on labour supply from demographic changes may be aggravated by a continuation of a number of labour market trends. These are:

- A long-term trend reduction of the average age of retirement. Since 1965, the average age of withdrawal from the labour force has declined from around 67½ years to 62 years for women and from 69½ to 63½ years for men (Figure 1.17).

Figure 1.16. **Demographic impact on employment**
20-64 year olds



Source: Statistics Sweden; Ministry of Finance; OECD calculations.

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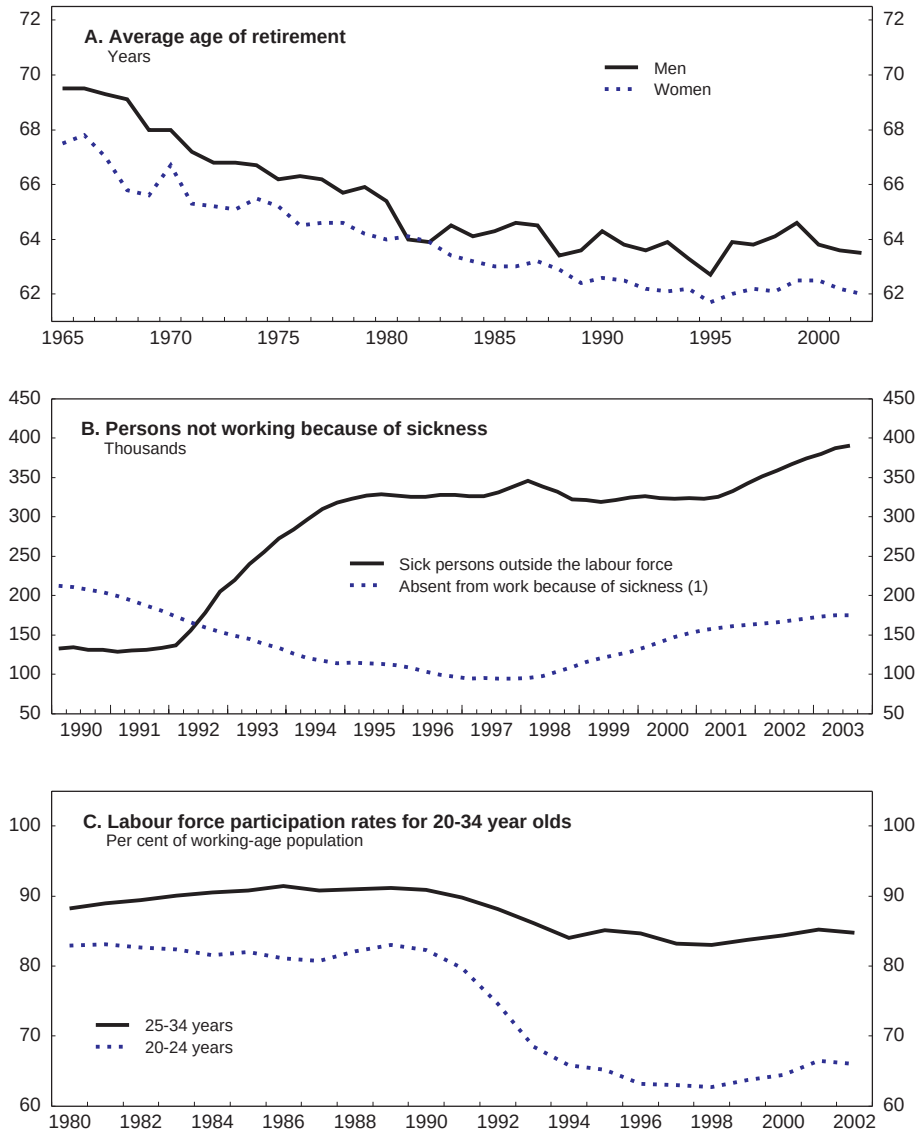
Figure 1.17. **Trends in labour supply**



1. Persons absent from work for the entire survey week because of sickness. In 2002, additional 105 000 persons on average were absent for part of the week because of sickness.

Source: Statistics Sweden; National Institute of Economic Research; OECD calculations.

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- A marked rise in persons not working because of sickness in the last five years. Since 1997, the number of employed persons reporting sick in the labour force surveys has increased by 80 000 to around 175 000, while the number of sick persons outside the labour force has risen by 60 000 to around 390 000. These increases sum to around 3¼ per cent of the labour force.
- Later entry into the labour market, partly due to postponed/prolonged higher education. The participation rates for people aged 20-34 have on average fallen by 10 percentage points from 1980 to 2002, with the major part of the decline taking place between 1990 and 1997.
- Declining average annual working hours. Despite some increase in the last 20 years, which might be attributable to a larger number of especially women shifting from part-time to full-time jobs, annual working hours per employed person have fallen by 0.4 per cent per year on average since 1960.

While the demographic changes, including the assumed net immigration, could raise total hours worked by almost 5 per cent between 2002 and 2050, continuation of trends towards lower participation rates and fewer annual working hours might instead leave total working hours nearly 9 per cent lower than in 2002 (Table 1.3). On the other hand, better integration of immigrants in the labour market could make for a substantial positive contribution to the total number of hours

Table 1.3. **Scenarios for total hours worked**

Per cent

	Changes from 2002		
	2010	2020	2050
Total demographic impact on total hours worked, basic scenario	0.1	0.6	4.9
of which:			
Size of native working age population, 20-64 years	–0.4	–2.1	0.4
Size of immigrant working age population, 20-64 years	3.1	5.6	9.2
Composition effects from employment rates (age, gender and origin)	–2.4	–2.7	–4.5
Composition effects from working hours (age, gender and origin)	–0.2	–0.1	–0.3
Additional change from prolonged trends			
Trend towards lower participation continues until 2020	–4.3	–9.4	–10.1
Trend towards lower working hours continues until 2020[1]	–1.6	–3.5	–3.5
Additional change from better integration of immigrants[2]	2.1	4.2	8.9

1. An annual decrease in average working hours per year of 0.2 per cent is assumed. That is only half of the average annual decrease between 1960 and 2002, reflecting that a further shift from part-time to full-time employment might take place in the future.
2. Half of the difference in employment rates to native Swedes is assumed to be eliminated by 2020.

Source: Statistics Sweden; Ministry of Finance; OECD calculations.

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worked. These scenarios indicate that the development in labour input will be determined to a large extent by the behavioural patterns of the population, and that the potential gains from effective policies to raise labour supply could be substantial.

Future growth paths for GDP *per worker and* GDP *per capita*

The steep rise in the dependency ratio suggests that maintaining the recent growth in GDP per capita will prove increasingly difficult in the future. The direct effect on labour supply from the demographic changes alone points to a slowdown of average annual growth in GDP per capita to around 1½ per cent, which is almost ¾ percentage point lower than the average over the last 40 years. Thus, although Swedes are unlikely to see a reduction of living standards, they will probably have to adjust their expectations about the rate at which these will materialise. If trends towards lower participation and working hours are prolonged, average annual growth in GDP per capita might even be ¼ percentage point slower than what the pure demographic changes would suggest.

While the calculation above is based on a trend growth rate in private sector (hourly) labour productivity of 2¼ per cent per year, demographic changes might also affect GDP per capita through changes in the growth rate of labour productivity. If the average immigrant continues to have lower productivity than the native Swedish counterpart, average productivity growth could be reduced as immigrants constitute an ever increasing part of the workforce. In contrast though, some lower skilled, older Swedes will move out of the working-age population.

Another aspect of the demographic changes is that the demand for labour-intensive services, such as health and elderly care, would increase with the rising share of elderly in the population. As these are still extensively provided by public sector employees, a relative shift of employment could probably be expected. In fact, the Ministry of Finance has estimated that an increase in public employment of 450 000 people (11 per cent of current employment) would be required if the production of public goods and services were to take place with the same labour intensity as today (Regeringen, 2003). That in turn would imply an absolute reduction of private sector employment by almost 240 000 people (8 per cent), if the labour force develops according to the baseline scenario. As national accounts in general do not measure productivity increases in the non-market sector, the relative shift towards employment in the government sector would dampen recorded productivity increases, all other things equal. However, where these services are exposed to competition (even if funding remains public), gains in productivity can be expected and are also easier to measure using market pricing.

One aspect that often gets less attention when assessing the development of living standards is the extent to which households take out part of the

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gains from labour productivity increases as leisure. Thus, a measure of living standards based solely on GDP or national income per capita would neglect that leisure may be just as valuable for the individual as market consumption possibilities and therefore naturally should increase along with productivity improvements. However, the choice between employment-based income and leisure when reaping productivity gains is significantly distorted in favour of leisure when an economy to a large extent is based on redistribution of resources and collective consumption that requires high tax ratios.[5] This in turn makes it harder to attain a certain level of redistribution and collective consumption financed by tax revenues. The reduction in average annual working hours in the last 40 years should be seen in that perspective, as taxes have increased rapidly to finance the expansion of the welfare state. This also gives an indication of what can be expected in the future: if the fiscal pressures from the demographic changes give rise to increases in tax rates, households may try to shift from work to leisure to an increasing extent. A vicious circle could result, which would ultimately endanger the welfare state.

Thus, while choosing more leisure is a perfectly valid way of reaping productivity gains for the working individual, it needs to be recognised that this may be incompatible with expectations about future tax-financed public service provision. Intergenerational distribution is the core issue. Workers in the future might not stand ready to pay higher taxes to support a growing non-working part of the population. For people in the working-age population at any one time, it is therefore necessary to take into account that deciding to work less implies a higher risk of facing lower tax-financed provision of transfers, services and consumption goods later in their lives (when they are most needed). This is obviously a very complex issue for the individual, but clear communication of this link between work now and welfare later in life might influence behavioural patterns of individuals as well as of their various representatives in organisations and political parties.

Public finances coming under pressure again

If public service standards are to be maintained in the future and public debt is to stay at a sustainable level, initial preparation in the form of significant debt reduction is required over the next 10 years or so before the demographic deterioration becomes more significant. However, the development of public finances in recent years has revealed strong underlying spending pressure coming from a widespread perception that earlier consolidation efforts were necessary only to restore public finances, while the generosity of the welfare state could be revived once the economy got back onto a stable footing (see Chapter 4). This raises some concern for the sustainability of the current position of public finances and especially the prospects of again meeting the 2 per cent structural surplus tar-

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get, as well as for labour supply when the ageing of the population starts to become visible.

Even though strains may already be building up, Sweden is in many ways in an enviable position to meet the future demographic challenges. F*irst* of all, gross and net public debt and thus net interest payments are at relatively low levels (Figure 1.18). S*econd*, general government finances are in a surplus position, in contrast to many other countries facing similar demographic difficulties. T*hird*, estimated increases in public expenditure related to the ageing population are somewhat less than elsewhere (Figure 1.19). And, *finally*, early recognition of the likely pressure on public pension expenditure from demographic changes has prompted the country to reform its public pension system, making it largely robust to changes in average expected lifetimes (Box 1.4).

Notwithstanding its relatively good initial position and the stated intention to prepare for the demographic pressures through fiscal prudence in the years to come, the country still faces significant challenges to public finances. Not only is the ageing of the population likely to generate substantial demands for public expenditure increases, as the main part of individual public consumption is directed to the elderly in terms of care and health services, but the provision of public services with intensive use of skilled labour might also prove to be difficult if labour and skills shortages appear: increases in wage premia for relevant skills might be necessary, putting further upward pressure on expenditures. While expenditure could be trimmed in areas where the demographic developments are expected to relieve spending pressure in the next decade, notably services directed to children and youth, that has typically proven difficult because of general inertia in public expenditure on individual spending categories and problems involved in redirection of labour to different tasks.

In addition, unless new immigrants become better integrated in the labour market, they too would add substantially to the fiscal burden as net recipients of public income benefits as well as public services on a lifetime basis. A recent study estimates that an average immigrant to Sweden represents a net government cost of SEK 175 000 over his/her remaining lifetime (Storesletten, 2003). Thus, on current patterns of labour market performance, immigration would not help to ease the impact of ageing on public finances. However, young immigrants (aged 20-30) are estimated to yield a net gain for public finances of more than SEK 200 000, suggesting that a carefully designed policy aiming at attracting immigrants of these ages could alleviate fiscal pressures, assuming that accompanying family members would also contribute positively to the economy over their lifetimes.[6] These calculations emphasise that current and future immigrants can be a valuable resource for raising living standards and curbing the pressure on public finances from the ageing population. But it requires a significant improvement in labour market integration; otherwise public finance problems would be aggravated.

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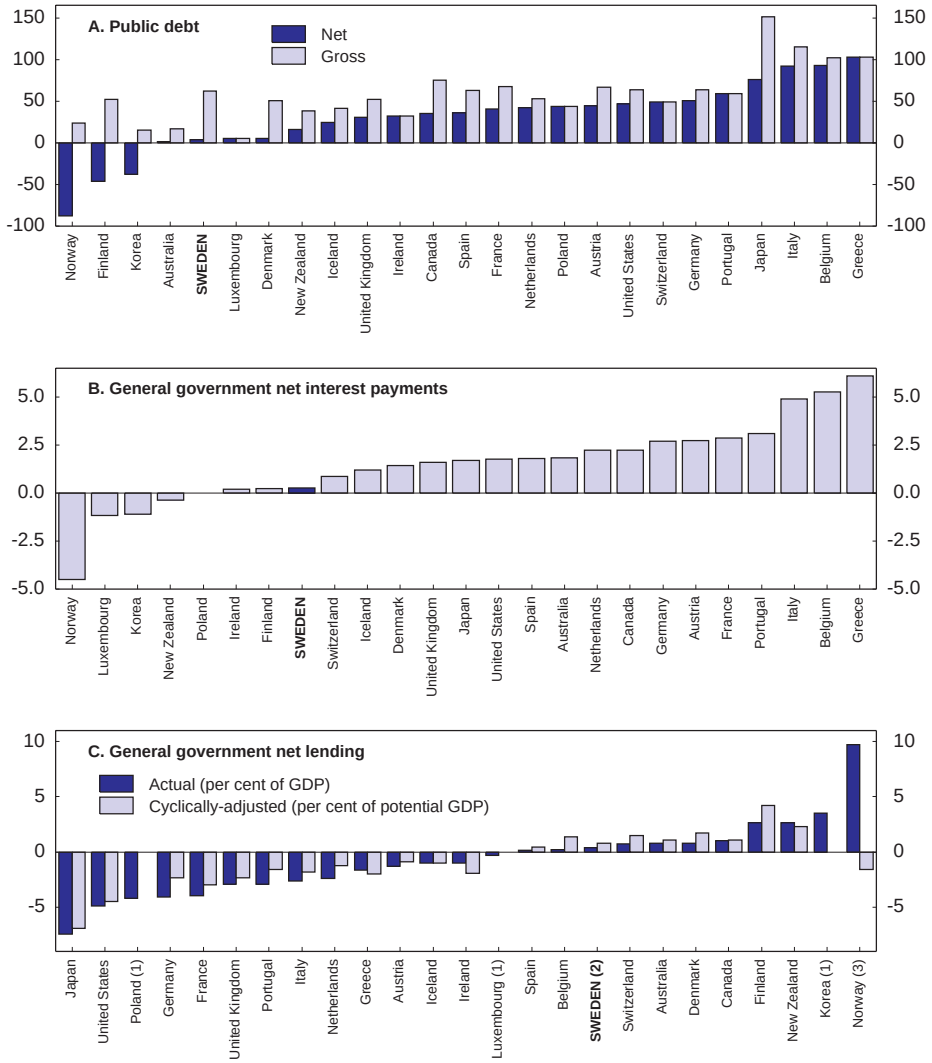
Figure 1.18. **Public finance position**

Per cent of GDP, 2003



1. No account of cyclically-adjusted net lending.
2. Data for Sweden differs from *Economic Outlook 74*, as account is taken of accrual taxes.
3. For Norway, actual refers to total economy and structural to "mainland" only.

Source: OECD.

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Box 1.4. The public pension system

Sweden introduced a new public old-age pension system in 1998 to be fully phased in by 2020. The reform turns the previous earnings-related defined-benefit system into a contributions-based system, with the larger part of contributions going to a pay-as-you-go (PAYG) system operating with notional accounts (see previous *Survey*). The system is actuarial neutral, and entitlement, which can be drawn (partly or fully) from the age of 61, depends on the number of years in work, lifetime earnings and the expected remaining lifetime at the age of withdrawal.

The reform was partly initiated to control the pressure on public pension expenditure from an ageing population. While the nature of the PAYG component implies that the balance of the system is affected by demographic changes, economic growth and employment rates, an automatic balancing mechanism ensures that the indexation of pensions and notional accounts is adjusted to reinstate balance, if the ratio of total assets (consisting of an initial buffer fund and contribution-based assets) to pension liabilities falls below unity. Also, annual pensions are calculated by dividing individual notional pension assets by a factor that is positively correlated with life expectancy, thereby reducing annual pension payments for given pension accounts if expected remaining lifetime at the age of 65 increases. These mechanisms should make for a more sustainable pension system in comparison with most other OECD countries. Nevertheless, it is still not entirely demographically neutral:

- A minimum guaranteed pension, which is financed solely by tax funds, is also provided for low income earners. Thus, some pressure on central government finances will arise from the demographic changes, and incentives for low income earners to stay active on the labour market beyond the age of 65 are very low because of high replacement rates. It should be mentioned though, that guaranteed pensions are indexed to the consumer price index, in contrast to the indexation to average wages of notional pension accounts in the PAYG component. Over time, the pension gap between people working full time and those with more sporadic attachment to the labour market would therefore widen, thereby increasing incentives to work harder and longer. However, given the traditional attachment to income equality at the political level, discretionary hikes of the guaranteed pension remain a risk to the improvement of work incentives for low income earners.
- The new pension system on its own probably makes for only a slight increase of incentives to later retirement compared with the old pension system (OECD, 2003b), as long as sickness compensation (disability pension), sickness insurance, occupational pension schemes and other systems continue to provide alternative exit routes.

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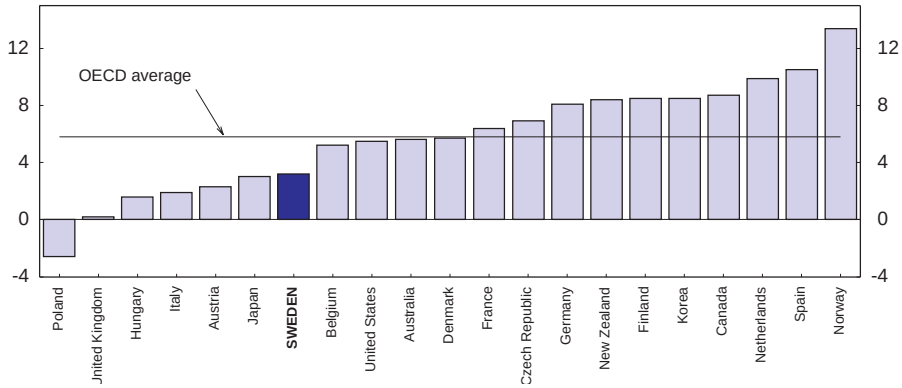
Figure 1.19. **Projected change in age-related government spending**

Per cent of GDP, 2000-2050[1]



1. This study does not incorporate the latest population projection by Statistics Sweden, which was published in May 2003. Thus, the increase in age-related government spending could be somewhat closer to the OECD average, as indicated by calculations in Regeringen (2003).

Source: Casey *et al*. (2003).

A long-term scenario for public finances based on the latest demographic projections was presented in the central government budget for 2004. Under assumptions that would ensure an improvement of the general government structural surplus to the target level of 2 per cent of GDP in 2007, the scenario projected an increase in primary revenue as a share of GDP by 1¾ percentage point between 2003 and 2050, while primary spending, as a ratio to GDP, would increase by 3¼ percentage points (Figure 1.20). Higher spending on health and elderly care is the main reason for the rise in primary spending, with public pension spending rising somewhat less.

Maintaining general government surpluses over the next 25 years would reduce government gross debt substantially, from almost 53 per cent of GDP in 2000 to nearly 28 per cent of GDP in 2025, and bring the government into a solid net asset position. This would in turn reduce net interest payments. Beyond 2025, general government gross debt may rise again as a primary deficit is sustained, and in 2050 it would be back to 48½ per cent of GDP – almost the same as in 2006. Thus, Sweden would fulfil the Maastricht debt criterion throughout the projection period and the general government would remain in a net asset position. However, the developments do not stabilise at the end of the projection period. Net capital income is insufficient to cover the primary deficit on general government finances, and the debt ratio would increase further in the years beyond 2050.

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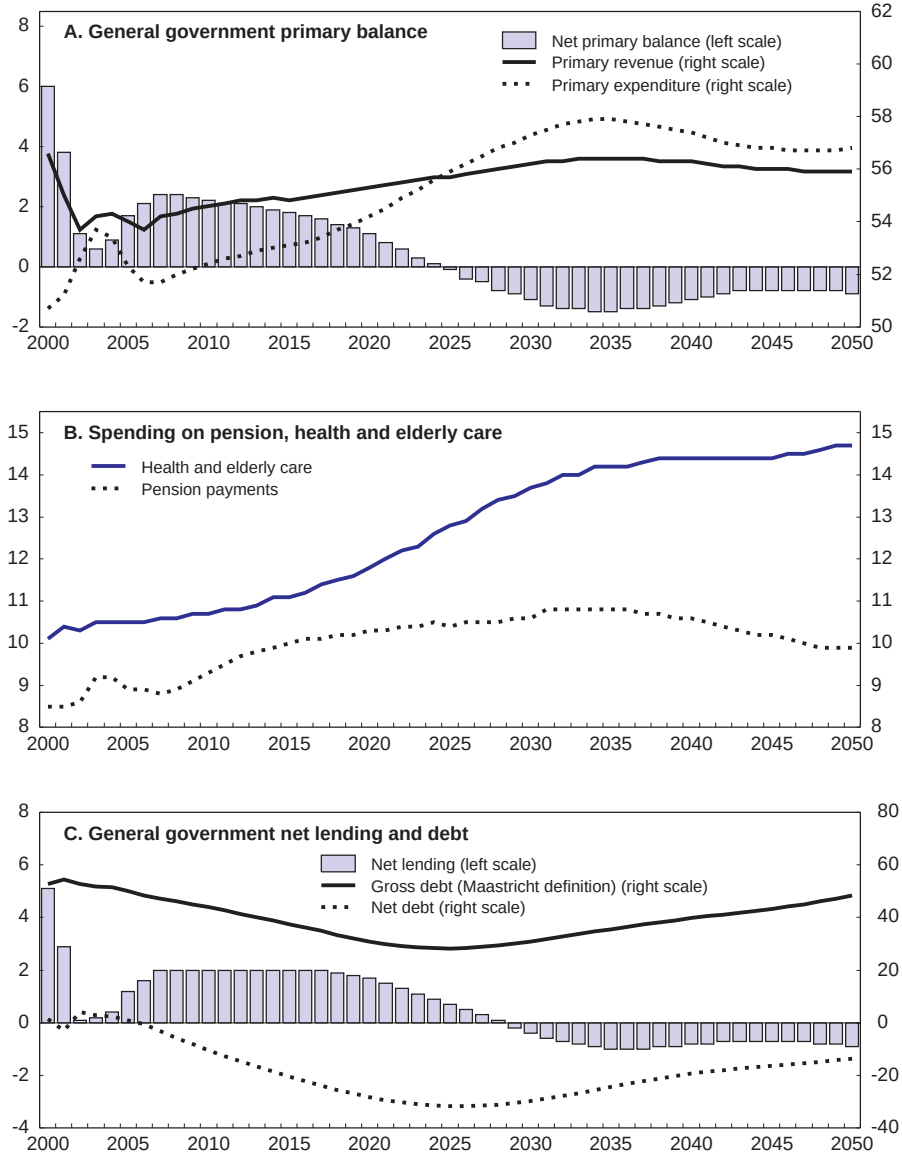
Figure 1.20. **Public finances in the longer term**

Per cent of GDP



Source: Statistics Sweden; Ministry of Finance.

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Another recent study estimates that an immediate and permanent improvement of the general government primary balance of 2-3 per cent of GDP would be necessary for public finances to be sustainable (Frederiksen, 2003).

The government's projection relies on several crucial assumptions. *First*, and most importantly, the integration of immigrants in the labour market is assumed to improve, so that one-third of the differences in the employment ratio and in unemployment between immigrants and native Swedes is eliminated in the period until 2020, while the remaining difference is sustained through the rest of the projection period. This raises employment by 2¼ percentage points compared with a pure demographic scenario. *Second*, the projection also assumes an annual reduction of average weekly working hours of 0.2 per cent, reflecting a rising demand for leisure as household income and consumption possibilities increase. That alone would reduce growth in total hours worked by almost 10 percentage points. The net effect of these two labour market assumptions is a reduction of total working hours by almost 3 per cent between 2002 and 2050. However, for public finances the positive effect from fewer people drawing on public income support is probably larger than the negative effect on tax revenue from lower average annual working hours. *Third*, half of the increase in life expectancy is assumed to lead to more "healthy" years, so that elderly care is shifted to older age groups.[7] The associated postponement of expenditure reduces the projected increase in total spending on health and elderly care by around 1¼ percentage points of GDP in 2050.

It should also be noted that most of the projected increase in the structural surplus until 2006 is due to a substantial fall in the expenditure ratio, partly because some temporary spending schemes are not prolonged. Given the promises made by the government on raising various benefit levels and increasing spending on welfare services, this might be a somewhat optimistic assumption (see Chapter 4). On the other hand, deferred taxes on pension contributions have not been included in the scenario. These are likely to give significant contributions to revenue in the future.

An assessment of fiscal sustainability based on long-term scenarios of this kind is of course sensitive to changes in the basic assumptions. The pattern of future immigration is very uncertain, as is the behaviour of households in terms of labour force participation. The 2004 Budget provides one alternative scenario involving an even larger improvement of labour market integration of immigrants: if three-quarters of the gap in employment rates between native Swedes and immigrants were closed, instead of just one-third in the basic scenario, the employment rate in 2050 would be 2¼ percentage points higher than in the baseline, and the expenditure ratio would be around 1½ percentage points lower. Studies for other countries suggest that deviations from the assumption on average life expectancy and changes to the average retirement

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age can also give rise to a significantly different assessment of public finance sustainability (OECD, 2003c).

The internationalisation of the economy is another aspect of uncertainty concerning the prospects of maintaining high tax ratios to finance the welfare state. Sweden has a relatively high level of taxation on most tax bases, with corporate taxes being one important exception. This might suggest that some harmonisation towards tax rates in other countries and more emphasis on immobile tax bases, such as property, could be necessary to ensure sufficient tax revenues in the future. However, to the extent that other European countries face even larger demographic challenges to public finances, any harmonisation of tax rates might rather be upward, thereby reducing the pressure on Swedish tax bases.[8]

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Notes

1. This means that if employers have employed temporary workers to replace those absent for long periods, or have maintained a larger workforce to cover high absentee rates, then policies that succeeded in reducing absence rates might, in the short term, drag down the recorded employment rate.
2. In November 2003, Statistics Sweden started to publish national accounts on an accruals basis. As the OECD *Economic Outlook* 74 was finalised before the release date, significant differences exist between data published in the *Outlook* and the new national accounts, especially in sector accounts. The new national accounts are used in this *Survey*, with projections corrected where appropriate. For instance, the projected net lending of general government is raised by 0.1-0.2 per cent of GDP in accordance with the Ministry of Finance's estimate of the accrual effect of tax revenue (which constitutes the major part of the difference between using accrual accounting or not).
3. A wage agreement for the next two years was reached in municipalities in 2003. An initial agreement was rejected by members of employees' organisations leading to a series of strikes before a final agreement was approved in June. The eventual outcome was a rise in compensation by 4 per cent in 2003 and 2.5 per cent in 2004, somewhat below the employees' initial claims.
4. In manufacturing, working hours were reduced by 0.5 percentage point per year in the previous collective agreements that predominantly covered the period until the first quarter of 2004. For the business sector as a whole the annual reductions were around 0.3 percentage point, whereas working hours in the public sector were unchanged.
5. More than half of the population get practically all their income from the government, either as employees in the public administration and service production or as transfer recipients (Lindbeck, 2003).
6. These results are based on assumptions of perfect assimilation of second-generation immigrants, which might overestimate the net fiscal gains. On the other hand, relatively good labour market performance of immigrants during the recent upswing – if sustained – suggests that fiscal gains could be higher, as the estimates rely on labour market outcomes from the mid-1990s.
7. A similar assumption has not been made for health care, as medical advances are presumed to lead to improved quality of health care and/or new treatment methods for different groups of patients.
8. A recent government commission concluded that public finance pressure from international interaction would probably be of limited scope, thus not posing any imminent risk to sustaining high revenue and expenditure ratios to GDP (SOU, 2002). One important factor in this assessment is that some other EU countries find themselves in much more difficult positions in terms of coping with the future pressure on public finances from the demographic changes.

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Annex 1.1

Progress in structural reforms

This annex provides a compendium of recommendations carried forward from the previous *Economic Survey*, and records what action has been taken, along with any other major policy changes announced since the previous *Survey*.

Labour markets

Previous recommendations

- Lower replacement rates in unemployment insurance to allow the wage structure to adapt to differences in unemployment risk.
- Identify and implement ways of limiting the damage to third parties from industrial action.
- Ease employment protection legislation further.
- Reduce generosity of part-time unemployment benefits.
- Enforce the Activity Guarantee rigorously, including firm sanctions for non-compliance. Add a time limit to the Activity Guarantee.
- Significantly reduce the use of active labour market programmes outside the framework of the Activity Guarantee, and ensure a sizeable overall downscaling. Evaluate programmes rigorously, and eliminate poorly performing ones.
- Strengthen basic-skills acquisition at compulsory school level, and make further efforts to stimulate quality improvements without increasing unit costs.
- Encourage quicker completion of tertiary education through less generous grants and promoting the acceptability of higher wage premiums for increased educational attainment.
- Resist reductions in working time.
- Reconsider the proposed introduction of individual learning accounts; encourage greater wage dispersion for skills instead.

Action taken

- Pilot schemes for lower working hours are being introduced in 2004, and a sabbatical year is to be implemented from 2005, with the condition that the person on leave is replaced by someone who is currently long-term unemployed. Parental leave has been prolonged and parental insurance benefit levels raised (although not the remu-

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neration rate). The government intends to present its position on the proposals from the government commission on new rules for working hours and annual leave in 2004.

- Secondary study grants have been extended from 9 to 10 months of the year. Subsidies for short-term study have been increased, giving around 10 000 more people the opportunity to study for a limited period while continuing to work.
- The government has appointed a special expert to examine ways of modifying university selection rules and admission procedures in order to reduce incentives for prospective students to take municipal adult education courses to improve their upper-secondary grades. The government has proposed that students with a completed three-year upper-secondary education should no longer be eligible for the high study grant for refresher or supplementary courses.
- Active labour market programmes involving employment training have been cut back.
- Additional public funds have been allocated to primary schooling, and pre-school classes were made available to 4 and 5 year-olds in 2003.

Social benefits

Previous recommendations

- Improve the incentive structure for sickness benefits by placing more responsibility on employers and employees.
- Introduce upper limits to the duration of sickness and unemployment insurance benefits.
- Integrate sickness and disability schemes, strengthen administrative controls, and place more emphasis on rehabilitation and returning beneficiaries to the workforce.
- Reduce the effective replacement rate on sickness and disability that arises *inter alia* from housing supplements and payment of pension contributions on top of benefits.

Action taken

- The general replacement ratio for sickness benefits has been lowered from 80 to 77.6 per cent, and the benefit levels for unemployed persons reporting sick have been aligned with unemployment benefits.
- The employers' period for paying sickness benefits has been extended by one additional week (employers now pay benefits from the second to the twenty-first day of sickness). Protection against excessive sickness costs has been provided for employers with few employees. Subsidies for employers who employ persons from long-term sick leave have been introduced, as has compulsory listing of sickness absence rates in companies' annual reports.
- Sickness administration has been strengthened. Meetings between the person on sick leave, the doctor, the employment agency and the regional social insurance board are to take place on a more regular basis; more emphasis is put on part-time sick leave; sick-leave certificates from doctors are to provide a better assessment of ability to work; fewer people are to be awarded sickness benefits for longer than one year.
- Introduction of time-limited (three years) award of sickness compensation (previously denoted disability pension) has been announced in the 2004 budget. The budget also announced further efforts to combat fraud and periodic reassessment of work ability of people receiving sickness or activity compensation.

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Tax policies

Previous recommendations

- Remove wealth taxation.
- Average marginal tax rates should continue being reduced, as part of a broad package of measures covering taxation of earned and capital income. In the current conjuncture, net reductions should be matched by expenditure cuts.
- Raise the threshold at which the state earned income tax rate applies.
- Broaden the tax base by restoring, and eventually increasing, the property tax level.
- Examine ways to minimise potential distortions to the allocation of capital embodied in the tax system.
- Simplify the basic allowance by making it the same for all income ranges. Consider replacing it with a tax credit or an initial zero tax bracket.
- Reduce the adverse effects on marginal effective tax rates arising from the housing allowances for families with young children and young people.
- Raise the effective rate of VAT to make room for greater income tax cuts.

Action taken

- The fourth step of the multi-year programme to reduce the number of people paying state income tax and to provide deductibility for employee social security contributions has been postponed. Meanwhile, the annual adjustment of the state tax threshold has been reduced for 2004, implying that a larger number of people (40 000 or 0.8 per cent of the working age population) will pay higher marginal taxes. Further "green" tax shifts have been introduced, involving higher basic allowances and a reduction in income taxes for income earners along with lower social insurance contributions for employers to compensate for higher environmental taxes.
- Average local government tax rates were raised by 0.7 percentage point in 2003.
- A tax subsidy to new construction and rebuilding of small and medium-sized rental and student dwellings has been introduced for the period 2003-06. In effect, it corresponds to a reduction of the VAT on construction from 25 to 6 per cent.
- The capital gains tax on limited liability companies with unlisted shares has been abolished.
- A change of the 3:12 rules, applying to closely-held companies, has been announced implying a (long-run) tax relief of SEK 1 billion.
- A rule for limiting the wealth tax has been announced. It will have effect for the income year 2003 and apply to people who are already covered by the rule to limit real estate taxes and pay wealth tax on their permanent residence, thus reducing wealth taxes for 15-20 000 property owners. In addition, a rule to smooth property tax payments in the event of rising tax assessed values on property has been announced.

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Product markets and competition

Previous recommendations

- Develop a privatisation programme aimed at significantly downsizing state ownership of market activity within a reasonable time-frame.
- Allow owners to charge market rents for new housing, and progressively strengthen municipalities' incentives to supply suitable land sites for housing construction, for example through instituting a municipal property tax.
- Continue efforts to dilute incumbent power in electricity and telecommunications, and in particular, achieve "unbundling" of the local loop for telecommunications.
- Strengthen the scope of the Competition Act at the expense of the Municipal Act. Follow through on earlier recommendation to ensure that effective competition applies to municipal entities.
- Ensure that block exemption is abolished. Resolve the outstanding obstacles to implementing the EU directive for building materials, and encourage municipalities to facilitate competition in construction.
- Address the factors that are inhibiting construction activity, including weak competition.

Action taken

- A parliamentary committee has been appointed to review planning and building legislation and report its findings no later than end-2004.
- The government is currently reviewing the Construction Committee's proposals to improve the functioning of the construction sector.
- Carrier pre-selection is extended to cover local telephony and is thereby fully implemented. The National Post and Telecom Agency (PTS) renewed its oversight of the observance of the EU regulation on unbundled access to the local loop in early 2003, and a mapping of perceived problem areas has begun that will serve to direct the focus of the PTS' supervisory activities. The authority of PTS to intervene in markets with ineffective competition has been expanded *via* new legislation that implements new EC directives on electronic communication networks and communication services.
- To increase mobility in the electricity market and improve its functioning, monthly reading of electricity meters will begin from mid-2009. A commission has been appointed to propose ways of incorporating the new revised EC directives on electricity and natural gas markets and regulation on cross-border trade of electricity into Swedish legislation.

Promoting environmentally sustainable growth

Previous recommendations

- Ensure that rigorous cost-benefit analysis is systematically used to evaluate environmental policy options.

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- Reinforce the economic incentive to install NO_x abatement equipment on Baltic shipping by raising fairway dues on those ships operating without the equipment to cover some of the cost of the subsidy to convert.
- Restructure energy and carbon taxes so that marginal abatement costs are equalised across the economy while taking appropriate measures to address competitiveness concerns, such as tax credits or grandfathered tradable emission rights.
- Consider the possibility of using a domestic cap-and-trade approach to meeting Kyoto targets, instead of, or as well as, using environmental taxes. Use emissions-trading, Joint Implementation and the Clean Development Mechanism provisions under the Kyoto Protocol to meet targets in the most cost-effective manner.
- Re-examine energy efficiency promotion to ensure that measures generate net social benefits, and do not simply strive to achieve maximum engineering efficiency.
- Examine carefully the policies to promote renewable energy sources to ensure that they do not overstep the mark and become, *de facto*, industrial policy. Allow market signals, adjusted for environmental externalities, to play a greater role in determining the appropriate mix of energy supply. Build sunset clauses into research and development projects.
- Ensure that the competition drawbacks of encouraging a switch from individual to district heating are fully addressed.
- Align energy taxes with energy content, and apply uniformly across all energy consumers.
- Examine the merits of either adopting a minerals accounting approach and taxing nitrogen losses (*i.e.* residual balances) to provide an economic incentive to reduce the damage from agriculture or shifting to an environmental-outcomes-incentive charge for farmers. Evaluate costs and benefits compared with present arrangements.
- Reconsider the emphasis placed on recycling, and identify which items are worthwhile recycling on the basis of cost-benefit analysis. Evaluate the relative merits of extended producer responsibility in the light of international experience with alternative approaches to recycling.
- Consider moving to a weight-based general waste collection system to provide a stronger economic incentive to reduce waste.
- Undertake cost-benefit analysis before proceeding with the planned ban on organic waste.

Action taken

- A system for sale of certificates for electricity produced from renewable energy resources has been introduced, aiming to increase the share of electricity produced from these resources.
- The "green" tax shift has implied increases in a wide range of environmental taxes.
- Following the proposals from a committee reviewing the rules on tax reductions for the business sector presented in 2003, the government aims to have a new regulatory regime in place by the beginning of 2005, which should improve environmental management in the business sector.
- The zero tax rate on electricity used in industry is being abolished, and a new tax of SEK 0.005/kWh will be introduced in mid-2004.

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- A commission has been appointed to present proposals on the design of a tax on waste incineration by mid-2004.
- The 2004 budget included a proposal for increasing the tax on pesticides.
- The government is preparing a bill on implementation of the EU-wide emission trading system to be introduced at the beginning of 2005.

Fiscal framework and efficiency of public expenditure

Previous recommendations

- Strengthen the expenditure ceilings by operating the ceiling for operational and non-entitlement transfers as a hard budget constraint on a three-year ahead basis, and building in an explicit safety margin on a separate ceiling on entitlement spending.
- Rebalance the budget envelopes, and bolster the incentives for prioritising expenditures within and between expenditure areas.
- Strengthen the accountability of the state agencies, and ensure that they have clear and measurable performance targets to achieve.
- Identify and pursue cost-effective ways of improving the performance of the education system, particularly in mathematics and science.
- Look at ways of truncating the years spent in school by raising the number of hours spent in instruction each year towards international averages, and re-examining the balance between instruction time and after-school care.
- Invest in developing better quality indicators of school performance both to increase transparency for parents, municipalities and the national educational authorities, and to enable the identification of factors influencing outcomes.
- Strengthen the voucher element, and institute a system of tuition fees for the tertiary sector, allowing institutions to offer whatever courses and places they wish, while encouraging students to complete their studies in a shorter time frame.
- Undertake more rigorous evaluation of the costs associated with outcomes in the health sector, and consider integrating detailed cost data into the national registers on clinical outcomes.
- Consider whether the annual maximum for patient co-payments could be raised to provide additional resources for the health sector, and examine the advisability of allowing private insurance markets to assist in dealing with the upward pressures likely to arise as the population ages further.

Action taken

- The government has announced the establishment of a commission of inquiry on social insurance, which will look at ways of financing future expenditure on unemployment, sickness and parental benefits, including a possible combination of basic public financing, collective contributions and individual insurance.

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Annex 1.2

Macroeconomic stability

Sweden enters the projected upswing without any underlying macroeconomic imbalances. A stable surplus on the current account of around 4 per cent of GDP since 1998 reflects relatively small variations in private financial savings and general government net lending compared with the earlier part of the 1990s. Both domestic sectors have remained in a surplus position in the last five or six years, except for 2000, where large corporate tax payments following substantial earnings made for a temporary rise in government net lending and a drop in measured private financial savings. This was reversed in the following two years, mainly as a result of large personal income tax cuts and a rise in the households' savings ratio, which led to a significant increase in households' financial savings (Figure 1.A2.1).

Uncertain economic prospects made households react cautiously to tax cuts

Households' prudent spending behaviour in recent years should be seen against the background of a collapsing stock market and rising unemployment, although some element of Ricardian equivalence when the central government lowered income taxes in 2001 and 2002 cannot be ruled out. Indeed, local governments raised taxes in 2003 and 2004, thereby offsetting part of the state tax cuts. Developments on the housing market, on the other hand, have continued to underpin household spending during the economic slowdown. The real price of owner-occupied housing has continued to rise, although at a slower pace during the last year, providing security for additional take-up of loans. While the ratio of debt to disposable income is approaching historical highs, the ratio of interest payments to disposable income nevertheless remains very low.

The steep rise in household debt to disposable income since 1997 has fuelled concerns that indebted households would be vulnerable to higher interest rates. However, one recent study suggests that interest expenditure would still remain at a relatively low level if interest rates rose by 1 percentage point on average, even with a continued increase in the debt ratio (Konjunkturinstituttet, 2003b). Also, the ratio of household debt to the market value of owner-occupied housing is quite low by historical standards, and household net lending is slightly positive, implying that households on average are not consuming on credit. A sharp drop in house prices therefore seems to be the main risk for household spending. However, the development of house prices has generally been in line with that of construction costs, indicating no fundamental imbalance. Furthermore, a steep rise in real interest rates, which provoked the crash of the housing market in the early 1990s, does not seem likely.

Troubles in pension companies may affect consumer confidence

One recent concern for confidence in the household sector is the scaling-back of pension payments to policy-holders in some life-insurance companies. A common technique in these

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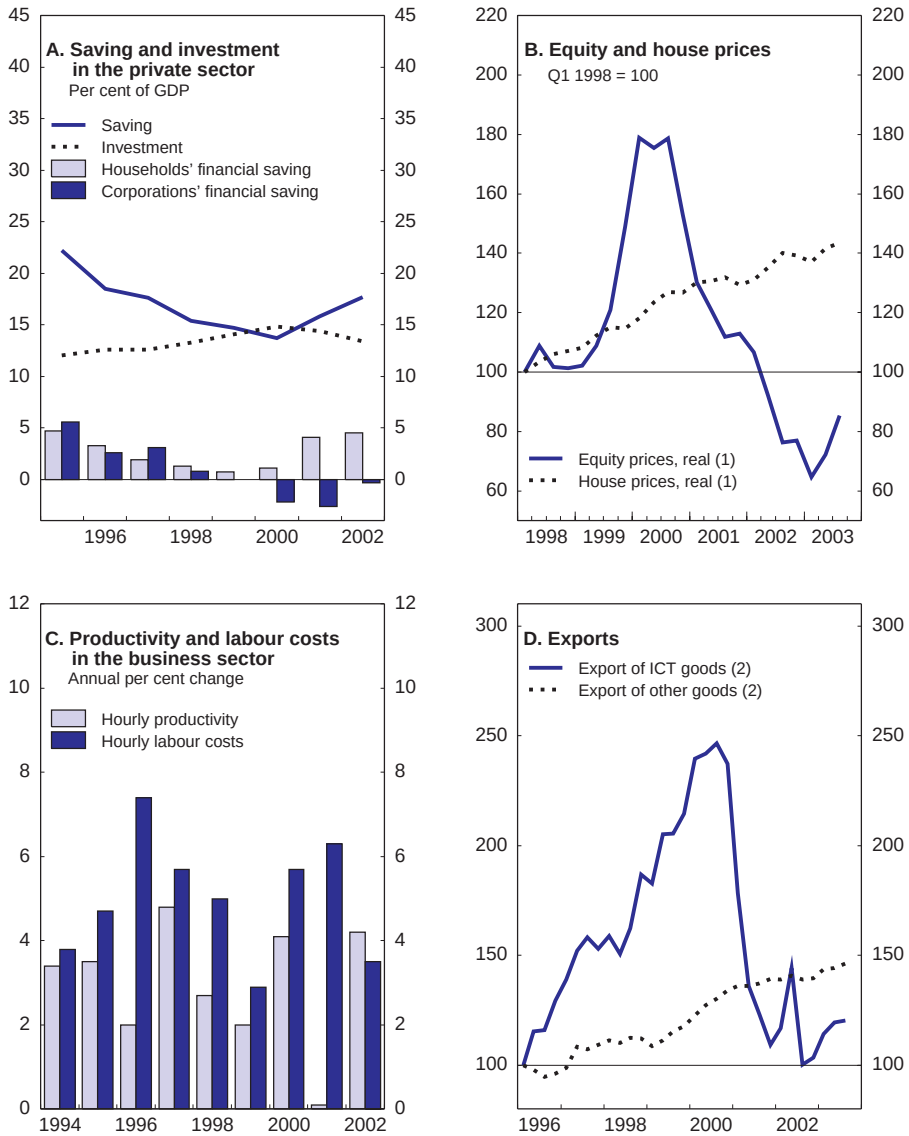
Figure 1.A2.1. **Factors affecting macroeconomic stability**



1. Deflated by the private consumption deflator.
2. Index, Q1 1996 = 100, current prices.

Source: National Institute of Economic Research; Swedish Labour Market Board; Statistics Sweden; OECD.

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companies is to smooth distributed returns, *i.e.* returns in excess of what is required to fulfil the guaranteed return. While this practice can delay distribution of realised returns, it can also imply that non-realised returns are distributed in periods with low or negative returns on assets. This is possible because life insurance payments in general constitute a minor part of total assets. In the last three years, distributed returns to policy holders have not been fully adjusted to the negative development on stock markets, implying that collective consolidation (the difference between the market value of assets and the sum of guaranteed undertakings and additional distributed returns) has fallen to levels below the targets set by life-insurance companies in their stated policies. The Financial Supervisory Authority (FSA) has in general advised that deviations from the target level should not go on for longer than 36 months, and some firms have – or will soon – reach that limit (Finansinspektionen, 2003). As a first step, firms have set distributed returns at zero. But as the time limit has approached, it has become necessary to take back previously distributed returns, including for people currently receiving pension payments, even though it implies a breach with the standard practice in the sector (the so-called Allen rule). However, it is in line with the recommendation from the FSA that adjustments be borne by all policy holders. The FSA has further required that company policies on reallocations be communicated clearly, so that any misperceptions among policy holders can be eliminated.

While a fair division of recent losses for the companies and clear communication of policies certainly should be advocated, reductions of pension payments do constitute a risk to confidence and spending behaviour. However, even though insurance saving has increased rapidly in the last 10 years and represents around 40 per cent of households' total financial savings at present, the reductions in pension recipients' disposable income is relatively small. Thus, the financial situation of households and comfortable average increases in real disposable income should imply a stabile basis for private consumption in the near term.

Restructuring in the business sector

The business sector reacted with a certain lag to the slower demand when the economic slowdown set in. Many companies refrained from reducing the number of employees until later in the downturn, leading to almost zero hourly productivity growth in 2001. But as prospects of an upswing were continuously postponed, both investment and labour input were adjusted.

While the manufacturing sector in generally has suffered from the economic slowdown, production of telecommunications equipment was especially hard hit as the ICT bubble burst and demand plummeted. Substantial declines in production and exports of telecommunication equipment sent the dominant producer, Ericsson, to the verge of collapse as the value of its shares plunged. The company survived thanks to heavy restructuring and downsizing efforts; although demand is still weak, it reported positive operating profits in the third quarter of 2003.

The stability of the financial sector has not been altered during the downturn, as banks in general have managed the weak business cycle without significant difficulty. It should be stressed that while the troubles in life-insurance companies indicate an imbalance within the insurance collective and might constitute a credibility problem for the companies, it does not affect solvency, as both distributed and non-distributed returns are part of the companies' risk capital. Thus, it also does not pose any significant risk to financial stability.

Monetary policy has underpinned stability

The generally good balance of the Swedish economy is also attributable to the strong macroeconomic framework, which continues to include a floating-exchange-rate regime with

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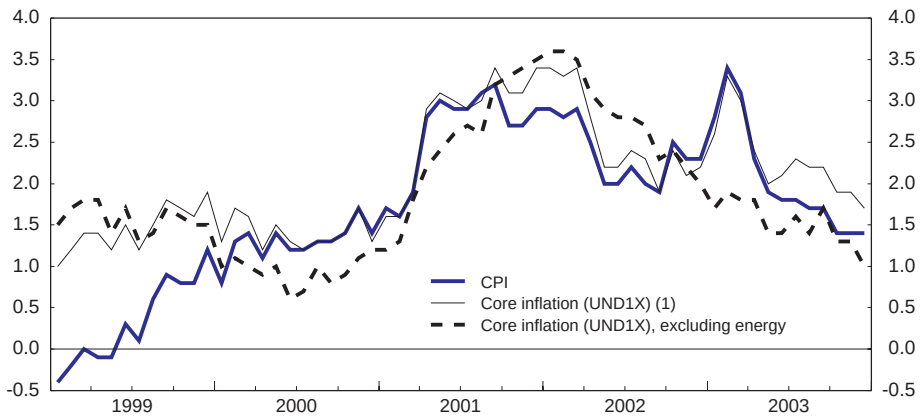
the independent central bank, the Riksbank, setting its policy rates to steer inflation towards the target rate of 2 per cent over a one to two year horizon, while the government is aiming at a 2 per cent surplus for general government finances over the cycle. The Riksbank has successively lowered the repo rate by 1½ percentage points since November 2002, leaving it at an unprecedented low level of 2¾ per cent. The monetary easing came against a background of further postponement of a recovery in Sweden and abroad and persistent projections of inflation below the 2 per cent target.

In several instances the bank made direct reference to its measure of core inflation (UND1X*) while specifically emphasising also the effects from fluctuations in energy prices. The latter became relevant when energy prices jumped markedly in late 2002 and the early months of 2003, as electricity prices were driven up by supply restrictions on hydro-power (due to cold weather conditions) and oil prices reacted to the prospect of a war on Iraq (Figure 1.A2.2). Consumer price inflation consequently rose sharply, peaking at 3.4 per cent in February 2003. The subsequent reduction of oil prices and a partial reversal of electricity prices made for a rapid decline in consumer price inflation during the following three months. Additional deceleration of the consumer price index (CPI) was provided for by continuously decreasing contributions from non-energy elements, which were partly attributable to an appreciation of the krona since the second half of 2001. As the effect from energy prices was seen as temporary, the Riksbank chose to put more emphasis on developments in UND1X excluding energy when motivating its policy, as this measure was believed to give a clearer indication of underlying inflation pressure in the economy. That implies also that

Figure 1.A2.2. **Consumer price inflation**

Year-on-year percentage change



1. CPI excluding mortgage interest expenditure on owner-occupied housing and direct effects of changes to indirect taxes and subventions.

Source: Statistics Sweden; The Riksbank.

* UND1X is the consumer price index excluding mortgage interest expenditure on owner-occupied housing and direct effects of changes to indirect taxes and subventions.

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the likely drop in 12-month inflation rates in the early months of 2004, as the base effects from the jump of electricity prices disappear, will not be regarded as requiring any policy reaction.

The Riksbank's references to different price indices during 2003 made some observers claim that it was unclear which inflation measure the target was actually related to. In its inflation report of October 2003, the Riksbank explicitly noted that inflation excluding energy was expected to be around the target at the end of the projection period, *i.e.* two years ahead, thereby signalling that this was the relevant inflation measure for policy decisions at the current state (Sveriges Riksbank, 2003). But the bank also made clear that its operational target was a rate of 2 per cent for CPI inflation with a tolerance interval for deviations of ±1 percentage point. However, when considering CPI to be affected by disturbances that the bank neither "needs" nor "fully wishes" to offset through changes in the repo rate, the simple monetary rule is set aside and developments in various relevant sub-indices of CPI are emphasised. Disturbances not "needing" offsetting policy reactions were characterised as being either of a temporary nature or coming from specific factors, which would not have any substantial permanent impact on inflation or inflationary processes, although the CPI might be affected beyond the two-year forecast horizon. The bank would not "fully wish" to counter disturbances that create substantial deviations from the inflation target for a limited period of time, if the monetary policy response required to meet the target would affect the stability of the real economy negatively. The emphasis on overall macroeconomic stability has probably been strengthened over the last decade, as an increasing part of owner-occupied housing is financed by floating-rate mortgages, thus making households' spending more sensitive to changes in interest rates.

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Annex 1.3

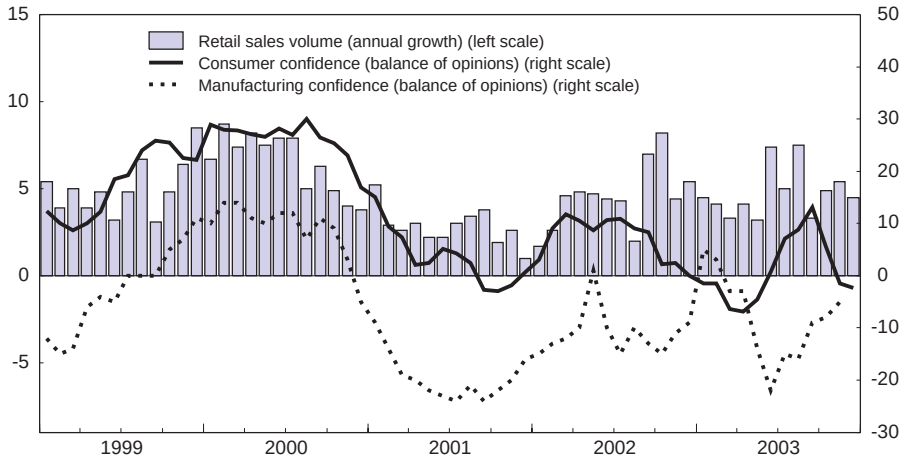
Outlook to 2005

Output growth picked up somewhat in the first half of 2003, primarily reflecting higher export growth. Rising unemployment has probably been restraining household spending, but buoyant retail sales and a marked improvement in consumer sentiment over the summer point to some acceleration of private consumption (Figure 1.A3.1). However, business confidence remains subdued despite some recent firming, which along with relatively low outstanding orders indicates that a moderation of activity took place in the second half of the year. Growth for the year as a whole is likely to have been around 1½ per cent. Consumer price inflation has eased substantially after the up-tick in the early months of 2003, mostly reflecting lower underlying inflation but also some reversal of earlier energy price increases.

While households reacted cautiously to buoyant disposable income growth in 2001 and 2002 by saving more, a gradual reduction of the saving ratio is projected to boost private

Figure 1.A3.1. **Retail sales and confidence indicators**

Per cent



Source: Statistics Sweden; National Institute of Economic Research.

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consumption in 2004 and 2005. Large central government tax cuts and other discretionary easing of fiscal policy have made significant contributions to household income growth in recent years, but fiscal policy is assumed not to add further stimulus over the projection period, as higher local government taxes broadly offset a slight central government easing. Automatic stabilisers and falling revenues from corporate and capital gains taxes reduced the general government surplus to less than ½ per cent of GDP in 2003, while the closing of the output gap should contribute to bringing it back to around 1¼ per cent of GDP by 2005 (Table 1.A3.1).

Table 1.A3.1. **Summary of short-term projections**

	2000		2002	2003	2004	2005
	SEK billion, current prices	Per cent of GDP	Per cent			
Demand and output						
Private consumption	1 078.4	49.1	1.5	1.7	2.5	2.6
Government consumption	583.4	26.6	3.2	0.7	0.7	0.7
Gross fixed investment	389.0	17.7	–3.0	0.4	2.7	6.1
Final domestic demand	2 050.8	93.4	1.1	1.2	2.1	2.7
Stockbuilding[1]	16.5	0.8	–0.2	0.1	0.0	0.0
Total domestic demand	2 067.3	94.2	0.9	1.3	2.1	2.7
Exports of goods and services	1 012.1	46.1	0.2	5.0	5.0	6.6
Imports of goods and services	884.4	40.3	–2.5	5.2	4.8	6.9
Net exports[1]	127.7	5.8	1.0	0.5	0.6	0.7
GDP	–	–	1.9	1.5	2.3	2.7
GDP at current prices	2 195.0	100.0	3.4	3.5	4.5	5.6
Price, costs and employment						
GDP deflator	–	–	1.4	2.0	2.2	2.8
Consumer price index	–	–	2.4	2.1	1.4	2.2
Total employment	–	–	0.1	–0.1	0.3	0.7
Unemployment rate	–	–	4.0	4.8	4.7	4.4
Financial variables						
Short-term interest rate	–	–	4.1	3.0	3.1	4.1
Long-term interest rate	–	–	5.3	4.6	5.0	5.6
General government balance[2]	–	–	0.1	0.4	0.6	1.2
External transactions						
Trade balance[2]	–	–	6.1	5.9	6.4	6.8
Current account balance[2]	–	–	4.1	3.7	4.3	4.9
Memorandum items:						
Output gap[3]	–	–	0.3	–0.5	–0.4	0.1
Household savings ratio[4]	–	–	8.8	8.2	6.5	5.7

1. Contributions to changes in real GDP (percentage of GDP in the previous period), actual amount in the first column.
2. Per cent of nominal GDP. Projections for general government balance differ from *Economic Outlook* 74 because of correction for accrual accounting (adding 0.1-0.2 in each year).
3. Per cent of potential GDP.
4. Per cent of household disposable income.

Source: OECD.

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The Riksbank provided additional monetary stimulus when lowering its policy rate by 75 basis points in June and July 2003, leaving the repo rate at a long-time low of 2¾ per cent. Although annual inflation is likely to drop temporarily below 1 per cent in the early months of 2004, as large rises in electricity prices in early 2003 drop out of the base, a cautious approach to any further interest rate reductions is warranted. Once the economy gathers steam and approaches its potential, rate hikes would be appropriate.

Output growth is projected to climb to around 2¼ per cent in 2004 and 2¾ per cent in 2005. Private consumption and business investment should accelerate through 2004 as households and firms react to previous policy stimulus. The unemployment rate is projected to fall back as businesses start to increase hiring during 2004. Wage increases should be moderate for a time, before rising again in 2005 as the labour market tightens. This could push up consumer price inflation to around 2¼ per cent in 2005.

With the telecommunications sector seemingly on a more stable footing, the main risk centres on the collective wage negotiations in 2004. Though the recent agreement for the municipal sector may provide a yardstick for achieving a moderate wage outcome, the signals in the 2004 state budget concerning lower working hours and "sabbatical leave" may inspire demands for more leisure. Along with a still growing number of sickness compensation recipients (disability pensioners), this could generate additional pressure on capacity and more wage drift.

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2. Product market competition and economic performance

As shown in Chapter 1, Sweden has seen strong GDP growth in the past decade, labour productivity has increased at rates well above the OECD average, and employment has improved markedly after the earlier labour shake-out. Developments in the telecommunication equipment sector have been especially impressive, whereas other sectors have lagged behind somewhat. To some extent, this slower rate of expansion in certain sectors is probably linked to insufficient pressure to compete and a resulting lack of dynamism.

The country has been engaged in a major effort to strengthen competition in the last 10-15 years. New competition legislation and monitoring institutions were introduced, and a number of sectors were deregulated, including previously largely state-owned network industries. Accession to the European Union in 1995 was a crucial step in this process. These measures have arguably increased the overall performance of the economy – the requirement to replace Sweden's liberalised agriculture policies with the Common Agricultural Policy (CAP) being one important exception.

Though the country was early in liberalising a number of important sectors, progress towards a more competitive economic structure seems to have lost momentum in recent years. This is especially evident for publicly provided services, where there is still too little recognition of the potential benefits to consumers from adopting pro-competition policies. These include higher cost effectiveness, better quality, more freedom of choice and greater responsiveness to clients.

This Chapter identifies the main areas where impediments to competition are likely to keep the country from realising the highest output from available resources, and where a shift to pro-competition policies would help to boost growth potential. The first section briefly reviews the link between competition and growth and presents a number of general indicators on the stance of competition. The overall framework of competition legislation and its enforcement is then assessed before selected network industries and various other sectors are reviewed. Next, the stance

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of competition within the public sector is examined, and the Chapter ends with a wrap-up section of conclusions and recommendations.

Product market competition and macroeconomic performance

Competition is generally considered to contribute to stronger economic growth by generating static and dynamic gains in labour and multi-factor productivity. The link between competition and static gains is well established empirically for OECD countries, although the connection between competition and dynamic gains is not yet completely clear (OECD, 2002). In general, positive spillover effects from product market competition to employment have been identified (Nicoletti *et al.*, 2001), although in Sweden, these benefits might not be fully realised, due to obstacles slowing adjustment in the labour market (see Chapter 3). Indeed, labour market characteristics can have a significant and complex impact on industrial structure and innovation, as stricter labour market regulations have been shown to generally inhibit the full realisation of gains from greater competition.

Sweden has been a leader in reforming and deregulating a range of sectors that had previously been shielded from competition. Examples are the financial market (mostly in the 1980s), rail transport (in successive steps; 1990, 1992 and 1996), taxis (1990), domestic air traffic (1992), the postal and telecom markets (1993) and electricity generation and distribution (1996). Alongside these changes, a new competition law was introduced (1993), and the Swedish legislation has since been harmonised with EU rules. The country has also been a steady advocate for freer international trade. Nevertheless, according to a recent study only 32 per cent of total household and public consumption was acquired on markets fully exposed to competition in 1999 – well below the European average of 45 per cent (Table 2.1).

Notwithstanding these efforts in the 1990s, the picture of the overall stance of competition is mixed when looking at various individual indicators (Box 2.1). Although price level differences are normally linked to variations in GDP per capita in international comparisons, even when corrected for this effect, the overall price level – measured on the basis of purchasing power parities – does seem to be relatively high in Sweden (Figure 2.1). Comparatively high indirect taxes certainly explain part of the deviation, and quality differences possibly also contribute, but more sophisticated studies in Sweden have suggested that prices are indeed somewhat above what can be reasonably explained by such factors, despite the relative openness to international trade. So while significant efficiency gains have been realised and prices reduced on certain deregulated markets, with one study concluding that deregulation of the electricity, telecommunications and railway markets has resulted in efficiency improvements of around 5 to 10 per cent (Bergman, 2002), there are still indications of some effects from weak competition

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Table 2.1. **Consumption exposed to competition**[1]
1999

	Sweden		European Union		Sweden's relative position[2]
	Per cent of:		Per cent of:		
	Total consumption	Category fully exposed to competition	Total consumption	Category fully exposed to competition	
Food and non-alcoholic beverages	8.1	20.0	9.6	50.0	<<
Alcohol and tobacco	2.7	0.0	3.2	80.0	<<
Clothing and footwear	3.5	100.0	5.0	100.0	=
Housing, water, electricity, gas and other fuels	19.7	33.3	15.7	50.0	<<
Furniture, appliances and household maintenance	3.1	100.0	5.2	100.0	=
Health care services	1.6	20.8	2.4	20.0	=
Transport	8.4	20.0	10.4	15.0	>
Communication	1.9	76.7	1.8	32.3	>>
Recreation and culture	7.1	50.5	6.9	50.0	=
Education	0.1	50.0	0.7	74.9	<<
Restaurants and hotels	3.3	96.1	6.2	80.1	>>
Other goods and services	4.8	100.0	7.2	100.0	=
Total private consumption	64.4	46.5	74.3	59.5	<<
Public consumption	35.6	5.1	25.7	5.0	=
Total consumption	100.0	31.8	100.0	45.5	<<

1. The assessment of exposure to competition is based on indicators of concentration and other impediments to competition in the various sectors. The focus is on domestic factors impeding competition, thus not taking into account the possible effects of, for instance, special import restrictions, which would lead to lower competition exposure in both Sweden and the EU.
2. >> (<<) indicates that Sweden is at least 10 percentage points more (less) exposed to competition than the European Union. > (<) indicates a difference of less than 10 percentage points.

Source: Braunerhjelm *et al.* (2002).

due to static inefficiency in general. Moreover there are major differences in prices of different categories of goods and services (Table 2.2).

Although the comprehensive reforms in network industries in the 1990s have on the whole yielded positive results, the benefits of the earlier reforms cannot be clearly detected in the growth rate for GDP per capita, despite the relatively good growth performance since 1994. Labour productivity increases have indeed been higher than the OECD average, more than offsetting below-average increases in employment (Table 2.3). But a closer examination indicates that the evolution of labour productivity in different sectors is not obviously correlated with deregulation, especially for the sector comprising electricity, gas and water supply. On the other hand, product market deregulation seems to have contributed significantly to the high R&D intensity, which is generally believed to raise

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Box 2.1. The stance of competition in Sweden

A number of indicators can be used to assess the state of competition in the economy, although they are subject to some measurement problems and should be interpreted with caution.

First, the price level is often used to gauge possible static inefficiency. The debate on competition in Sweden has focussed on this, with a number of research reports trying to explain the apparent higher prices when compared with other countries. The prevailing view is that only part of the comparatively high price level can be attributed to various special characteristics for Sweden. A number of studies commissioned by the Swedish Competition Authority have concluded (Konkurrensverket, 2003a):

- Prices are indeed higher in Sweden, and around half of the difference between Sweden and EU may be attributed to lack of competition pressure.
- Product markets with remarkably high price differences are characterised by high concentration, strong brand names, and/or little competition from imports. Also non-tariff barriers were fairly common.
- High food prices overall can partly be attributed to the planning process in municipalities, which impedes competition by favouring the three dominant actors on the retail food market.
- Substantial regional price differences in some sectors, notably food items and building materials, cannot be explained by differences in the economic and regulatory environment.

Second, profit mark-ups have not been high in Sweden, despite relatively high prices (Oliveira Martins *et al.*, 1997), Estimates show mark-ups well below average in a range of industries, indicating a relatively competitive environment, and average in other sectors. Also, wage *premia* have not been found to be especially high in Sweden, *i.e.* sharing of product market rents with workers does not seem to take place to an unusual extent (Jean and Nicoletti, 2002). This result may reflect wage bargaining arrangements and/or small rents to be shared between labour and capital.

Third, concentration indices for Sweden are found to be high in a number of industries including a range of chemical products, pharmaceuticals, food and beverages, pulp and paper, and construction materials (Konkurrensverket, 2002a). In non-manufacturing, relatively high concentration is also found in construction, retailing, water utilities and land transport. However as a small country, Sweden could be expected to have concentration indices generally above average, whereas its companies are more likely to be exposed to competition from abroad.

Fourth, Sweden has a relatively high import ratio, once corrected for structural factors such as country size (population), GDP per capita levels and natural barriers to trade associated with transportation costs. Large foreign direct investment flows relative to GDP are another indicator of openness. Johansson (2001) finds a positive effect from import competition on Swedish productivity growth in sectors

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Box 2.1. **The stance of competition in Sweden** (*cont.*)

where companies have very different technological levels. Sweden has been one of the most diligent in implementing EU directives that facilitate making the single European market a reality (European Commission, 2003). However, adoption of the EU Common Agricultural Policy has reduced the agricultural sector's exposure to competition from imports from third countries.

Fifth, entry and exit of businesses, which could be interpreted as indicators of dynamic competitive pressures, give a somewhat mixed picture. By international standards, Sweden has a relatively low number of entrepreneurs, which might to some extent reflect a higher degree of risk aversion (see Chapter 3).

growth, although in Sweden's case this does not seem to have translated into substantially higher growth (see Chapter 3).

The apparent absence of links between pro-competition policies and the overall price level, and between innovation and growth, has recently been described as the Swedish growth and competition paradox (Braunerhjelm *et al.*, 2002; Andersson *et al.*, 2002). However, one difficulty in assessing the gains from pro-competition policies is the absence of a "counter-factual", namely, what outcomes would have been in the absence of reforms. Another explanation could be that various aspects of the economic structure still prevent the country from realising some of the typical gains from pro-competition policies: the existence of some dominant sectors still shielded from competition (especially the public sector), regulations on the labour market, the tax structure and state subsidies. These issues are dealt with in the remaining part of this Chapter and in Chapter 3.

Competition legislation and enforcement

An effective legal framework governing competition is a critical factor in shaping the way markets function and in boosting economic performance through product market competition. In 1993, Sweden adopted a modern Competition Act, modelled on the EU rules prohibiting restrictive agreements and abuse of a dominant position and including merger control. Legal reform has kept Swedish law in line with EU developments, and some changes have aimed at enhancing its effectiveness, such as strengthening the leniency rules, which provide for protection from prosecution for the first company that comes forward with information on undetected cartels. With the EU modernisation programme becoming effective in spring 2004, harmonisation and international co-operation in Swedish and Community competition law enforcement will be further enhanced. The Competition Act applies to the production and trade of

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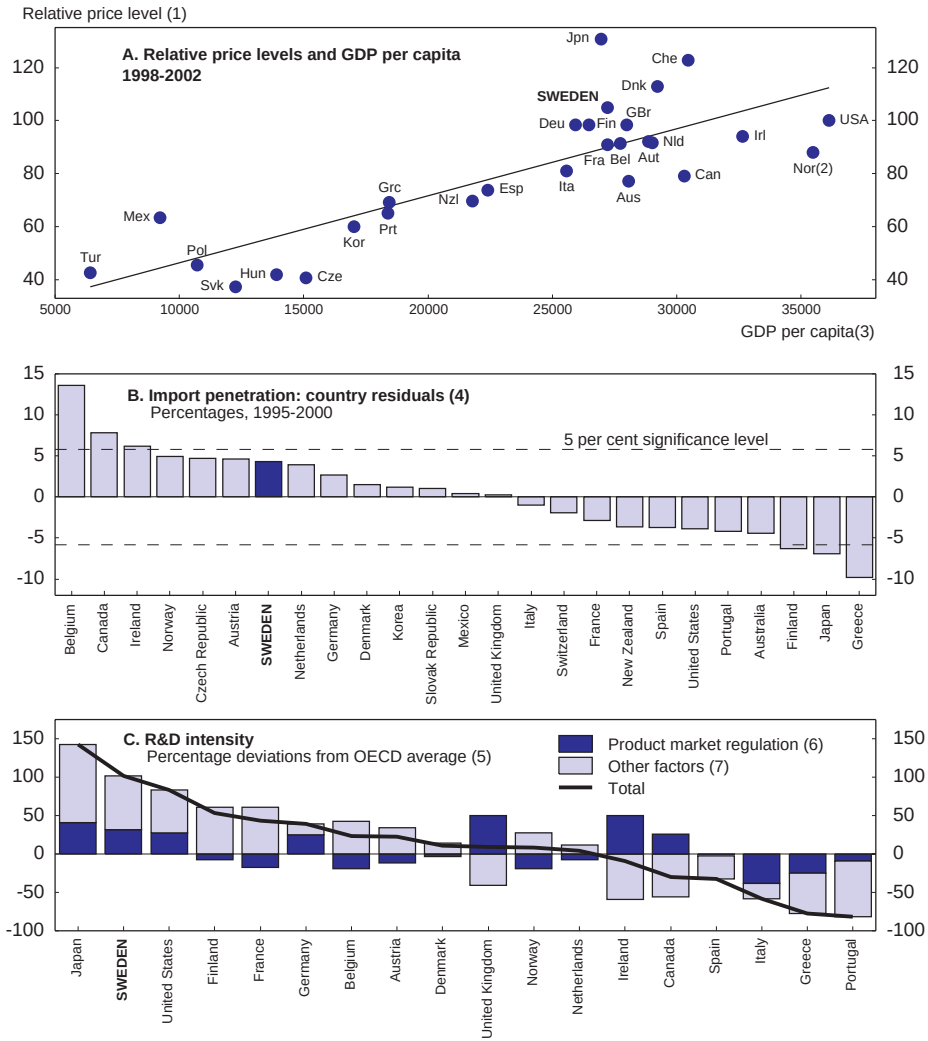
Figure 2.1. **Indicators on competition**



1. Purchasing power parities divided by the exchange rate, USA = 100.
2. 2001.
3. In US$, converted with the PPPs.
4. Residuals after control for effects of country size, GDP per capita and transportation costs.
5. The total deviation of R&D intensity from OECD average (adjusted for industry composition) is divided into estimated contributions from product market regulation and from other factors.
6. Includes administrative and economic regulations.
7. Includes EPL, other controls and country-specific effects.

Source: OECD.

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Table 2.2. **Relative price levels**

1999 EU-15 = 100

	Sweden	Denmark	Finland	Germany	United Kingdom	Italy	France	Netherlands
Food	**118**	127	113	101	101	97	109	94
Non-alcoholic beverages	**126**	143	123	105	120	92	88	90
Clothing and footwear	**107**	99	103	106	105	95	98	101
Rentals for housing	**125**	107	133	132	105	65	111	90
Construction	**126**	128	79	108	104	80	127	120
Electricity, gas and other fuels	**86**	158	90	106	86	103	102	125
Medical products and equipment	**119**	119	126	126	105	88	94	85
Purchases transport services	**163**	121	121	125	146	67	99	109

Source: Eurostat.

Table 2.3. **Output, employment and productivity**

Average annual percentage change, 1994-2002

	Sweden	Denmark	Finland	United States	Germany	United Kingdom	European Union	OECD
Average GDP growth	3.1	2.8	3.8	3.3	1.6	2.9	2.4	2.7
of which:								
Productivity per worker	2.3	2.0	2.2	1.8	1.2	1.9	1.3	1.7
Employment	0.8	0.7	1.5	1.4	0.4	1.0	1.1	1.0
of which:								
Unemployment[1]	0.5	0.6	0.9	0.1	–0.1	0.6	0.3	0.1
Labour force	0.3	0.1	0.6	1.3	0.5	0.4	0.8	0.9
	1994-2001[2]							
Growth in labour productivity per worker in selected industries								
Manufacturing	6.5	4.2	5.4	4.0	2.8	1.7	..	..
Manufacturing of radio, television and communication equipment[3]	42.9	8.5	17.5	11.7	10.8	4.8	..	..
Electricity, gas and water supply	0.5	2.6	5.7	1.3	6.2	10.8	..	..
Construction	0.8	0.5	–2.0	–0.0	0.3	1.8	..	..
Wholesale, retail trade, restaurants, hotels	3.7	1.7	3.0	4.3	–0.9	2.0	..	..
Transport, storage, communication	4.1	5.5	4.8	2.4	8.7	4.7	..	..
Memorandum item:								
MFP growth[4]	1.6	1.0	2.9	1.0	1.0	1.2	..	..

1. A positive sign indicates that unemployment has declined and helped to boost output growth.
2. For Sweden and Germany: 1994-2000.
3. For United States: Manufacturing of machinery and equipment; for United Kingdom: Manufacturing of electrical and optical equipment.
4. Total economy: For Sweden and Finland: 1994-2000.

Source: OECD.

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all goods, services and other products and to all institutions engaged in such activity, except for two sectors; primary agricultural associations and certain types of co-operation between taxi companies enjoy a legal exemption from the prohibition of restrictive agreements. The Competition Authority (CA) has the competence to enforce the Competition Act in all economic sectors covered by the law although, under the new Electronic Communications Act, the CA will share its traditional role of sole expert on competition analysis with the telecommunications regulator. Co-ordination with sectoral regulators and other similar authorities has reportedly worked smoothly, and competition concerns have also been a high priority in regulated sectors.

In practice the Competition Act is difficult to apply on certain anti-competitive practices in the public sector, most notably abuse of a dominant position. While the competition rules, in principle, apply to all enterprises whether private or public, irrespective of legal or organisational status, and non-profit as well as profit-making, in practice they are based on concepts designed to apply to private profit-making enterprises, and have proven difficult to apply when bodies such as municipally-owned enterprises rely upon public finances to exclude competitors through pricing below costs. The CA is also empowered to apply the Law against Undue Behaviour in Public Procurement, but this Act has proven to be ineffective as a remedy against current problems related to public procurement.

The CA has the power to take certain enforcement decisions on its own, but more important decisions are subject to ruling by Stockholm City Court with scope for appeal to the specialised Market Court. Private actions in competition cases may be brought before all civil courts, but they are rare. The reasons often cited include: the costs involved; expectations of meagre damages; and prevailing cultural traditions. The process can be rather drawn out: for example, the CA initiated the Uponor hard-core cartel case in 1997 and the case finally concluded with the Market Court ruling in January 2003 (Table 2.4).

Table 2.4. **Major competition cases ruled upon by the Market Court**
1998 to January 2003

Case	Initiated at the CA	Petitioned to the Stockholm City Court by the CA	Stockholm City Court's first instance ruling	Market Court's last instance ruling
Optiroc (merger)	May. 1997	Sep. 1997	Mar. 1998	July 1998
Nitro Nobel (fine, abuse of dominance)	Dec. 1994	Mar. 1997	June 1998	Sep. 1999
Scandinavian Airlines Systems (fine, abuse of dominance)	Oct. 1994	Oct. 1996	June 1998	Nov. 1999
State Railways (fine, abuse of dominance)	Dec. 1993	Jan. 1996	Dec. 1998	Feb. 2000
AGA Gas AB (fine, abuse of dominance)	Nov. 1994	Apr. 1999	Dec. 2000	Sep. 2002
Uponor Sverige and others (fines, cartel)	June 1997	Aug. 1999	Dec. 2001	Jan. 2003

Source: Annual reports to the OECD on competition policy developments in Sweden.

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So far, court proceedings in cartel cases have not resulted in clear signals deterring such behaviour. In such proceedings, defendants have engaged highly qualified legal experts, commensurate with the economic interests at stake. In some cases, very significant quantitative and qualitative resources have been put up by the defendants and the CA has consequently failed to prove its case. One example is the petrol cartel where leading companies had agreed on prices and reduced rebates. In court, the CA claimed that customers whose rebates were reduced suffered from an increased net price, but the Court found that price levels were not raised as a result of the concerted action and imposed fines less than 10 per cent of the level proposed by the CA (Table 2.5). Fines at this level provide little deterrent to other businesses from engaging in cartel activities. Furthermore, only one merger case has been brought to court since the CA lost the Optiroc case[1] in 1998, and in that instance, the defendant, Svenska Girot, decided not to proceed with the acquisition. This development may partly be due to the fact that major mergers mostly fall under the competence of the European Commission, but the standards set by existing jurisprudence are also perceived to discourage the CA from bringing other merger cases to court.

Table 2.5. **Fines claimed and imposed in major competition cases**

	Fine claimed by the CA	Fine decided by the Stockholm City Court	Fine decided by the Market Court
	(SEK 000s)		
Petrol companies (cartel)	651 000	52 000	Pending
State Railways (abuse of dominance)	30 000	8 000	8 000
Plastic pipe manufacturers (cartel)	17 500	10 600	10 600
Scandinavian Airlines Systems (abuse of dominance)	10 000	1 000	1 000
Nitro Nobel (abuse of dominance)	5 000	200	400
AGA Gas AB (abuse of dominance)	3 000	600	0

Source: Annual reports to the OECD on competition policy developments in Sweden and supplementary information from the Swedish Competition Authority.

Experience of the last 10 years indicates a number of areas where improvements could be sought. Public awareness of the detrimental effects of cartels has increased, but a competition culture is still not sufficiently strong in business circles, and the CA has so far unveiled conspiracies in such product markets as plastic pipes, petrol and asphalt. Some cartels even include state bodies, a prime example being the SNRA Construction and Maintenance subsidiary of the National Road Administration. Court rulings to date are unlikely to generate a strong deterrent effect given the potential economic rewards for companies participating in a cartel, and the sanctions

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actually imposed have not begun to approach the maximum of 10 per cent of annual turnover authorised by the law. Producing a clear demonstration effect through successful prosecution might be helped by providing the CA with the resources to employ more experts or to engage private counsel for litigating the most complex cases. Streamlining the process and shortening the time from initial investigation to final judgement might also help to reduce costs and make enforcement more effective. One option might be to replace the present two-tier court structure by establishing a Council within the CA for first-instance decisions, with appeal to one specialist court, following models applied in some other OECD countries.

The CA's capacity to act might also be helped by stronger incentives to "whistle-blowers' and cartel participants who offer full co-operation with the enforcement authority. Since 2002, the Competition Act has allowed for total immunity from fines in cartel cases on conditions specified in the law, but at the discretion of the CA. Given OECD experience, the effectiveness of leniency programmes might be enhanced by increasing the risk of disclosure, increasing the gravity of sanctions, and improving the transparency and certainty of amnesty terms. Present sanctions do not include criminal or other sanctions applicable to natural persons. Although criminalisation has recently been rejected by a government commission, many OECD countries have found that more severe sanctions may be needed in order to deter cartel behaviour effectively, and a new commission is revisiting this issue. Reconsidering alternative ways to make sanctions applicable to individuals may be a way for the government to signal its firm rejection of anti-competitive attitudes where they may still prevail.

Merger control under national legislation does not at present have a demonstrable effect in preventing the development of anti-competitive structures in the economy, despite the resources spent, both in the enforcement agency and by the companies concerned. If present legislation does not provide sufficient support for such action, a reform of the merger provisions of the Competition Act may need to be considered.

Serious competition problems are found in the public sector – or at the private/public interface – where the enforcement of competition law often fails to be an effective remedy. These problems have been observed for a long time, but very little progress has been made in dealing with them. The Competition Act may need to be modified so that it applies to anti-competitive activities by municipalities or other public institutions in practice as well as in principle. The Government's Bill to Parliament (1999/2000:140) Competition Policy for Renewal and Manifoldness stated that "the share of the total economy where there is competition should increase". But no major initiatives have been taken since the mid-1990s to expand the area exposed to competition and thus subject to competition law enforcement. An additional problem is that court rulings on anti-competitive practices have proved difficult to enforce, if the local government refuses to comply.

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Regulatory policy and competition issues in specific markets/sectors

The Swedish case provides some useful lessons on liberalisation of certain markets, particularly network industries, where it took bold steps to inject competition, but where the results were not always quite those initially expected. The following section reviews some of these sectors where further changes have been necessary, but also where other countries at an earlier stage in the process might benefit from the Swedish experience. It then moves to discuss several sectors that are apparently resistant to efforts to strengthen competition and where the obstacles to competition are not always easy to pinpoint, let alone dismantle. The section concludes with a review of two state-owned retail monopolies, where the authorities are not seeking to increase competition.

Electricity generation and distribution

Sweden was an early mover in opening up its electricity markets to competition. Electricity generation was opened up to international trade, and prices are now entirely determined by supply and demand through Nordpool, the power trading exchange established in 1996. Analysis of price data for 2001 and 2002 indicates that prices were equalised over the whole Nordic region only one-third to one-half of the time (Nordic Competition Authorities, 2003). Bottlenecks in the grids and weather conditions mean that at other times, the relevant geographic market is a shifting combination of sub-regions, although it has been rare for the Swedish part of the market to be completely isolated from its neighbours.[2] Simulations by the Nordic Competition Authorities indicate that there are several periods during the day when market power could be potentially exercised within the Nordpool area.[3]

Market concentration is a traditional measure of the degree of power that owners can conceivably exercise over a market. In Sweden, the state-owned company, Vattenfall AB, is the fifth largest electricity company in Europe and owns one-fifth of the generating capacity in the Nordic market and half of Sweden's. Nevertheless the Herfindahl-Hirschman Index (HHI) indicates that the Swedish market is less concentrated than many other European countries, and the Nordic market as a whole should be relatively free of market dominance (Figure 2.2). However, Vattenfall's management has already indicated its desire to buy hydroelectric generation in Norway as soon as that country eases ownership rules, which would enable the company to increase its market share in the Nordic region.

Sweden's experience with electricity price movements after market liberalisation provides a salutary lesson on interpreting the benefits from reform. Prices fell dramatically on Nordpool between 1996 and 2000, although prices to consumers fell less sharply (Table 2.6). Given the co-incidence with the timing of market liberalisation, it would be tempting to attribute this to increased competition, as many Swedes apparently did. The large price movements since 2001 thus left many consumers feeling dismayed, especially those who had opted for

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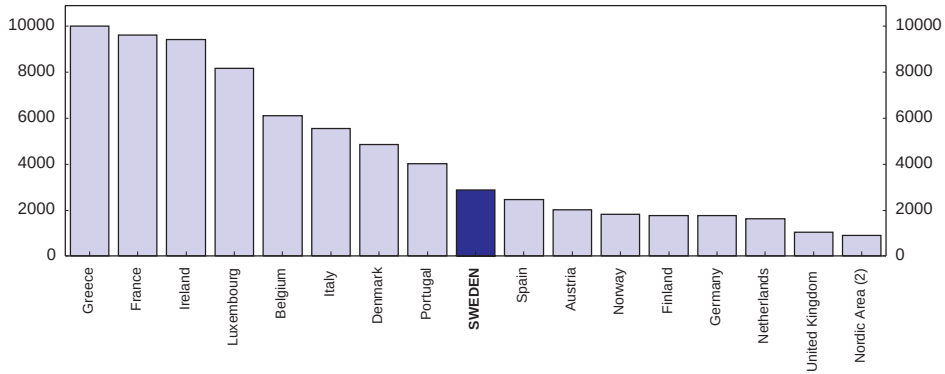
Figure 2.2. **Market concentration in electricity**

Herfindahl-Hirschman Index[1]



1. This is defined as the sum of the squares of percentage market shares of all firms in the relevant market. The more concentrated a market is, the closer the index will come to its maximum of 10 000.
2. Denmark, Finland, Norway and Sweden.

Source: Nordic competition authorities.

Table 2.6. **Electricity prices**

Excluding taxes, mean values, öre per kWh As at 1 January

	1996[1]	1997	1998	1999	2000	2001	2002	2003
Apartments	28.2	29.2	29.0	27.1	25.8	27.0	35.6	51.9
Single family dwelling								
Without electric heating	26.7	27.6	36.8	26.3	23.4	24.2	31.6	47.1
With electric heating	24.7	25.9	25.1	24.4	21.8	22.5	29.6	44.7
Agricultural and forestry	23.7	24.9	24.1	23.1	21.4	22.1	29.3	44.5
Commercial operations	n.a.	25.8	24.5	23.3	21.0	22.1	28.8	43.6
Industrial plant								
Small	24.0	25.6	24.1	22.8	20.4	22.0	28.5	44.3
Medium-sized	22.3	24.4	23.1	21.6	19.6	21.7	28.3	44.8
Electricity-intensive	22.0	23.7	22.7	22.5	19.7	22.6	28.3	48.0
Large electricity-intensive	n.a.	23.4	22.0	22.0	19.2	22.7	28.3	48.8
Average Nordpool system prices for full year	26.6	14.6	12.3	11.8	10.8	21.3	24.6	..

1. 1 July 1996.

Source: Swedish Energy Agency (2003).

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electricity contracts which rapidly adjusted to the changes in wholesale prices. In fact, 1996 turned out to be a dry year, requiring greater recourse to more expensive forms of generation, while 1997 to 2000 were years of abundant precipitation, followed by another dry year in 2001. Thus, the very real benefits of competition were largely obscured by weather-related fluctuations in prices.

It should also be noted that since the market was deregulated, installed capacity has fallen as uneconomic plants were decommissioned and replaced by imports. At the same time demand has risen. In the short term, with infrastructure limitations, this might lead to a lack of capacity on very cold days, with consequent rises in prices. But these increases would provide the necessary market signal to potential producers to expand capacity, while the reforms put in place should ensure that the new installations use the most efficient technology available within the Nordic region as a whole. Indeed, the full benefits of competition, *i.e.* the most efficient production of electricity possible in volumes that match consumer demand, will be realised through just such channels.

Sweden was also early in opening up retail distribution of electricity to competition, giving consumers the right to change electricity suppliers. Several remaining obstacles were eliminated in 1999, most notably the requirement for households to install expensive hourly metering equipment. In addition, the notification period was reduced from half a year to one month, and switching fees were no longer allowed. Around 45 per cent of Swedish electricity consumers have taken advantage of this opportunity to either switch supplier or to renegotiate their contract with their existing supplier, including virtually all industrial users. The legal requirement on distributors to publish their tariffs has helped consumers to compare rates and identify where gains might be possible.

Telecommunications

Sweden's experience as one of the first countries to liberalise its telecommunications market provides some insights into the practical issues that need to be addressed to generate effective competition (Annex 2.1). Some expected improvements did not materialise, and the 1993 Telecommunications Act was amended 15 times before being superseded by the Electronic Communications Act in July 2003. The new act strengthens the role of the National Post and Telecom Agency in several areas by giving it the powers to instruct the dominant operator to allow competitors access to their networks or limit their prices to the end-customer to what is reasonable. While the thrust of these provisions indicates a stronger *exante* regulatory approach than before, Sweden's experience suggests that it may not be sufficient to rely on the interested parties to negotiate contracts that provide effective competition. Nevertheless, Sweden already enjoys among the lowest prices for composite residential and business telephone charges in the OECD while mobile phone charges are around the middle of the range (Figure 2.3).

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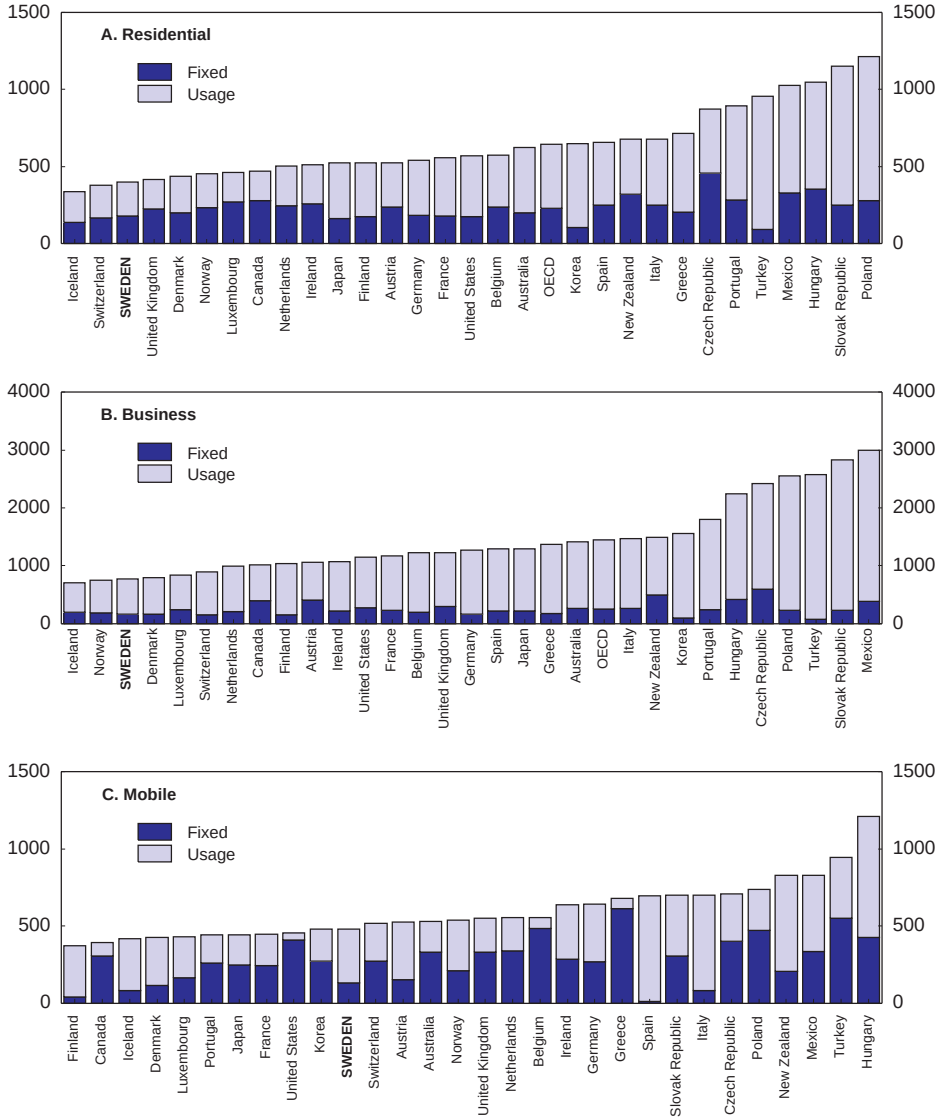
Figure 2.3. **Telephone charges**[1]
US dollars at current PPP, August 2002



1. Composite basket of services for residential and business charges and “average user” basket for mobile charges.
Source: OECD (2003b).

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Air transport

Various parts of the transport sector have been deregulated over the years, but getting vibrant competition to emerge has proved more difficult. Domestic air transport was liberalised in 1992, but SAS group still dominates the market, with 74 per cent market share, while Skyways, 25 per cent owned by SAS, has another 13 per cent of the market. Only seven routes provide any choice of airline, not counting the 10 routes that are provided on social grounds and for which tenders were called. The Nordic Competition Authorities rate SAS as a high-cost airline, partly because of its "business traveller profile", but also because it has a larger share of its income coming from its dominant position in its "domestic"[4] market than other European airlines (Nordic Competition Authorities, 2002).

Two main obstacles have been repeatedly identified as hampering more effective competition in passenger airlines. F*irst*, SAS's evident dominant position makes the use of loyalty programmes (and fidelity discounts) an effective tool in inhibiting others entering the market. Indeed the Swedish market court ruled that, from October 2001, SAS could not award air-points on the seven routes currently subject to competition on grounds of abuse of dominant position. S*econd*, in line with the EU Slots Regulation, landing slots are "grandfathered" to incumbent airlines, as long as they have actually used it 80 per cent of the time in the previous year. This process could make it significantly harder for a new entrant to operate at a congested airport, although new entrants are entitled to be allocated half of any spare slots. Practically speaking, this only affects Stockholm-Arlanda airport, where serious congestion arises at both ends of the working day. Four-fifths of all domestic flights start or end there, and since 1997 almost half the available landing slots have been allocated to SAS, which gives them an advantage in offering flights at peak times.

Rail transport

Despite an early start on liberalising the rail market, effective competition is still patchy. The first steps towards liberalisation were taken in 1988, when the national railway was separated into a public enterprise, Swedish State Railways (SJ), and a government agency, the National Rail Administration, which is responsible for the infrastructure. In addition, passenger services were deregulated for local services on the county lines. Further liberalisation took place in 1996, when county public transport authorities were given the right to operate passenger traffic on the main trunk network within the counties. However, SJ still retains a monopoly in long-distance passenger services. Where private providers were able to enter the market, either winning contracts for non-commercial services, or in providing inter-modal competition through bus services, the results have been beneficial for consumers. Prices have been reduced, especially for those in lower-income brackets such as students and pensioners (Nilsson, 2002).

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Part of the original motivation for liberalising the rail sector in 1988 was to redress persistent operating deficits, despite state subsidies from the early 1960s onwards. But SJ found itself again in need of a capital injection and financial restructuring in 2002. The company had competed aggressively in tendering for non-commercial activities, beating other bidders by offering services at prices 20 to 30 per cent less than the costs for the same activities prior to competition (OECD, 1998a). Subsequent operating deficits would seem to confirm that these prices were set at unrealistically low levels. The CA investigated a complaint against the behaviour of SJ and found that it had used predatory pricing techniques, a conclusion subsequently confirmed by the courts. Such behaviour was possible because the government was willing to underwrite it, giving the company a "soft budget constraint". Indeed, the government approved a new injection of capital in spring 2003 to stave off liquidation, while acknowledging that costs remain well above a level where it could compete with others (Regeringskansliet, 2003).

Construction

The construction sector is composed of several sub-markets with a high level of concentration, and significant vertical and horizontal integration. Sweden's two largest construction companies, Skanska and NCC, offer a comprehensive range of construction and infrastructure services and rank among the 20 largest companies worldwide in this field. Other large construction companies also have significant international activities. Despite the sector's evident competitiveness in international contracting, building costs in Sweden remain high, albeit with significant regional differences (Konkurrensverket, 2002b). Full implementation of the European single market has been hampered by slow progress in resolving the technical issues of the Construction Products Directive. However, with strong established networks, the industry as a whole tends to operate in a way that effectively excludes new suppliers, and several suspected cartels are under investigation or facing legal proceedings. Furthermore, the architecture and engineering consulting segments of the market have also become increasingly dominated by a few large companies. With their common practice of specifying Swedish materials and products, it will be difficult for imports to make inroads, even with the Construction Products Directive. Currently only 25 per cent of building materials are direct imports.

Rental housing

There are some unusual competition dimensions in the Swedish rental housing market. It is a market characterised by comprehensive rent regulation and extensive tenure rights (OECD, 2001). But this approach pre-empts the role of the price mechanism in helping the market to allocate efficiently the housing that is available, as well as to provide signals to expand the housing stock. The result is a very segre-

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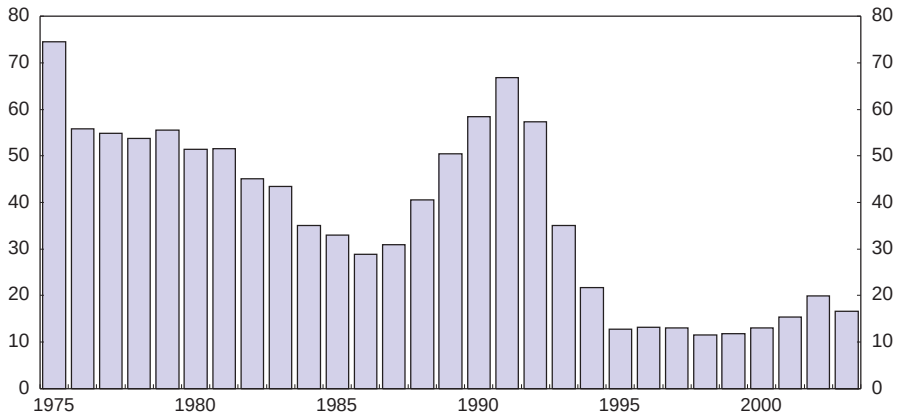
gated market and limited turnover as tenants hold on apartments with low rent, even if they no longer suit their needs or preferences. In the Stockholm area, for example, older people often retain long-held central city apartments, while younger generations are left with high-priced apartments in new buildings and rental housing in the suburbs. The rationing system that has emerged for allocating apartments that do become available, along with widespread black-market activities, tends to particularly disadvantage low-income households (Andersson and Söderberg, 2002a). These outcomes raise the question whether the lottery-like characteristics of the rental housing market are compatible with the stated equity objectives of the current rent-setting system. Such doubts are reinforced by the strong incentives for private landlords and existing tenants to circumvent the rules by accepting cash payments or sub-letting at a profit to those willing and able to pay the higher prices.

The rent negotiation system restricts the ability of rents to respond to changes in supply and demand, since in the collective rent negotiation system they are based on the so-called "use value" of the dwelling, which by and large corresponds to the cost-determined average rent (see previous *Survey*). With the rent structure effectively shielded from potential new tenants' willingness to pay, investors face a low return on construction of new rental dwellings, especially in the growth regions where land prices and construction costs are relatively high. This impedes the supply response to demand changes, which is manifest in the very low level of construction over a number of years (Figure 2.4).

Figure 2.4. **Completed rental dwellings**

Thousand units



Note: The number for the year 2003 is estimated, using the annual growth rate for total completed rental dwellings for the first three quarters as a proxy for the whole year.

Source: Statistics Sweden.

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In 2003, the major actors on the rental market presented the government with joint proposals for changes to the Rent Act (see Annex 2.2). They would make the rental market work better by allowing more efficient ways of adjusting rent structures, including an adjustment of rents for new tenants to make investment in the housing sector more profitable. But they are defined within the current collective negotiation system, without opening the way to any individual negotiation over rents. Following these proposals, the government has initiated an inquiry on the possibilities for providing for special rental-setting rules for dwellings in newly constructed buildings and on possible improvements for tenants by allowing them to choose various sorts of extra services. The prerequisites for better adjusting the rents of existing dwellings, the so-called "use value", are also to be reviewed. However, broader terms of reference might have been more appropriate. Two recent studies indicate that a total deregulation of the market would have positive overall welfare effects and that everybody could benefit from such a reform if suitable direct compensation were implemented (Anderson and Söderberg, 2002a and 2002b). This suggests that further empirical examination of the full equity implications of a wider range of different reform options could help to guide policy choices.

Meanwhile, in spring 2003 Parliament passed legislation introducing a subsidy to new construction and rebuilding of small and medium-sized rental and student dwellings in the period 2003-06. This was intended to help the government reach its stated goal of building 120 000 new dwellings in that period. It is designed as a tax expenditure (resembling a reduction of the VAT on construction from 25 to 6 per cent) and therefore technically avoids adding to pressure on the central government expenditure ceilings (see Chapter 4). The bill was partly motivated by recognition that construction costs are too high. But given the identified competition problems in the construction sector, subsidies of this kind may help contractors to maintain excessive profits, despite the subsidy being conditional on "acceptable" rent levels, and it would be more efficient to devote government efforts to lowering costs by encouraging more competition into the sector.

Food and groceries distribution

The food and groceries part of the distribution sector is widely recognised as lacking effective competition; three national chains between them control 90 per cent of sales and are vertically integrated with wholesale distribution (OECD, 2001). Surprisingly, although there are no constraints on opening hours, average opening hours in the whole retail sector (*i.e.* supermarkets and all other retail businesses) are very short – less than 10 per cent of all retail outlets are open for more than 65 hours per week, compared with around half in France, Belgium and Greece (OECD, 2003c). Swedish consumers thus face the double bind of higher prices and lack of convenience, both of which can be attributed to a lack of competition.

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Currently, the municipal planning process often impedes competition by favouring the three dominant supermarket chains. Results of the CA's examination of 16 000 planning decisions suggested that in areas with restrictive planning, shop space per inhabitant also tended to be smaller. That suggests that the decisions not only determine the location of shops, but also affect how many and how large they are. Municipalities often call in experts to assess the likely effects of providing land sites for food retailers and their inquiries almost always focus on the negative consequences and put too little emphasis on the positive implications of greater competition. Thus, it can be a challenging task for municipalities to take an informed decision based on these assessments.

Planning is also often slow and costly, including for the applicant, who often has to share the cost of consultancy studies. It also involves a high degree of uncertainty, as the process might lead to a rejection of a proposed new location for an outlet. Therefore the process favours the three established actors, as they have the experience and strong financial resources to carry them through a lengthy process and may even be well acquainted already with the planning staff. These factors impede entry by others and especially for smaller entrepreneurs (Konkurrensverket, 2003a). The CA has proposed: *i*) improving knowledge among city planners of the benefits from competition in terms of welfare; *ii*) better tools for assessing the positive effects of entry; and *iii*) clearer emphasis in the Planning and Building Act on competition.

Notwithstanding these difficulties, the German supermarket chain, Lidl, finally opened 11 stores in September 2003 and immediately provided one illustration of how the lack of competition had disadvantaged Swedish consumers. For example, the chain was able to sell fresh milk shipped from Germany at up to 20 ore per litre cheaper than the price of Swedish milk sold by the other chains. Swedish dairy suppliers complained about the competition, despite having declined to respond to Lidl's invitation to tender to supply Lidl with "own brand" label milk. It could be noted that there are no rules that would prevent the established supermarkets from selling German milk as well.

Pharmaceuticals

The retail distribution of pharmaceuticals (including those sold without a prescription) is in the hands of the state monopoly, Apoteket. The company's overall objectives are to ensure a "rational use … and a good provision of pharmaceutical products in Sweden … supplied at the lowest possible cost" (Regeringskansliet, 2003). The Swedish government has opted to maintain this monopoly approach, in large part because it wishes to ensure uniform sales prices in all areas of the country. While this may appear highly anti-competitive, pharmacies in other countries are generally highly regulated, and it has proved very difficult to establish effective competition as long as pharmaceutical purchases are almost fully reimbursed by the state[5]

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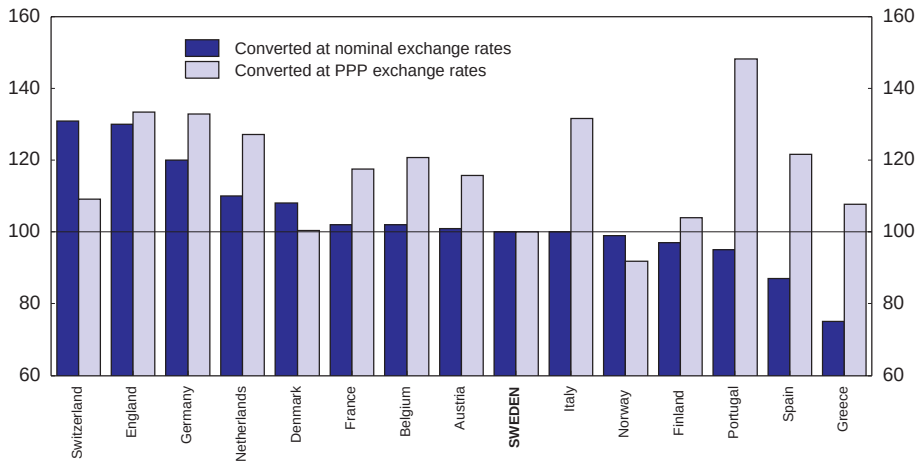
(OECD, 2000). The key difficulty is establishing the appropriate retail mark-up in the face of asymmetric information about the individual pharmacy's costs. Applying the same reimbursement rate to all retail pharmacies means that the mark-up is generally set high enough to keep the marginal pharmacy in business (often found in rural areas). However, since this implies that some (possibly many) pharmacies will be over-compensated, countries generally impose further controls on entry or on profits.

The Swedish approach does indeed appear to have delivered pharmaceuticals at lower relative prices than in many other European countries (Figure 2.5), with part of these gains coming through being able to effectively implement a strategy favouring generics and parallel imports. Nevertheless, a state monopoly provider does carry risks that the absence of competition may over time result in costs higher than necessary. One alternative would be to introduce competition by tendering the right to operate a certain number of pharmacies across the country: this would reveal information about local cost conditions which could then be used to set appropriate reimbursement rates. All other pharmacies could set their own prices and freely enter and exit the market. However, while this approach is theoretically appealing from a competition and efficiency point of view, it has not been tested in practice.

Medicines that are available without a prescription are not reimbursed even when prescribed by a doctor. Since consumers pay the full retail price

Figure 2.5. **Relative pharmaceutical prices**
Sweden = 100



Source: Riksförsäkringsverket (2002) and OECD calculations.

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(although there is no VAT on pharmaceuticals if the patient has a prescription), the retail mark-up can be determined by what the consumer is willing to pay, and competition would put downward pressure on such prices. Restricting sales of these items to Apoteket inhibits this process. Allowing over-the-counter medicines to be sold through other retail outlets would expand competition and the case is even stronger for non-medical personal hygiene products that are also sold under the Apoteket monopoly.

Alcohol retailing

Sweden also sells alcoholic beverages to the public through a state-owned monopoly, Systembolaget. It was allowed to retain this structure when it joined the European Union in 1995 because of concerns about public health, although monopolies in production and wholesaling were removed. However, concentration in wholesaling is still relatively high: the previous monopoly company, V&S group (still entirely state-owned), supplies more than a quarter of all beverages sold by Systembolaget (Konkurrensverket, 2003b). While there seems to be no discrimination against foreign suppliers, it has been argued that Systembolaget's system for selecting products to be offered in its outlets favours – and has been abused by – large incumbent suppliers (see Annex 2.3). Furthermore, Systembolaget is currently under police investigation for suspected bribery, and the company has already fired 10 managers on this account. Also, the European Commission has asked formally that the current prohibition of private imports *via* other companies than Systembolaget be lifted, as this ban is found to conflict with EU rules.

Since 1995, Sweden has successively increased the quantity of alcoholic beverages that individuals may import for private consumption, and from 1 January 2004 restrictions will be in line with the general rules in the EU. As taxes on alcohol remain the highest in the EU, the reduced import restrictions has led to higher imports for private consumption, and smuggling to avoid excise duties altogether has also increased (Table 2.7), while Systembolaget has experienced falling market shares especially in the regions closest to its southern neighbours. The incentives for Swedes to engage in cross-border trade were strengthened even further when Denmark reduced duties on alcoholic beverages from 1 October 2003. Indeed, Systembolaget's sales have plummeted in border regions, and similar initiatives in Finland in 2004 will add to the pressure.

This calls into question whether retaining a state-owned retail monopoly will continue to be a viable way of controlling alcohol consumption. Furthermore, the main instruments through which Systembolaget ostensibly exercises control over the availability of alcohol, namely location of stores, opening hours, age limits and right of refusal to serve (Regeringskansliet, 2003) are embedded in regulations that apply to competing retail outlets in many other OECD countries.

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Table 2.7. **Total consumption of alcohol**

	1996	1998	2000	2002
	Litres per person aged 15 and above, 100 per cent pure alcohol			
Spirits, wine and strong beer	6.7	7.1	7.4	9.1
Systembolaget	3.9	3.9	4.2	4.9
Restaurants	0.9	0.9	0.9	1.1
Import for private consumption	1.1	1.5	1.6	1.9
Smuggling	0.2	0.2	0.2	0.5
Home production	0.7	0.5	0.4	0.5
Light beer[1]	1.3	1.2	1.0	0.8
Total consumption	8.0	8.2	8.4	9.9
	Per cent			
Share of market for spirits, wine and strong beer				
Systembolaget	58	55	57	55
Restaurants	13	13	12	12
Import for private consumption	16	21	22	21
Smuggling	3	3	3	5
Home production	10	7	5	5

1. Almost exclusively purchased in food retail stores.
Source: Leifman and Gustafsson (2003).

Anyway, the apparently minor gap in consumption levels compared to, for instance, Denmark, where there is free entry to the market for retail of alcoholic beverages, suggests that retail competition and lower prices might not necessarily lead to an increase in quantities purchased; it might instead lead consumers to choose a wider range of products.

Competition and the public sector

The general government sector constitutes around 30 per cent of the Swedish economy, measured as a share of employment, while the public consumption-to-GDP ratio is the highest in the OECD area (Figure 2.6). In addition, fully or partly owned public enterprises produced almost 14 per cent of total business-sector value added in 1998 (Figure 2.7), which was the highest in the EU. In 2002, seven government-owned companies each had a net turnover (roughly corresponding to value added) of more than SEK 10 billion and 16 more showed a turnover between SEK 1 billion and SEK 10 billion (Table 2.8) including in several cases significant earnings from activities abroad. Although the public sector has been pared back significantly since the early 1990s, it remains one of the largest in OECD countries and therefore of particular interest, as inefficiencies may have

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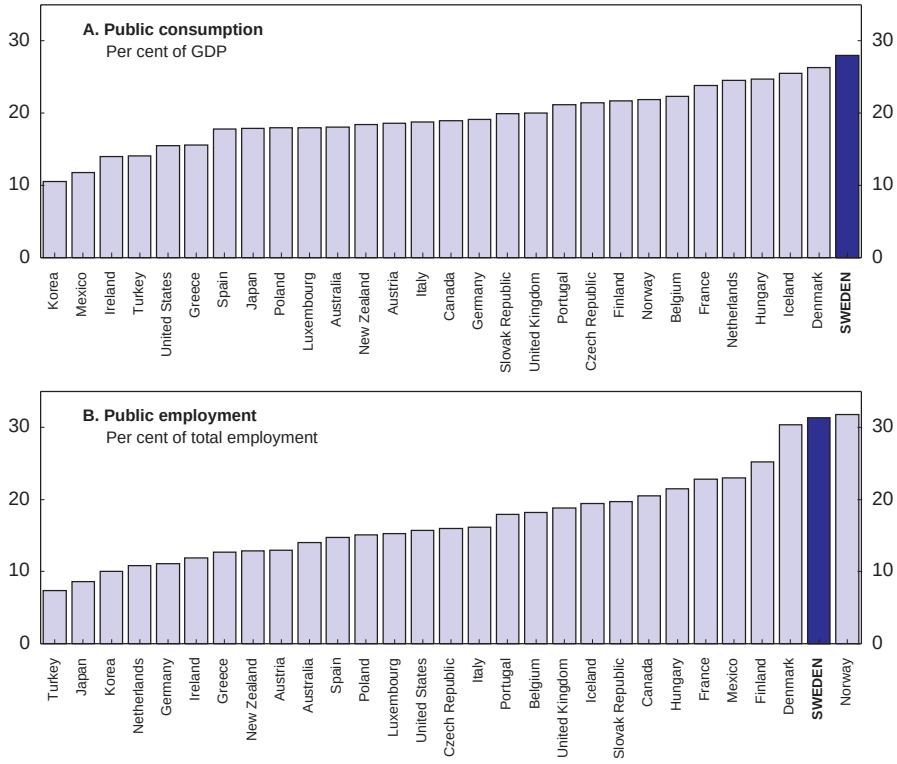
Figure 2.6. **General government consumption and employment**
Per cent, 2002



Source: OECD.

developed as a result of the usual absence of competition in the production of publicly funded services.

Deregulation, privatisation and procurement of publicly provided services have been on the agenda in the last 10-15 years. The objectives and operations of state-owned enterprises have become much more transparent, and various user-choice arrangements as well as contracting out have been introduced. Sweden has also applied an enterprise structure to some government activities that remain within the general government sector in other countries. But major areas remain where competition is weak or non-existent. Although some local governments have taken a lead in opening their activities to competition, some parts of the public domain still appear unwilling to recognise the role that competition

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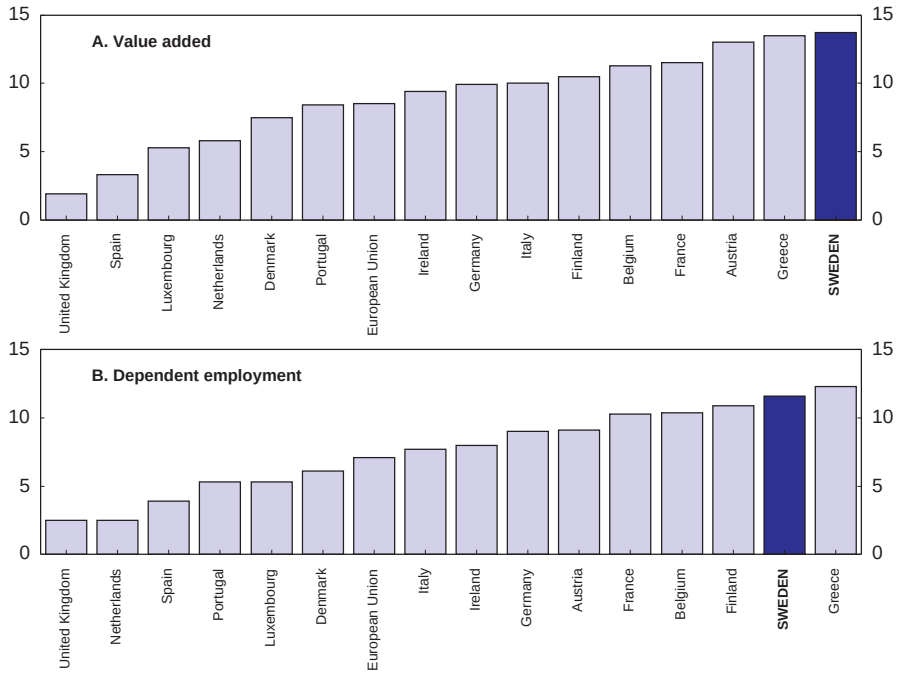
Figure 2.7. **Public enterprise sector in EU countries**[1]
Per cent of total, 1998[2]



1. The figures are likely to underestimate the size of the public enterprise sector, since they include only enterprises with majority public participation. Thus, enterprises in which government had a controlling influence *via* large minority holdings or golden shares are not covered.
2. Total comprises the non-agricultural business sector.

Source: CEEP (2000).

can play in ensuring more efficient production of public services, while in some circumstances also providing greater choice for consumers.

A number of studies on the effects of exposure to competition in local government show that the production and provision of public goods and services has become significantly more efficient, while quality has been maintained or improved (Konkurrensverket, 2002c). One particular effect identified is that the preparation of a call for tender has required a precise specification of the services required, which has allowed redundant functions to be identified and eliminated. However, smaller enterprises have sometimes found it difficult to get established in the public sector market, primarily because of lower ability to spread risks and relatively high fixed costs when engaging in public tenders. Some concentration of

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Table 2.8. **State-owned enterprise sector**[1]

	2000	2001	2002
Income statement			
Net turnover	247	294	337
of which:			
Vattenfall (electricity)	32	69	101
SAS (air)	48	51	65
TeliaSonera (telephony)	54	57	59
Apoteket (retail pharmacy)	28	30	32
Posten (postal services)	25	22	24
Systembolaget (alcohol retailing)	17	18	19
Svenska Spel (lotteries)	14	16	18
Operating profit	32	33	16
Net profit after tax	20	17	9
Dividends paid	9	9	10
Balance sheet			
Assets	860	963	977
Liabilities	698	787	782
Shareholders' equity	162	176	195
Memorandum items:			
Net capital investment	69	95	24
Government grants	22	23	24
Return on equity (per cent)	12.9	10.3	4.6
Employees (thousands)	197	201	199

1. Includes only state enterprises included in the government's annual report on State-owned companies. It does not include companies set up as subsidiaries of state agencies or those owned by counties and municipalities.

Source: Regeringskansliet (2003).

contractors has taken place, for instance in health and disability care, as minor enterprises have been bought out by bigger actors. However, this may partly reflect the scope for private contractors to benefit from economies of scale and scope.

Some Swedes have expressed concern about whether exposure to competition through external procurement would reduce democratic control and management of activities. One examination of experience in the city of Stockholm revealed that objectives were better targeted, but some politicians observed that it was more difficult to change objectives and reorganise activities once the contract was established (Konkurrensverket, 2002c). A central issue is the design of processes and agreement terms, *i.e.* whether the services in question and level of quality had been clearly identified in discussions involving the main interested parties. In those cases where an adequate design had been worked out, politicians – as well as relevant consumer representatives – would have substantial influence on outcomes. Another concern has been continuity in the supply of services, especially health and disability care. Frequent changes of supplier could

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reduce the perceived quality of the services and make it more difficult to find firms willing to invest in developing expertise or making associated capital investments. Stockholm City has introduced consumer choice among certified suppliers in elderly care, in order to reduce uncertainties for both consumers and employees once activities have been procured and to improve the incentives for suppliers to develop their services.

The effective opening up of publicly funded services to competition can be hampered by the tax system. A special grant providing compensation for VAT on inputs for private producers of tax-exempt services has helped to alleviate the cost disadvantage faced by private providers competing with municipal producers, thereby ensuring a more level playing field. However, the effectiveness of this system has been called into question, and the Competition Authority has found it insufficient to remedy the cost disadvantage for private providers of dental care. Furthermore, both the level and the structure of taxes affect competition more generally, especially through their impact on business creation and small, closely held companies (see Chapter 3).

Public activities in competitive markets

It has become increasingly common for various levels of government to operate in areas where private companies already exist (Konkurrenskommissionen, 2002; Statskontoret, 2000). To some extent, this might reflect state ownership of companies that were previously monopolies on now deregulated markets, for instance in the telecommunications and postal sectors. On the local government level, it might also be a result of policies to deal with regional differences, for example in rural areas ostensibly lacking potential private providers. Examples can be found of municipalities operating bakeries, gymnasiums, garden centres, sun-bed centres and privately financed health care, either as part of the municipal administration or provided through a company owned by the municipality.

While there may be legitimate reasons for publicly operated activities on competitive markets, the extent to which it is happening in Sweden has been challenged. The Swedish Agency for Public Management has concluded that in a number of cases there are no valid motives for the state to engage in activities on competitive markets, and that distortions to competition from such activities in general have a negative effect on the ability of these markets to function effectively (Statskontoret, 2000). Meanwhile, the National Commission for Competition on Equal Terms[6] has registered what it considers an unnecessarily large number of conflicts between private and public providers, while the Competition Authority has received complaints that the public provider cross-subsidises the competition-exposed activity, thereby distorting competition. Other examples of recorded problems are:

- Some non-profit public entities have a statutory right, although no obligation, to carry out competitive operations. For example, this is the case

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with commissioned courses provided by universities and upper secondary schools and municipalities running hotels, holiday villages, camping sites and ski slopes.

- Public agencies pursue competitive operations on the premises of the agency's primary activities, for example hospitals providing restaurant facilities and public swimming pools providing gymnasium or solarium facilities.
- Services developed for internal purposes, for instance in transport, cleaning, laboratory or carwash services, are sold to others to make use of spare capacity.
- Official authority in activities protected by monopoly is used to gain advantages in a related competitive market. An example is the municipal rescue service, which also sells fire-control equipment in competition with private retailers.
- Public companies have engaged in unlawful behaviour, for instance by participating in cartels, as already noted above.

Several Acts regulate competition between private and public producers. The Competition Act includes restrictions on predatory pricing, but those provisions apply only to companies with a dominant position. The practical application of that law has not prevented public agencies operating in competitive markets and non-profit entities from trying to exclude others. Local government legislation imposes some restrictions, but not, for instance, on activities in competitive markets that do not fully cover costs. Also, central government authorities are subject to certain restrictions, but there are large grey areas here as well as in the local government domain. In a few sectors, special legislation does clarify competition issues, but the general picture is that legislation on public activities on competitive markets is non-transparent, lacks coverage and provides little room for private actors to address competition problems through the legal system. Even though any citizen can take the municipality in which s/he resides to court for an alleged infringement of the Local Government Act, there are no effective sanctions to ensure enforcement of any subsequent court rulings. Furthermore, the legality of local government companies cannot be tested in court.

Public procurement

Outsourcing *via* public procurement is one way of introducing greater competition for the supply of goods and services in the public sector. Sweden has adopted the EC Directive on public procurement, but it has proved difficult to ensure full compliance with its requirements. Nearly 5 per cent of GDP was openly advertised public procurement in 2001, which is higher than the EU average (Figure 2.8). However, public consumption is also higher in Sweden, and the scope

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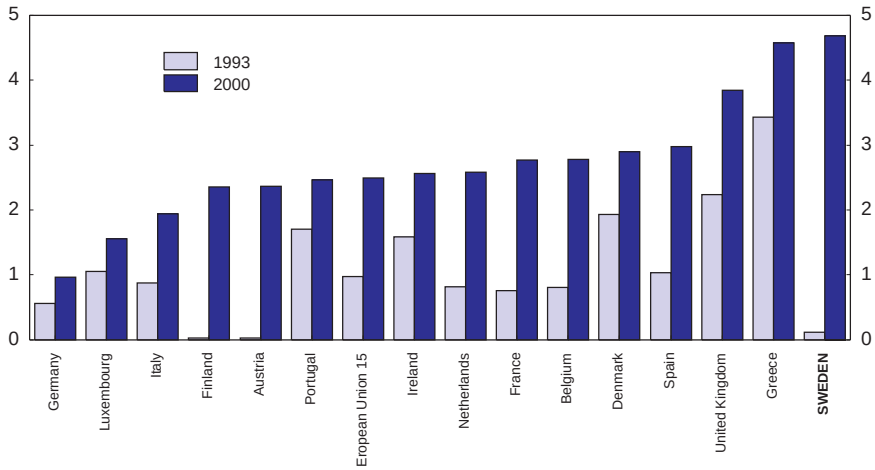
Figure 2.8. **Openly advertised public procurement**
Per cent of GDP



Source: Eurostat.

for further public procurement (or other forms of exposure to competition) has been estimated at 11-12 per cent of GDP (Konkurrensverket, 2002c). Thus, even though experience to date has been generally positive (Box 2.2), it seems that Sweden is still a long way from using all the possibilities in this area. This could be for a number of reasons, including weaknesses in the legislative and institutional framework, as well as reluctance by some municipalities and state institutions to put activities out to tender.

Almost all public procurement in Sweden is regulated by the Public Procurement Act (LOU), which was implemented in 1992. The rules vary for procurement of services according to the amount involved (see Annex 2.4) while the regulations applying to tenders larger than the threshold value reflect EC directives. The National Board for Public Procurement, an independent public authority under the Ministry of Finance, is responsible for observance of the Act. The government's National Commission for Competition on Equal Terms performs regular monitoring of the Act as does the Competition Commission (a private initiative).

These institutions have identified several impediments to effective procurement, including: unlawful direct procurement; distortion through vague design of the tender documents or specifications aimed at a particular company; awarding contracts based on factors other than those stated in the contract documents;

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Box 2.2. Experiences with procurement at the municipal level

Most of the surveys conducted on the experience of exposure of public activities to competition have focused on municipalities, as state activities have primarily been exposed to competition through deregulation of various markets. The Competition Authority draws the following general conclusions from the existing studies in Sweden (Konkurrensverket, 2002c):

- In most cases exposure to competition has initially resulted in cost reductions and/or quality improvements for the local government and individual consumers – even if an activity continued to be produced by the local government itself after exposure to competition.
- The procurement process has led to lower costs on goods and services and more focus on quality issues, one reason being increased incentives (through the Public Procurement Act) for municipalities to provide a detailed description of the service to be produced in the invitation to tender. However, careful preparation is required when introducing competition in order to reduce the risk of default, which could leave clients stranded.
- Municipal production can become as cost-effective as that of private sector suppliers. There are some cases where exposure to competition has raised the efficiency of in-house production units competing alongside others.
- Sentiment in consumer polls is neutral in most case on whether a service is produced by municipal or private actors. However, frequent changes of providers can undermine the continuity of a service, which is a highly valued quality for consumers, especially in service areas such as elderly and child care. Employee morale has often been lifted by exposure to competition, although offset by a perceived heavier workload.
- It is important for municipalities to separate the role as purchasing and financing authority from that of producer in order to minimise the risk of inconsistency or conflict, although some interaction between these various roles will sometimes be almost unavoidable in order to improve the service.

and local governments purchasing goods and services directly from their own companies (total or part ownership), even though they are obliged to undertake a public procurement process. Examples of the last are, for instance, found in the areas of cleaning, management of housing buildings, street maintenance and laundry services. In many cases, contracts are prolonged automatically, and the municipal companies are often at the same time competing with private companies by supplying surplus production to the private sector. While it might be natural to allow government-owned production units to compete with private firms for procurement contracts in a transition period following the opening up of a market, there are examples of these companies being favoured by discontinued procure-

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ment procedures[7] as well as better conditions in terms of specific requirements in the contract documents and the lack of a required return to capital in general (Konkurrensverket, 2002c).

Unlawful direct procurement seems to be significant (Konkurrenskommissionen, 2002). To some extent, this might be explained by insufficient knowledge about the legislation as well as lack of internal management and control of procurement in the public institutions, which at least partly reflects insufficient guidelines in the present legislation. However, the absence of effective sanctions against such behaviour might also play a role. Although this has been pointed out by a number of organisations and government committees for several years, no proposal on sanctions has yet been introduced to Parliament. Nevertheless, the government is making efforts to improve the public procurement process and is considering introducing special sanctions where the rules are breached by signing a contract directly with a supplier, although corresponding legislative amendments have not yet appeared.

Until recently, it was almost impossible for private suppliers to bring an alleged infringement of the Public Procurement Act to court. But in 2002, the Act was amended to make it easier for tendering firms to request a legal review of the tendering institution's decision. Now, the tender decision must be made subject to review for at least ten calendar days, and the secrecy requirement on submitted tenders now expires on the decision date instead of when the contract is signed. While this is obviously an improvement, the ten days minimum reviewing period is still very short. In addition, court practice has implicitly required that a supplier must have submitted a tender or an application to tender in order to be able to bring the matter before the courts. Thus, any company wanting a court ruling on an alleged unlawful procurement process must be able to document a financial loss. Along with the short minimum reviewing period, this constitutes a significant impediment to an effective application of the legal framework to the procuring institutions.

In certain cases, public authorities have deliberately ignored court rulings. This is made possible by a legislative framework that lacks mechanisms or sanctions to ensure that court rulings are actually observed by public institutions, although new rules implemented in 2002 gave further scope to municipal auditors to investigate and report on observance of court rulings. Unlawful decisions in procurement processes are typically not considered to be the exercise of authority under the penal code, thereby preventing prosecutions for misconduct in office, and members of municipal executive committees are not subject to disciplinary responsibility regulated by either law or collective agreements (Konkurrenskommissionen, 2002). Thus, it seems that Sweden does not currently fulfil EC directives on public procurement that require national legislation to guarantee observance of court rulings. Also, some municipal and state companies and a few other public

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organisations apparently believe that they are not subject to the Public Procurement Act. This suggests that the coverage of the Act also needs to be clarified.

Public enterprises and privatisations

The Swedish state is the corporate sector's largest shareholder, and the government has made strenuous efforts over a number of years to provide a sound governance structure for these activities. One key element is the distinct treatment of companies deemed to be operating under market conditions versus those with special societal interests. State ownership is clearly a choice made on political criteria (and any sale of shares in a company more than half owned by the state must be approved by the Swedish Parliament). But as already indicated above, some state-owned enterprises have very strong positions in their markets, and the capacity of the government to underwrite any eventual losses could add to their market dominance or increase their capacity to see off competitors (OECD, 1998b). At the same time, international empirical evidence indicates that privatised firms almost always become more efficient, more profitable, increase capital investment spending and become financially healthier, although the impact on employment is unclear (Megginson and Netter, 2001). These benefits could be interpreted as unexploited dynamic gains from competition that are stifled by continued public ownership.

Subsidies and state aid

State aid paid by the Swedish state amounts to a small share of GDP compared with most other EU countries (Figure 2.9). Around one third is directed to the manufacturing sector and one third to the transport sector, almost all of which is provided to the railroad track company (Näringsdepartementet, 2002). It is difficult to assess the extent to which this might distort inter-modal competition, especially with road transport, given the public provision of the road network. More generally, the numbers reported according to the European Commission criteria may not reflect the full extent of economic subsidies. In some cases, these are made indirectly or are disguised, for example, as lower VAT rates. Also, local governments provide significant amounts of aid in the form of cheap loans and guarantees (Konkurrenskommissionen, 2003). Other aid to Swedish producers is delivered by the EU, in particular through the Common Agricultural Policy, which provided Swedish farmers with half their gross receipts in the form of direct payments and market price support in 1999 (OECD, 2003d).

Subsidies have often been used to maintain employment in the short term, especially in specific sectors experiencing severe downturns, but the effect is to impede structural adjustments. Some local government subsidies might also conflict with the prohibition in the Local Government Act against subsidies to individual companies. Currently, residents in the municipality may take the local authorities'

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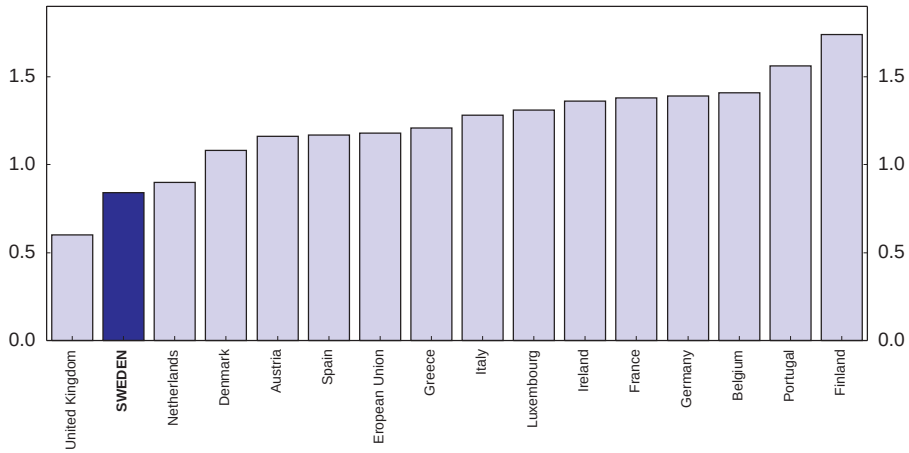
Figure 2.9. **State aid to enterprises in the EU**
Per cent of GDP, average 1997-1999



Source: European Commission (2001).

decisions on subsidies to the courts. However, it is very difficult, and sometimes impossible, for a company to contest the subsidies given to a competitor. The company claiming an economic loss can only make a case if it is registered in the municipality. Even if an appeal can be filed, the court can go no further than finding the municipality guilty of subsidising in an unlawful manner, as no sanctions are provided. In the situation where a municipal company decides to provide a subsidy, a court appeal cannot even be filed. Also, it is often difficult for a company to detect that the municipality has decided to make a subsidy, and an appeal might then be refused on formal grounds, for instance, because the time limit has expired.

Concluding remarks and priorities for policies

There are many areas where competition is already working well, or steadily improving, but the analysis in this Chapter indicates a number of areas where adjustments to policy settings could facilitate the emergence of greater competition and lead to a more dynamic and thriving economy. The authorities themselves have pinpointed a number of areas where competition is still weak and have set in train a range of measures designed to make markets work more effectively (Ministry of Finance, 2003). But it should be admitted that in some areas it is no easy matter to generate effective competition; in several sectors, the government has removed all apparent obstacles and has still seen disappointingly little improvement in results.

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This perhaps indicates the importance of also shifting the expectations of both businesses and consumers so that they value and more actively seek out the benefits of competition. It could also be helped by more widespread recognition of the need for more flexible labour markets, so that the economy can adjust more rapidly and easily to the greater changes at the firm level that more intense competition implies.

The first area where competition could be enhanced is through strengthening enforcement. While the competition legislation contains the right framework for underpinning competition, it is handicapped by several gaps that need to be addressed:

- The Competition Authority and the courts need to demonstrate their capacity to break down hard-core cartels through successful prosecutions, and fines that correspond to the economic damage inflicted. Despite efforts made by these institutions, success to date falls short of providing an effective deterrent to such anti-competitive behaviour, in part because of the asymmetry between the specialist resources available to the authority compared with those at the disposal of the defendants.
- The time taken for cases to reach a final ruling through the courts could be shortened by streamlining the legal process; reconsidering the need for two separate judicial reviews of the rulings of the CA might be one worthwhile option.
- Providing scope to penalise the individuals within companies for their anti-competitive actions, without resorting to criminal sanctions, could send a clearer signal that such behaviour is unacceptable, strengthen the recently introduced leniency provisions and help change the business culture in sectors where collusion remains common.

There are several sectors where further action is needed to ensure that effective competition occurs, even though considerable progress towards liberalisation has already taken place in some instances:

- *Electricity* is fully exposed to competition through Nordpool, but large generators are still able to exercise some market power in Sweden at certain times when grid bottlenecks occur. This could for example be dealt with by extending the grid infrastructure to reduce the scope and possibility for the companies to manipulate prices.
- Given the present EU regulation that "grandfathers" landing slots at Stockholm-Arlanda *airport*, the only option available to the Swedish authorities to reduce congestion at peak times and boost competition is to expand airport capacity so as to create more slots.
- The financial injection provided to the state-owned *railway* company, SJ, gives it an advantage against competitors. This playing field will only be

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levelled if SJ faces a hard budget constraint, in other words, if it is allowed to go out of business if it cannot compete. Other passenger transport companies would still compete to provide rail or alternative modes of transport, while market signals would be better able to establish the most appropriate services.

- Injecting greater competition into *construction* activities would be facilitated by implementation of the EC Construction Products Directive. The demonstration effect of successfully tackling cartels within this sector would help to reduce the pervasive co-operation within the market that shuts others out, despite their potential to supply at lower cost.
- While the *rental housing* market is oriented towards meeting equity objectives, a careful examination of the equity and efficiency implications of a less tightly regulated market might identify alternative ways to meet key social goals more effectively, while securing the economic benefits of a better functioning market.
- Increasing competition in the *food and groceries* sector would be helped by easier access to suitable building sites for new supermarkets. Although planning approvals are a local government responsibility, the Planning and Building Act could be modified to include a clear requirement to take the benefits of competition into account in making decisions.
- The present state-owned monopoly approach to *alcohol retailing* is designed to place limits on the consumption of alcohol, but such controls could instead be imposed through regulation of competing private-sector outlets as occurs in most other OECD countries.

The public sector is very large in Sweden and thus has a significant impact on competition in several different ways that need to be addressed:

- Where public companies compete in markets in which other suppliers also operate, greater efforts are needed to ensure that they are not benefiting from explicit or implicit subsidies. The legal framework governing such activities could be strengthened to reduce the likelihood of unfair competition emerging; these activities could be brought within the scope of the Competition Authority; and the opportunity to seek redress through the courts could be expanded.
- Despite Sweden's adoption of the EC Public Procurement Directive, and the government's efforts to expand public procurement, there is still significant scope to raise the degree of competition through such channels. Given that most of it takes place at the sub-national level, this would benefit from a multi-pronged approach:
  - Devoting more resources to education, information and supervision in the area of public procurement.

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- Rationalising the present supervisory structure into one agency and granting it the authority to impose fines in cases involving serious infringements of the law on public procurement, such as illegal direct procurement.
- Modifying the legislation governing public procurement to give identical status to tenders from internal units and external producers, while requiring the call for tender to specify whether internal offers will also be considered.

- Delegation and decentralisation of the activities of the 300-odd state agencies has made it possible for some agencies to diversify into commercial activities in ways that are unhelpful to competition. This suggests that government's mandates to agencies need to specify more clearly the core activities of agencies and place limits on possible expansion in other directions. More effective monitoring of these agencies' behaviour and their compliance with public procurement rules would also demonstrate the government's commitment to promoting competition through its own activities.

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Notes

1. The CA asserted that Optiroc's acquisition of Stråbruken would create a totally dominant position for product markets such as facing bricks, floor putty, mortar and dry concrete, with market shares of between 63 and 85 per cent. Optiroc and Stråbruken were also the only companies with anything approaching a full range of such products.
2. Over the past three years, this has only been the case for less than 0.1 per cent of the time. Usually Sweden and Finland or Sweden and East Denmark form the geographic market.
3. The extent to which market dominance can actually be exploited depends on the characteristics and nature of supply as spot electricity prices are relatively insensitive to changes in demand. The extent to which any supplier can respond to a potential excess of demand by increasing supply depends on whether the production technology is flexible, how close to capacity a plant is already operating and whether bottlenecks inhibit the distribution of any additional output (OECD, 2003a).
4. SAS's domestic market comprises Denmark, Norway and Sweden: SAS is half-owned by the governments of these three countries.
5. Sweden operates a scale of user co-payments that gradually reduces to zero depending on the individual's consumption of pharmaceuticals in any given year (see previous *Survey*).
6. This Commission was established in 1998 to resolve disputes concerning competition between public and private actors, as the competition law was considered an inadequate tool for dealing with such problems. The Commission has no judicial powers and can only express opinions. Its mandate ended on 31 December 2003, and the government has instead instructed the Competition Authority to pay attention to conflicts between public and private actors.
7. In some instances, public entities have cancelled a procurement process, once it was clear that external companies had submitted lower offers than in-house production units could match. Discontinuation of the procurement process would imply that internal production would continue.

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Annex 2.1
Telecommunications competition in practice

A recent review by the National Post and Telecommunications Agency identified the following areas where lessons emerged from the experiences of the last decade with deregulation (National Post and Telecom Agency, 2003):

- *Interconnection* charges were originally left to be negotiated between the relevant operators, with the National Post and Telecommunications Agency (PTS) empowered to act as mediator in disputes. This proved ineffective, and in 1997 the PTS was given the authority to make a decision if any of the parties requested it to do so. This amendment significantly rebalanced the powers between the larger and smaller operators and produced a major reduction in interconnection charges. However, the resolution of disputes can still take a long time, in part because of the appeals process.
- "*Pre-selection*" (enabling the consumer to automatically place calls with alternative operators) was implemented at the end of 1998, and approximately one-third of private telephone subscribers have opted to use this facility. In addition, consumers can now choose between around 30 operators. This reform has facilitated greater competition than the earlier "dial-a-prefix" arrangements were able to deliver.
- *Number portability* introduced for mobile phones from September 2001 was expected to make it significantly less costly for consumers to switch operators and increase competitive pressures. But to date it has not lived up to expectations, in part because of lack of public awareness. Operators have weak incentives to promote number portability because the switching costs are relatively high.
- *Provision of mobile network capacity* may require that operators be obliged to lease network capacity to those who do not have their own infrastructure. Several new service providers have entered the market, but their market shares remain tiny. Incumbents have very limited incentive to open up their networks, while the PTS can only mediate in disputes, and it faces significant informational asymmetries *vis-à-vis* the established operators.
- *National roaming* was introduced in 1999 to make it easier for new mobile operators (subsequently including UMTS licence holders) to enter the market without having to establish complete geographical coverage from the outset. In practice, it has proved difficult to conclude contracts to cover roaming, and the mediation efforts of the PTS have not yielded significant results.
- *Access to the local loop* is covered by EU regulation and requires operators to grant reasonable requests for access to the network for a cost-based fee. Telia has published a reference offer that complies with the regulation, but few access leases are in place. Access for ADSL connections are not covered by the regulation, and Telia has already established a strong advantage in the market.

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Annex 2.2

Proposals by housing organisations

Acknowledging that imbalances had been building up in the rental housing market, the main actors on the rental market initiated a discussion in 2001 on how to improve the system. In 2003, SABO (Swedish Association of Municipal Housing Companies), the Swedish Union of Tenants and the Swedish Property Federation presented a joint proposal (SABO *et al.*, 2003). Under the presumption that the collective negotiation of rental terms be retained, the organisations suggested various changes to the Rent Act:

- Special attention to construction costs should be allowed in the collective agreements negotiated on rents for newly constructed dwellings. Currently, a tenant can apply for a "fairness test" of the rent by a rental committee, based on a "use value" comparison with other rental dwellings after six months of rental. In many localities there is a big risk that it will lead to a reduction of the rent to a level below that required to cover production costs. If the organisations on the rental market could agree on rents that would cover construction costs, only very special circumstances should make the rental committee decide in favour of the tenant.
- Collective agreements on different rent levels for dwellings with the same "use value" should be allowed. Rents in some locations do not adequately reflect the use value of dwellings, leading to differences in rents that do not reflect variation in characteristics. One way of rearranging the rent structure would be to fully adjust rents for vacant dwellings, while rents for existing tenants could be raised gradually, but this is currently prohibited by the Rent Act. A relaxation of the rules, combined with transition regulations that protect existing tenants from excessive rent hikes, should make it possible to gradually adjust rents – and rent differences – to better reflect the real assessments of various dwellings. To prevent abuse, the transition regulation should not apply for dwelling-swapping or subletting.
- Multi-annual agreements on certain individual rental arrangements, such as special installations and services supplied to the tenant, should be allowed. If these arrangements involve additional constructions or rebuilding, costs could be higher than the rent based on "use value", and uncertainty about the outcome of a subsequent "fairness test" of the total rent – especially because a basis for comparison of a certain investment might be hard to find – has made owners reluctant to make such investments. In some instances, tenants have then chosen to bear the full costs themselves, even though the asset produced by the investment ultimately belongs to the owner of the dwelling. Multi-annual agreements could pave the way for a fairer division of benefits and costs as well as a wider range of services offered to tenants.
- The Act should include a clear requirement that rent committees provide a detailed description of the basis for decisions in cases where a comparison of rents is not pos-

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sible. While the "use value" system in general presumes that rent disputes are settled on the basis of a comparison with rents for similar dwellings, there has been a trend for rent committees to base their rulings on a "fairness assessment" rather than on comparisons, allegedly because there has been no basis for comparison. The motivation for the ruling in such cases has been found unclear and inadequate by the parties on the housing market, leaving no guidance for future negotiations.

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Annex 2.3

Alcohol retailing

Prior to Sweden's accession to the EU in 1995, retail and wholesale trade as well as production of alcoholic beverages was confined to state monopolies. The system has its roots back in the middle of the 19th century, when the first local distribution outlet, "Systembolaget", was formed by mine owners as a reaction to alcohol abuse among the mine workers. Similar outlets spread around the country (they were later merged in 1955), and various restrictions on production, trade and consumption followed (home-distilling was prohibited already in 1860). Throughout, the concern for public health has been the dominant motivation for developing the state monopoly (Systembolaget, 2003).

Health concern was also the argument put forward by the Swedish government, when negotiating the terms for Sweden's accession to the EU. While Systembolaget's monopoly in alcohol retailing was accepted by the EU (based on a balance of concerns between health and competition), the monopolies in wholesale trade (to restaurants and others with a license to serve alcohol) and production had to be given up. To monitor the risk that the monopoly might be abusing its dominant position, the Swedish Competition Authority prepares semi-annual surveillance reports to the EU, focusing on consumer choice of products and the possibilities of entry for new products.

Systembolaget is obliged to offer a wide range of various alcoholic beverages. After a strong increase in the number of products in the second half of the 1990s, there has been a decline in recent years. While the company itself emphasises changes in demand patterns and exhaustion of stocks as possible explanations, some have suggested that Systembolaget's new selection policy restricts product diversity. One of the main objectives of Systembolaget is to offer the products with the highest demand based on market surveys and various other contacts with consumers and suppliers, while at the same time ensuring that new products are introduced continuously to replace the less popular brands within different market segments.

Systembolaget's purchase procedures are as follows (Konkurrensverket, 2003b): *i*) when preparing the selection strategy for the following year, an introduction plan for purchases of new products is drawn up, which is then sent to all suppliers; *ii*) each month, tender inquiries are sent to all companies holding a right to wholesale trade of alcoholic beverages in Sweden; *iii*) the offers submitted by the companies are registered and investigated by Systembolaget and samples are requested for selected products; *iv*) the samples are blind-tested by Systembolaget and ranked; *v*) the products with the highest ranking in each category are purchased.

In general, no discrimination against foreign suppliers seems to be taking place in the selection of products offered. However, a number of concerns have been raised regarding entry of suppliers/products to Systembolaget's selection of products:

- Products that – based on Systembolaget's evaluation of the product – have not been admitted to the basic or supplementary range of Systembolaget's ordinary list of prod-

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ucts can be signed up to an evaluation by a consumer test panel, which then might recommend that the product is included on the ordinary list. However, there is currently a long queue of products waiting to be tested amounting to several years. Since wholesalers often have to place an order with a producer within a few months and are required to supply a certain quantity of the product, if and when, the product is admitted, they face a risk and significant costs when trying to introduce the product through this channel. While the queue is partly due to some companies signing up products but failing to deliver samples for the test, the capacity of the test panel is also an explaining factor. The CA has received complaints that a number of well-established wholesale traders/importers have blocked the testing of products from new companies by signing up numerous products for testing. Systembolaget has introduced a number of restrictions on suppliers to reduce abuse of the system.

- The ordering list – from which consumers can order products that have not qualified to Systembolaget's ordinary list of products – is not considered an attractive way for entry to the market for wholesale traders. By signing up a product to the order list, the wholesaler undertakes to deliver as little as a single bottle at the request of a customer of a Systembolaget outlet, and the distribution cost then often exceeds the wholesaler's margin. Also, since the delivery time could be up to a week, consumers will often tend to choose other products that are available straight away from Systembolaget's stocks.
- In general, suppliers indicate that Systembolaget's supply contracts place a large part of the delivery costs on them. For instance, a subsidiary company of Systembolaget, Lagena, that was set up to ensure distribution access for smaller wholesale traders and producers is charging basic fees for its services at a level that is regarded to be prohibitive by smaller companies.
- A wholesale trader/producer that has offered a product to Systembolaget but has been rejected can file a complaint at the Alcohol Selection Council (ASC). However, the ASC will not deal with the complaint until the product has been tried out for the test list. Given the waiting period in the test list system, this process is not a viable way of getting a product introduced to consumers, and wholesale traders/producers have, thus, to a large extent ignored this option.
- Smaller wholesale traders/producers find it difficult to meet Systembolaget's thresholds for minimum quantities to be supplied.
- There have been reports of purchase interventions by wholesale traders/producers in support of their own products to have them transferred from the order list to Systembolaget's ordinary selection or displayed in a larger number of outlets. Systembolaget claims to have developed statistical systems to reveal such behaviour.

Overall, these problems point to some discrimination against minor wholesale traders and producers. That might – at least partly – be reflected in the supply structure. Today, Systembolaget has around 200 suppliers, but the concentration among them is very high. The five biggest suppliers alone accounted for 50 per cent of Systembolaget's wine sales in 2002, the largest supplier being the government-owned V&S Group with 23 per cent (Konkurrensverket, 2003b). The supply of strong beer is dominated by Carlsberg Sverige AB with 39 per cent and Spendrups with 21 per cent, whereas V&S Group dominates the supply of other beverages with a supply share of 54 per cent.

In total, V&S Group account for a quarter of the beverages supplied to Systembolaget. The company is owned entirely by the Swedish state and was, prior to the accession to the EU, a monopolist in import, export, production and wholesale trade in Sweden and thus sole supplier to Systembolaget. In 2000, it was the world's eighth largest alcohol beverage com-

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pany in terms of sales volumes, producing, for instance, the brand Absolut vodka and owning a vineyard in France. Since 1995, when the Swedish state gave up the domestic monopoly, the company has expanded worldwide, buying up several companies, especially in Denmark and Finland. At the same time, the company's production plants in Sweden have undergone some rationalisation (V&S Group, 2003).

Swedes aged 15 and above consume on average 9.9 litres of pure alcohol per year, of which 1.1 litres is provided through smuggling and home brewing (Leifman and Gustafsson, 2003). That is only 12 per cent below Danish consumption, where there is a free market for alcoholic beverages. Around 25 per cent of the total consumption in Sweden is provided through legal and illegal private imports, mostly from Germany and Denmark.

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Annex 2.4

Public procurement*

In Sweden the EU Council directives on public procurement have been implemented by means of the Public Procurement Act (Lagen om offentlig upphandling, LOU). Contracting entities must procure goods, works and services in a businesslike, competitive and non-discriminatory (objective) way. LOU consists of seven sections: *i*) general provisions; *ii*) supply contracts over the threshold values; *iii*) works contracts over the threshold values; *iv*) contracts within the areas of water supply, energy, transport and telecommunications over the threshold values; *v*) service contracts over the threshold values; *vi*) procurement under the threshold values and "B-services" regardless of value, etc.; and *vii*) review and damages, etc.

Contracting entities are government agencies, local government agencies and other agencies, county councils and certain publicly-owned companies, foundations, societies and associations. In addition, certain private companies that conduct operations with special permission from the authorities can be contracting entities in the utilities sectors. The above-described organisations must decide whether they are covered by LOU. However, if a procurement is appealed, the courts will ultimately determine whether the organisation is within the jurisdiction of LOU.

The contracting entities shall calculate the value of each procurement in order to determine whether the procurement is over or under the so-called threshold values. The threshold values for supplies and services are around SEK 1.8 million (in the utilities sectors there are other higher values and for central government a lower value, around SEK 1.3 million) and for works just under SEK 46 million (the threshold values in Swedish currency are determined in a regulation and apply for a period of two years). The value of the contract is normally calculated for the entire duration of the contract, and options and prolongation clauses must also be included as if they were utilised. Extension of a contract can only take place if the contract includes a prolongation clause, and a new procurement must take place when the contract period has ended, after the invocation of a possible prolongation clause. Procurements may not be divided into smaller parts with the aim of avoiding values above the threshold. Procurements under the threshold values must follow the same basic principles as procurements over these values, but the tender procedure is not regulated in as much detail.

Various sorts of procurement processes exist, depending on whether the procurement contract is of a value above or below the thresholds. Three procedures for procurements either above or below the threshold values are stipulated in the Act:

Above the thresholds:

- Open procedure: All suppliers may submit tenders. After advertising the call for tender, the supplier requests the contract documents and these are dispatched as requests are received. Negotiations with suppliers are not allowed.

* The following is largely taken from *www.sweden.se* and NOU (2003).

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- Restricted procedure: The contracting entity publishes an advertisement inviting suppliers to submit tenders. The contract notice must specify the conditions to be fulfilled by suppliers, and these have to verify their ability to meet these requirements by submitting adequate certificates and apply to participate in the tendering (pre-qualification phase). Among the qualified suppliers the contracting entity selects between five and twenty (the exact number is to be stated in the contract notice), to whom the contract documents are sent simultaneously. Only tenders submitted by invited suppliers will be evaluated and negotiations with suppliers are not allowed.
- Negotiation procedure: The contracting entity invites certain suppliers to submit tenders and then negotiates with one or several of them. There is a pre-qualification phase similar to that in the restricted procedure, after which suppliers are invited to submit tenders or participate in negotiations. This procedure may only be used in a few situations stated in the Act, except in the utility sector.

Below the thresholds:

- Simplified procedure: This is somewhat similar to the open procedure, however, negotiations may take place with one or more suppliers, and under certain circumstances the contracting entity may send a written request for a tender to a supplier without publishing a contract notice.
- Selective procedure: All suppliers have the right to apply to tender, and the contracting entity invites some of the applicants to submit tenders. There are no rules on how the qualification of suppliers should be conducted, although the fundamental EC principles must be followed, but the invitation to apply to tender should be made in a publicly accessible electronic database.
- Direct procurement: Procurement without formal requirements may be used if the value of the procurement is very low or if there are other particular reasons. If possible, a comparison of prices should be made. Contracting entities with the intention of making direct procurements should stipulate ceiling amounts for these and indicate when direct procurement may take place.

The Public Procurement Act includes specific remedies that can lead to rectification of a procurement procedure, and it also regulates the award of damages to persons harmed by an infringement. During an ongoing procedure (until the conclusion of a contract), a supplier who believes he is harmed or risks being harmed may appeal to a County Administrative Court. Contracting entities are required to leave a period of at least ten days between awarding a contract – and announcing this to all applicants and tenderers – and final signing of the contract. The County Administrative Court may order recommencement of the award procedure or rule that it may not be concluded until the infringement has been remedied. The court can also make an interim decision pending a final decision. An appeal of the court's ruling can be lodged at the Administrative Court of Appeal. After the conclusion of an award procedure, a supplier can claim damages for alleged infringement against the contracting entity at a District Court. Appeal of the District Court's ruling can be filed at a Court of Appeal.

NOU is an independent public authority under the Ministry of Finance, charged with governing observance of the Act. It consists of a secretariat and a board. The secretariat is responsible for day-to-day operations and for contacts with contracting entities, other organisations and individuals. The tasks of NOU includes: *i*) to supervise observation of the Public Procurement Act (LOU) and the Government Procurement Agreement (GPA) under the WTO; *ii*) to work for efficiency in public procurement; *iii*) to spread information by telephone, newsletters, publications, seminars, etc.; *iv*) to give general comments on how the procurement regulations shall be interpreted; *v*) to follow developments in the area of procurement in the European Union and the WTO. The NOU reviews only cases of principle or general interest and is unable to execute legal sanctions against a contracting entity.

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3. Harnessing resources more effectively

Raising living standards depends on harnessing resources effectively and productively (see Chapter 1). This means not only that at any point in time inputs are allocated to their optimal uses, but also that through time, innovation and technical progress allow resources to be combined together to produce more output and ultimately a higher rate of growth. Furthermore, to realise the benefits of product market competition, resources need to be able to flow rapidly and smoothly from one use to another.

This Chapter looks at the policies that affect the development and utilisation of available resources. It starts by examining certain policies and practices that could impact on the mobilisation and the mobility of labour, before considering the impact of tax policies on savings and investment. The Chapter then considers the state of entrepreneurship, which is a key ingredient in spurring the rate of technical progress, and finally it discusses measures that might encourage innovation. Conclusions and policy recommendations are presented in the last section.

Labour supply and use

Boosting employment

As indicated in Chapter 1, the scope for mobilising additional labour resources in Sweden is more limited than in many countries, because the employment rate is already high. Nevertheless, the government is making efforts on a number of fronts, as outlined in Sweden's Action Plan for Employment 2003. Three particular groups stand out where there is more potential for adjusting policies to promote higher employment: addressing the high and rising numbers of Swedes on sick leave; working towards better integration of low-skilled immigrants; and getting young people qualified and into the workforce at a younger age.

Reducing the high numbers of Swedes absent from work because of sick leave or outside the workforce on sickness compensation (which replaces disability pensions) is a key priority for the government. In 2003, sick leave alone kept the full-time equivalent of 270 000 persons away from the work place for the whole year; while another 407 000 people were receiving sickness compensation –

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altogether, around 13 per cent of the working age population. The authorities announced a range of measures in 2003 to tighten the controls on access to income support as well as strengthen the incentives on employers to create a healthier working environment and encourage employers to find ways to promote rapid return to the workplace through rehabilitation efforts and greater flexibility (see Chapter 4). These measures go in the right direction, although it is too early to see their impact clearly. However, their success will depend to a considerable extent on whether the administrative measures signalled by the government are actually enacted at each and every local social insurance office. This would be easier to achieve if the National Social Insurance Board were given more authority over the local administrations. Benefit replacement rates have also been trimmed back, although with promises that this move is a temporary measure. If the measures taken thus far do not yield sufficient improvements, the policy choice would come down to either tougher controls on access and/or continuation of benefits or larger reductions in replacement rates.

Another challenge is to find policies that succeed in improving the situation for working-age immigrants. Foreigners in general participate less in the labour market than do native Swedes, and foreign men have a lower employment rate than their counterparts in all other OECD countries (Figure 3.1). Raising their participation will be particularly important in effectively responding to demographic pressures (see Chapter 1). Immigrants who are not currently in employment are more likely to be unskilled and may also have weaker Swedish language skills, but the Swedish labour market provides a diminishing number of unskilled jobs, for several reasons. F*irst*, relatively high negotiated wages at the bottom end of the pay scale have effectively eliminated low-productivity jobs. S*econd*, relatively high net replacement rates, especially for low income earners, generate meagre rewards for moving from income support into work (Table 3.1).T*hird*, the high tax wedges faced by Swedish households inhibit the development of jobs providing low-skilled household and personal services (see previous S*urvey*). All these factors together reduce the labour market opportunities available to low-skilled immigrants.

The employment rate has risen faster for immigrants than for native Swedes since 1997 but even so, the gap remains wide. In large part, this is because Sweden's immigration flows have primarily involved refugees (and associated family reunification), and overall, immigrants are a heterogeneous group. Experience from Denmark, which faces similar difficulties with economic integration of immigrants, suggests that there are no "quick fixes", and closing this gap requires a multi-pronged approach (OECD, 2003a). For well-educated immigrants, the main obstacles may be weak language skills and difficulties associated with obtaining recognition of foreign qualifications: some of these immigrants may only be able to find a job that is below their educational level. Low-skilled immigrants face an additional hurdle in countries such as Sweden and Denmark that have

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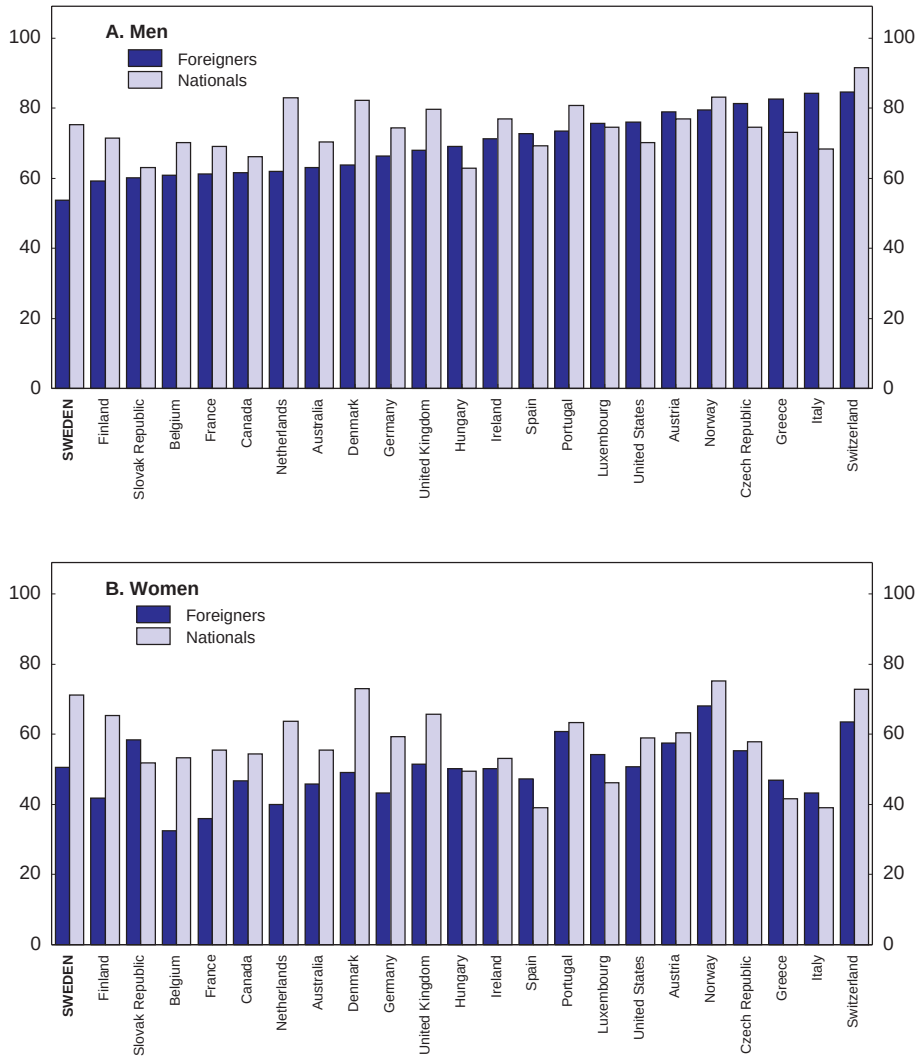
Figure 3.1. **Employment gaps for immigrants**

Employment rates of working-age nationals and foreigners[1] by gender
1999-2000 averages[2]



1. Nationals and foreigners refer to persons born in the host country and born abroad, respectively, for Australia, Canada, Hungary and the United States.
2. August 2000 for Australia; 1996 for Canada and March 2000 for the United States.

Source: OECD, *Employment Outlook* 2003.

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Table 3.1. **Net replacement rates**

Average over 60 months of unemployment for four family types[1]

	Single	Married couple	Couple 2 children	Lone parent 2 children	Overall average
Australia	39	34	69	53	49
Austria	57	59	75	72	66
Belgium	57	71	75	77	70
Canada	35	51	56	74	54
Czech Republic	46	71	88	83	72
Denmark	76	76	84	87	81
Finland	61	66	72	77	69
France	47	45	58	59	52
Germany	60	57	69	68	63
Greece	16	16	17	19	17
Hungary	36	37	47	48	42
Iceland	65	58	73	76	68
Ireland	36	52	62	72	55
Italy[2]	4	8	22	18	13
Japan	45	60	77	73	64
Korea	12	18	26	23	20
Netherlands	73	80	82	70	76
New Zealand	48	66	78	72	66
Norway	58	65	76	77	69
Poland	41	59	74	65	60
Portugal	57	62	65	65	62
Slovak Republic	49	75	87	79	73
Spain	44	46	55	53	50
Sweden	**69**	**84**	**95**	**70**	**79**
Switzerland	74	84	89	86	83
United Kingdom	56	67	81	74	69
United States	13	19	53	44	32
Average	**48**	**56**	**67**	**65**	**59**

1. Average of 100 per cent APW and ⅔ APW earnings levels.
2. Social assistance (*Reddito Minimo di Inserimento*) is not included in Net Replacement Rates due to its experimental character (on trial in 39 municipalities). Long-term NRRs are based on family benefits.

Source: OECD (2002a).

high benefit replacement rates and wage compression because there may be relatively few low-skilled jobs in the economy that match their productivity level. Helping these immigrants make a successful transition into work would require a combination of more effective programmes to build language and job-relevant skills, stronger activation policies and possibly also time-limited wage subsidies. All of these would have budgetary costs, but these need to be weighed against the alternative of higher long-term dependence on income support. Direct or indirect (so-called "statistical") discrimination may also be a contributing factor, although this is very difficult to ascertain. In any case, the Integration Board established in 1998 is charged *inter alia* with promoting equal rights, obligations and

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opportunities regardless of ethnic background and actively combating discrimination towards foreigners.

Speeding up the transition through education for young people could also boost employment. The typical Swedish 15 year-old can expect to spend another 7½ years in education before the age of 30, compared with the OECD average of less than 6½ years. To some extent, this probably reflects the later start for compulsory school, as well as relatively low hours of instruction per year (see previous *Survey*). However, it is also the result of a drawn-out approach to tertiary study. This is encouraged by study grants that are provided for up to six years regardless of the courses studied, and the absence of tuition fees. The provision of income support to students could usefully be tightened to encourage a more expeditious completion of qualifications, without needing to dismantle the overall framework for public funding of tertiary education. For example, student grants could be provided only for the minimum length of time required to complete the initially chosen study course.

Another feature of the Swedish transition to higher education is the waiting time and bottlenecks in gaining a place in a tertiary institution. This particularly affects 20-24 year-olds: in 2001, 7 per cent of this group were unemployed,[1] and another 5 per cent were neither in the labour force nor in education, although these results are still better than the corresponding OECD average of 18 per cent. The government has recognised these problems and is currently investigating ways to modify the entrance regulations for higher education. Going further, one way of handling the current tendency towards excess demand for places in some institutions and excess supply in others might be to adjust the funding arrangements for institutions. The present quasi-voucher arrangements do not allow funds to shift easily from the least popular institutions to those where students most want to study. Providing a "study voucher" instead directly to every student who matriculates, and then allowing the students to exercise them wherever they choose would be a more radical change, but might result in a simpler approach to getting an optimal allocation of education resources in the longer run. Together with tighter constraints on student income support, it should also facilitate transition through education and almost certainly deliver well-qualified young people to the labour force at a younger age.

Raising hours worked

Efforts to boost labour supply through a higher employment rate could be undermined by a reduction in working hours, and pressures to pare back average annual working hours appear to be mounting. As incomes rise, the value of leisure also increases, and an appropriate balance between work and family life is an important social objective in Sweden. But the attractiveness of taking productivity gains in the form of lower working weekly hours or additional days of paid leave

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instead of in higher wages is also heavily influenced by the tax wedge in Sweden. Much of this "leisure" time is actually spent doing home maintenance (DIY) and other non-market services, which in a lower taxed economy might be purchased instead from others through the market. Indeed a person earning the annual average wage would have to work 2½ additional hours to be able to pay a tradesman who earns the average wage for an hour's work, and more than 4 hours to pay for an hour of a professional who earns 1⅔ times the average wage (see previous *Survey*). This could especially discourage the second income earner from working longer hours or shifting from part-time to full-time. Furthermore, since a tradesman is generally more productive than a non-professional doing the job, this distortion not only reduces working hours, it can also trap households into less efficient non-market production.

The scope for addressing these tax effects on working hours is limited by the need to maintain high taxes to finance the welfare state (Annex 3.1). While any reduction in marginal earned income tax rates could be expected to generate positive labour supply effects, those paying central government (*i.e.* state) income taxes face the highest marginal tax rates and these could be reduced for relatively little fiscal cost. For example, reducing the higher state tax rate from 25 to 20 per cent would cost a maximum of around SEK 3.7 billion in lost government revenue, assuming the unlikely worst case of no supply response at all. Other options would include raising the thresholds for paying state taxes or reducing the rates for both brackets. Making such reductions would not only encourage high earners to maintain or even increase working hours, but also provide stronger net financial rewards for investing in human capital. It may also persuade older workers earning around the threshold for state taxes to stay longer in the workforce, by more closely aligning the before-tax and after-tax gaps between earned income and pension income. Such measures could be "paid for" by adjustments to other taxes, for example, by eliminating the lower rates of VAT, as proposed by the government's Tax Base Commission (SOU, 2002) or by increasing property taxes, which are currently lower than the OECD average.

Mobility

Utilising economic resources most productively involves ensuring that they can move easily to the situation where they can generate the highest output. For labour resources, this means being able to easily identify which jobs are the most productive and to move to them. This does take place through job shifts within the firm. But the Swedish labour market demonstrates low inter-firm job mobility. Although 5 to 10 per cent of workers change employer each year, the typical worker spends 10½ years in the same job and tenure rises more sharply with age than in other countries where data is available (Figure 3.2). This reflects a complex set of factors that discourage older workers, in particular, from changing

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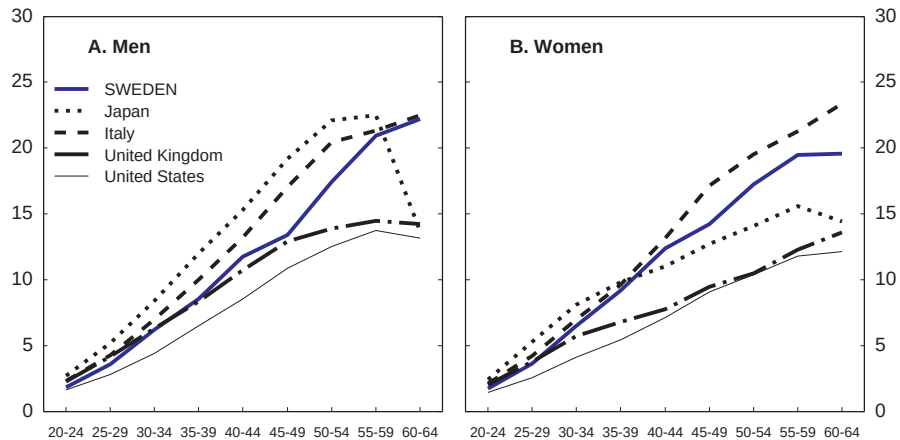
Figure 3.2. **Average job tenure by age**
2000[1]



1. Based on labour force surveys covering all workers, except for Japan.
Source: OECD (2003b).

jobs (OECD, 2003b). Sweden has a relatively strict set of employment protection rules that favour not only those who have longer job tenure, but also provide double points for years worked over the age of 45 (Figure 3.3). Employers are largely unwilling to hire older workers: some 70 per cent of employers surveyed would never or rarely hire a person older than 50 years (National Social Insurance Board, 2001). Against this backdrop, it is unsurprising that older workers cite job insecurity as the most important explanation why they are reluctant to switch jobs (National Institute for Working Life, 2001).

A key factor motivating a job move would be the expectation of earning higher wages, making the evolution of relative wages one of the key signals for labour demand at the enterprise level. Wage bargaining has gradually become more decentralised and now comprises seven different agreement models (Table 3.2). This has resulted in some increase in wage dispersion, but within-sector and between-sector variation is still much smaller than it was in the 1960s and overall it currently remains rather low by international standards (Hibbs and Locking, 2000). Thus relative wages are probably unable to fully convey information about the labour productivity of different jobs in the economy. This not only makes it harder for workers to spot better employment opportunities, but it also under-rewards human capital. Indeed the wage premium for someone with tertiary education is quite modest (Blondal *et al.*, 2002). This makes it more difficult for

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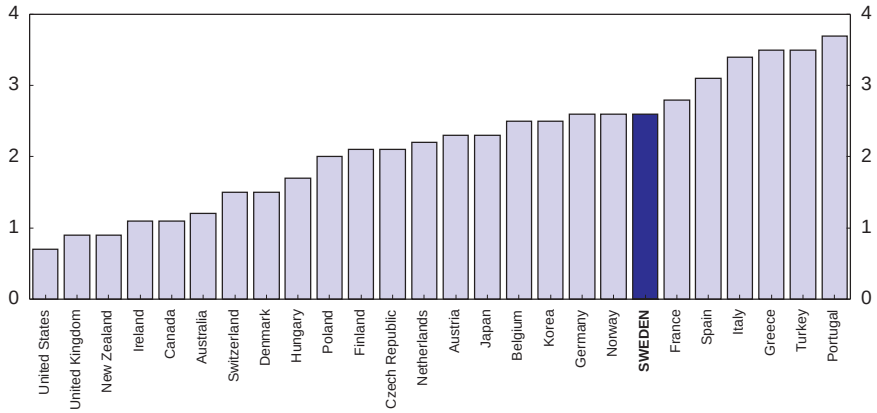
Figure 3.3. **Strictness of employment protection**

Late 1990s



1. This synthetic indicator is compiled from the detailed regulations applying in each country to regular employment, temporary employment and collective dismissals.

Source: OECD (1999).

Table 3.2. **Agreement models for wage bargaining**

Per cent

	Private	State	Municipal/ County Council
1. Local wage formation without nationally determined margin	7	32	28
2. Local wage formation with a cut-off regulating the size of the margin[1]	5		
3. Local wage formation with a cut-off regulating the size of the margin, plus some form of individual guarantee[1]	8	68	
4. Local wage pool without an individual guarantee	7		24
5. Local wage pool with an individual guarantee or alternatively a cut-off regulating the individual guarantee[1]	45		48
6. General pay increase and local wage pool	18		
7. General pay increase	10		

1. If two sides fail to reach agreement, these cut-off provisions specify both the amount by which pay at the company/ workplace is to be raised and a minimum pay increase for each individual. In some parts of the private sector and in the case of most agreements in the public sector, the cut-off mechanism involves a wage board/arbitration board headed by an impartial chair chosen by the social partners deciding how the wage margin is to be distributed.

Source: National Mediation Office (2003).

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young people to identify what skills are worth investing in. Low market returns for acquiring additional skills and undergoing training also weaken the incentive for workers to undertake employment-oriented lifelong learning, while the government's ambition is to increase lifelong learning.

Geographical mobility is another element of dynamic adjustment. The unemployment benefits rules have been modified to require job seekers to search more widely after the first 100 days. This should help encourage the unemployed to move, as long as it is effectively enforced at the local employment office. But relocation towards the large urban areas is made more difficult by the shortage of rental housing in these areas, which has resulted from long-standing rent controls (see Chapter 2). New arrivals in an expanding urban area may find themselves at the end of the queue and/or be expected to pay "black market" rents. A better functioning rental housing market would therefore also make it easier for people to move to a new region for work.

Improving the mobility and optimal allocation of capital

Physical capital is a fundamental element determining the potential output of a country, and fixed investment is the process by which it gets allocated to its most productive uses. The appropriate market mechanisms and structures operate to allocate capital, but in Sweden, as in all countries, the process is distorted by the tax system, which drives a wedge between the economic returns of an investment project and its after-tax yield. Estimates of the average corporate marginal effective tax rate (METR) indicate that fixed investment is to some extent discouraged overall,[2] but by less than in most European countries (Table 3.3).

However, variation in the METRs of different types of investments may lead companies to invest more heavily at the margin in certain types of capital assets rather than others. In Sweden's case, investment in intangibles and in machinery are taxed more favourably than buildings and considerably more so than financial assets and inventories. On some estimations, machinery even benefits from a negative METR (Thakur *et al.*, 2003). A neutral tax treatment of different types of investment would help to ensure that businesses were choosing the type of investment project purely according to the economic returns it would generate.

Another distortion to capital allocation and investment decisions comes from the tax treatment of different forms of financing. At the corporate tax level, debt-financed investments receive preferential tax treatment relative to retained earnings and new equity. This bias against equity may make it harder for growing firms to raise capital by reinforcing the position of large companies who have easier access to cheaper debt capital, rather than encouraging greater competition for available funds on a more level playing field. The position of these big corporations is further bolstered by the so-called "tax-exempt institutions",[3] which are in fact the largest owners of Swedish businesses, and who have the most favourable

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Table 3.3. **METRs for domestic investment**

Corporation taxes only, 1999

	Overall average	Intangibles	Industrial buildings	Machinery	Financial assets	Inventories	Retained earnings	New equity	Debt
Austria	20.9	15.2	18.0	15.2	31.5	20.6	33.3	33.3	–25.0
Belgium	22.4	3.8	28.6	5.7	37.5	25.4	37.5	37.5	–42.9
Denmark	21.9	–19.0	38.3	7.4	29.6	29.6	33.3	33.3	–13.6
Finland	19.9	18.0	18.0	10.7	26.5	26.5	30.5	30.5	–11.1
France	33.2	3.8	41.2	40.5	37.5	32.4	44.4	44.4	–8.7
Germany	31.0	7.4	30.6	13.8	50.0	36.7	48.4	35.5	–56.2
Greece	18.2	26.5	2.0	18.0	2.0	32.4	34.2	34.2	–47.1
Ireland	11.7	5.7	26.5	3.8	9.1	9.1	15.2	15.2	3.8
Italy	–4.1	–72.4	–8.7	–31.6	35.1	0.0	10.0	10.0	–38.9
Luxembourg	20.7	3.8	26.5	5.7	35.1	23.1	35.1	35.1	–35.1
Netherlands	22.6	2.0	27.5	15.2	32.4	27.5	35.1	35.1	–21.9
Portugal	22.5	25.4	19.3	3.8	35.1	23.1	36.7	36.7	–28.2
Spain	22.8	23.1	25.4	7.4	32.4	21.9	35.1	35.1	–21.9
Sweden	**14.3**	**0.0**	**16.7**	**0.0**	**24.2**	**24.2**	**25.4**	**25.4**	**–39.5**
United Kingdom	24.7	9.1	39.0	10.7	27.5	27.5	35.1	35.1	–25.0

Source: European Commission (2001).

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Table 3.4. **Effective marginal tax rates for various combinations of ownership and financing, 2001**

Per cent

	Debt financing	New equity issues	Re-investment of returns
Households, including wealth tax	29.7	61.0	44.1
Households, excluding wealth tax	24.7	51.0	34.1
Tax-exempt institutions	–1.4	23.6	23.6
Insurance companies	19.6	47.2	44.7

Note: Wealth tax on unquoted shares was abolished in 1992. The calculations are made under the assumption of a 10 per cent real return before tax using actual inflation in 2001. A 100 per cent tax implies that a 10 per cent real return before tax is 0 per cent after tax. A negative tax implies that the real return after tax is higher than before tax. The average investment period is assumed to be 10 years.

Source: Braunerhjelm *et al.* (2002).

tax position of all once taxes on owners are also taken into account (Table 3.4). These institutions have a more favourable position than other domestic owners, and in effect, they receive the same treatment as foreign owners, disregarding effects of double tax treaties, etc.

Generally speaking, the double taxation of income from equity also favours debt finance. As with corporation tax, there is an asymmetry in the tax treatment of returns on equity and debt, since returns on the former are taxed but interest expenditures are tax deductible. Taken together with a neutral taxation of interest income, dividends and capital gains within the taxation of personal capital income, this causes an overall additional tax burden on equity. Furthermore, shares generating capital gains tend to be encouraged over those paying mainly dividends, given the ability to defer taxes on gains until the shares are sold. As in other systems that provide for deferral of tax on capital gains, a "lock-in" effect is created, which can result in capital being less mobile than would be optimal. A critical issue is whether the asymmetry on the corporate tax side could be balanced by the introduction of a "reverse" asymmetry on the personal tax side, by applying a lower tax rate on dividend income and capital gains than on interest income. Here, the openness of any individual country's capital markets or, more precisely, the importance of foreign investors is a crucial factor.

With the internationalisation of Swedish capital markets, the negative effect of additional taxation of distributed profits on resource allocation is attenuated, although the extent to which this has happened in practice depends on whether the marginal investors are Swedish or foreign. Furthermore, since the mid-1990s, Sweden has provided some relief from double taxation to individual shareholders of unlisted companies, which typically make very limited use of international capital markets and might otherwise be more heavily penalised.[4] In

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any case, in a small open economy such as Sweden's, the presence of foreign investors may mean that further measures to mitigate the higher taxation on profits than on interest income for Swedish residents could be ineffective in reducing the bias towards corporate debt financing (Joumard, 2001). At the same time it would involve non-negligible tax revenue losses.

A separate but related issue on the contribution of capital to Sweden's future living standards is the high taxation on income and wealth. This discourages households from saving in the first place, depriving Swedish households of future income streams from capital assets. Of course, the best investment returns may not be within Sweden, and the country's integration with international capital markets facilitates an optimal portfolio mix. However, capital income, gift and inheritance taxes, and the still relatively low threshold for wealth tax, in particular, have almost certainly encouraged Swedes to hold a disproportionate share of their portfolios offshore to avoid paying taxes: the foregone tax revenue is estimated to be at least SEK 8 billion per year (SOU, 2002).

Entrepreneurial activity

Entrepreneurial activity is broadly accepted as another critical element contributing to multi-factor productivity growth and is also a potentially important source of employment growth. Entrepreneurship involves seeking out and identifying potentially profitable economic opportunities and taking the risk of trying them out (OECD, 1998). Furthermore, shifts towards outsourcing, the development of new services, and demand for more differentiated products all point to a greater role for small and new firms as engines of entrepreneurial change (Henrekson, 2002). The Swedish government has endorsed entrepreneurship as an important element of its Action Plan for Employment (*i.e.* the Lisbon Strategy). However, relatively few people in Sweden have ever considered the possibility of starting their own business, and a smaller proportion than elsewhere has experience of running a business (Figure 3.4).

Although Sweden has a low rate of start-ups and shut-downs, a very high proportion survives at least two years (Figure 3.5). But these businesses remain small, with around 1½ employees on average after two years, while start-ups in other European countries experience greater employment expansion. The contrast with the United States is even more striking; although turnover and survival rates are broadly similar to those in Europe, firms that survive in the United States grow significantly larger even early on (Bartelsman *et al.*, 2003). Sweden has not produced a significant number of "gazelles", *i.e.* very fast growing firms, either.[5]

The overarching lesson from the OECD's broad policy guidelines for promoting entrepreneurship (see Annex 3.2) is the importance of an overall institutional and economic framework that allows such activity to flourish: without this, specific programmes to boost entrepreneurship are likely to generate weak

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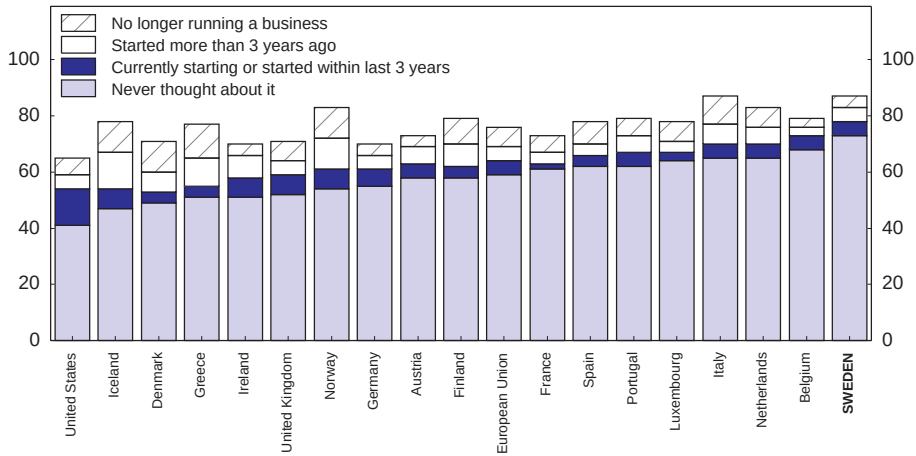
Figure 3.4. **Starting or running a business**
Percentage of answers, 2001



Source: Gallup Europe 2002.

results. In Sweden's case, several particular features of current arrangements may combine to inhibit entrepreneurship and could be addressed. F*irst*, vibrant product market competition (see Chapter 2) is clearly a prerequisite to expanding entrepreneurial activity, without which there would be neither the incentive nor the opportunity to take risks or set up new activities. Opening up a greater proportion of public services to competition could play a particular role that should not be underestimated, especially in sub-sectors where local knowledge can give start-ups an advantage over large established firms. International experience indicates a snowball effect: the presence of entrepreneurs encourages others to try it too, as well as providing a pool of experience and business expertise by acting as "business angels".

S*econd*, the types of capital best suited for entrepreneurial activity may be lacking in Sweden. Most entrepreneurs actually start out by drawing on their own savings, and/or the resources of family and friends. But the Swedish system discourages the accumulation of wealth outside of institutional savings such as pension plans. For obvious reasons, these assets cannot be drawn against and are thus "locked in". As noted above, debt financing has a lower effective tax rate, but it is generally unsuitable for start-ups because the entrepreneur often lacks collateral to counter-balance the higher risk of failure, and the lender faces significant information asymmetries. It also offers no opportunity for the lender to participate

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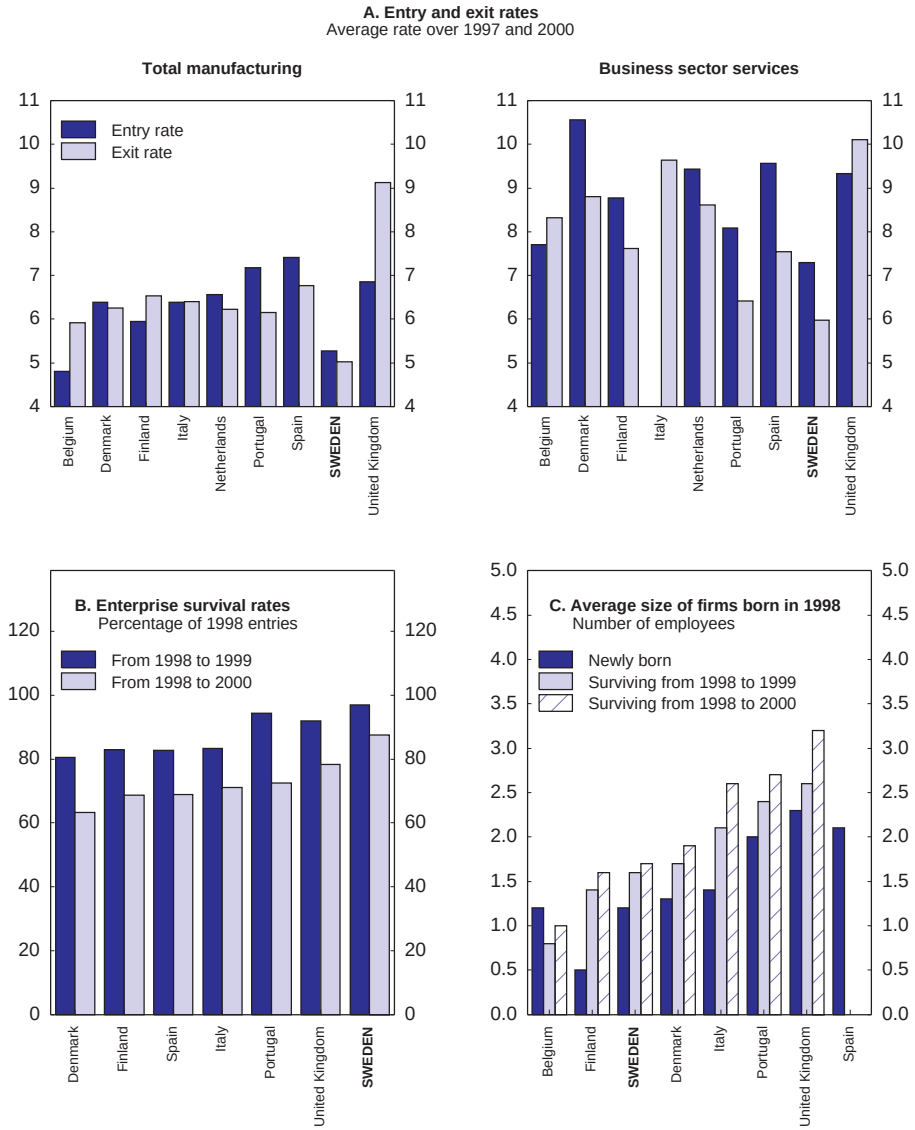
Figure 3.5. **Entry, exit and survival of firms**



1. Data on persons employed for Denmark, Finland and the Netherlands are expressed in full-time equivalents.
Source: Eurostat; OECD.

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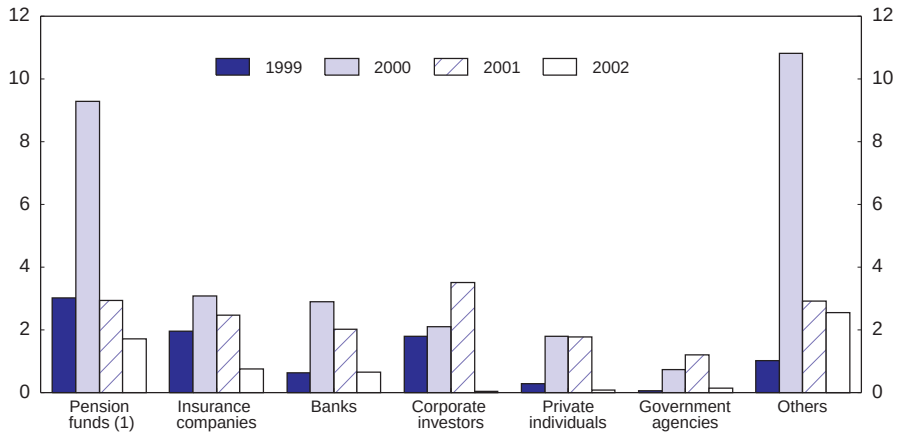
the "upside" of any investment, making it unsuitable for high risk but potentially high return projects.

Growing firms could expand by seeking private equity participation from "business angels" or venture capital firms. These usually inject important management expertise as well as funding. But the pool of potential "angels" (*i.e.* experienced entrepreneurs willing to invest their personal funds in a growing venture) is constrained by Sweden's relatively small number of successful entrepreneurs. And compared with the United States, the supply of venture capital in Sweden is also limited because pension and other institutional investment funds are generally still subject to quantitative restrictions on private equity (Baygan, 2003) although this is now under review. In contrast, the Sixth National Pension Fund was established in 1996 expressly to provide risk capital for small and medium-sized enterprises and is a major source of venture capital finance (Figure 3.6). However, although Sweden has the highest proportion of venture capital as a per cent of GDP, this is because of the significant proportion applied to buyouts and other activities[6] (Figure 3.7). Nevertheless, early stage and expansion activities are still higher than elsewhere in Europe, apart from Iceland though clearly behind North America.

Third, the tax rules governing closely-held companies are especially unfavourable to fast-growing firms set up by entrepreneurs. Companies are

Figure 3.6. **Swedish private equity**
SEK billion, 1999-2002



1. Including public funds.
Source: OECD; EVCA, 2003.

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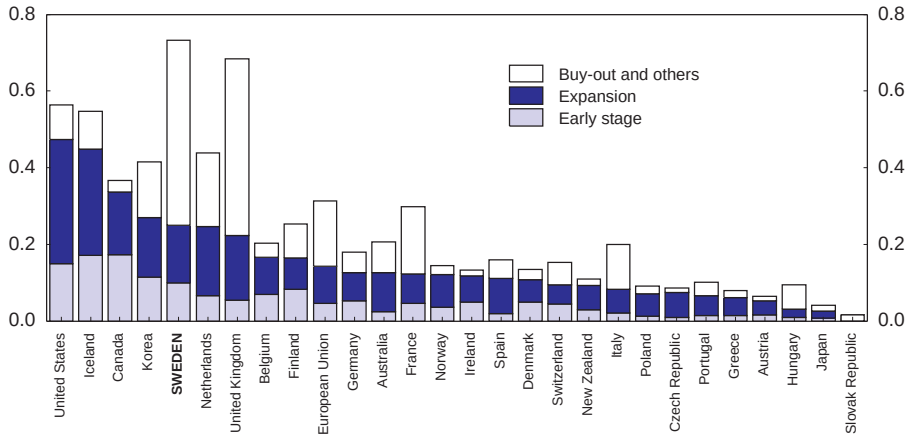
Figure 3.7. **Venture capital investment in OECD countries**
Percentage of GDP, 1998-2001



Note: The definition of private equity/venture capital tends to vary by country.
Source: Baygan (2003).

considered closely held if four or less natural persons hold at least half of the voting rights. Around 450 000 persons reported business income under this category in 2001 and of those, some 170 000 persons could be regarded as "genuine" businessmen (in terms of relative importance of income from the business). Income-splitting rules (the so-called 3:12-rules) are needed in any dual income tax system, in order to discourage reclassification of earned income to take advantage of lower capital tax rates. Dividend income and capital gains are split between capital income and earned income according to a fairly rigid proportion, basically determined by one single assumed *ex ante* average rate of return, including a presumed risk-premium, which is applied on the sum of the acquisition cost of shares and wages paid to employees. Thus, in practice, owners of highly profitable companies are assumed to earn higher wages than those of barely successful companies and are taxed accordingly. For those owners, the tax rules undermine the reward for taking risks although risks associated with having employees are partially accounted for by including wages paid in the base for the calculation of capital income. Parliament recently decided to increase the risk-premium applied by two percentage points from 2004, and further, as yet unspecified, proposals will be introduced in 2005. Together, those measures are intended to reduce the annual tax burden on owners of small enterprises by around SEK 1 billion. However, even after those measures, the limited rewards for risk-taking will still be an issue. The wealth tax may further raise the effective tax rate faced by individuals willing to

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take business risks although the large exemptions in the base for the wealth tax limits that risk.[7] Under certain conditions, business angels and venture capital firms may also be negatively affected and their efforts discouraged, although Parliament has just amended the rules for capital gains and dividend income from shares held by corporations "for business purposes", which would serve to make venture capital investments more attractive.

Fourth, a further employment-related factor concerns the risk of shifting from secure employment to self-employment. Swedish law does require employers to allow their workers to take up to six months leave to establish a new business under certain conditions. But overall, employment protection legislation gives increasing protection from redundancy with length of service. In contrast, starting an enterprise is a risky business and, even if they manage to survive, some ventures may generate very little net income in the initial years. Returning to dependent employment after being self-employed may leave the employee as one of the most vulnerable to subsequent retrenchment in any new position. The potential losses are exacerbated by the high earnings-related component of the unemployment insurance, which means that after an unsuccessful start-up attempt, the person may qualify for little more than the basic allowance.[8]

A general factor that can make people more willing to try setting up a risky business is the ease of exit. For successful businesses, this may involve selling the company either to another enterprise (through a take-over or merger operation) or floating the company on the stock market through an initial public offering (IPO). Only a very small proportion of venture capital investments lead to IPOs: Sweden's second-tier stock market saw 33 IPOs over the four years to 2002, compared with almost 1 000 in the United States over the same period. This may partly reflect relatively under-developed second-tier markets in the Nordic area (Baygan, 2003). For unsuccessful ventures, exit has become easier. Bankruptcy rules in Sweden have been modified and now the bankrupt person is excluded from further business activity only if such exclusion is considered to be in the public interest and if severe negligence has been established. In general, remaining debts can be discharged, although directors are liable if they deliberately or negligently caused damage to the company. Nevertheless, attitudes also matter. Swedes surveyed are more likely than other Europeans to believe that people should not start a business if there is a risk of failure. And while most consider that someone who fails in a business venture should be given a second chance, they would be less likely to buy goods from such a person, or invest in any of his or her subsequent business attempts (Figure 3.8).

The government is developing or has already put in place a number of measures to encourage entrepreneurship. These include the following:

- Business regulation is being simplified to reduce the administrative burden, especially for small enterprises, and agency services will be

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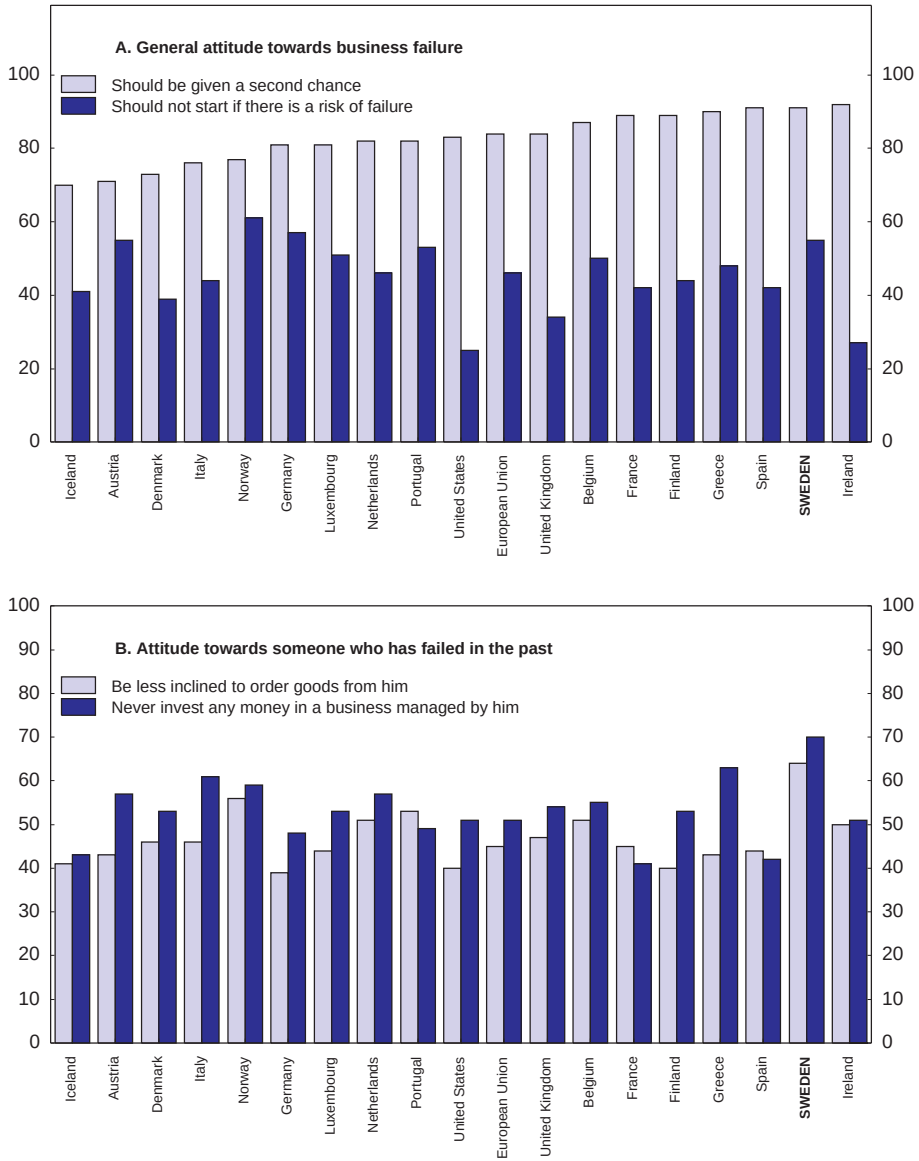
Figure 3.8. **Attitudes towards business failure**

Percentage of answers, 2001



Source: Gallup Europe (2002).

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improved. New businesses can be established more quickly through electronic registration.

- Some SEK 1 billion has been provided in grants to people either unemployed or employed in regional-development support areas who wish to start their own businesses. Special programmes are being developed to target entrepreneurship by immigrants and women, with SEK 42 million allocated for 2002-04.
- A general micro-loan scheme was introduced at the beginning of 2003, and other forms of government sponsored financing such as providing seed capital for projects assessed to offer high growth potential are being considered.
- Courses in entrepreneurship or business skills for small business owners are being offered, while researchers are being encouraged to commercialise their work (see below).

These measures may indeed help to boost entrepreneurial activity but, as the policy guidelines stress, they will generate better results if they are embedded within a framework that actively encourages such activity as a natural response to economic incentives throughout the economy.

Innovation

Innovation is perhaps the key element determining the growth rate of multi-factor productivity, that is, the amount of output that can be produced by all inputs combined (see Chapter 1). However, innovation in itself is a rather difficult thing to detect, much less measure, especially as it is the outcome of complex interactions and policy settings. It also involves the development of such key capacities as vision and strategy, knowledge management, creativity and ideas management, and it is crucially affected by the culture and climate within the enterprise (OECD, 2002b). Despite the difficulties that these factors imply for pinpointing exactly which government policies could make a significant difference, most OECD countries endeavour to find specific strategies to stimulate innovation.

Sweden stands out in international comparisons as spending a greater proportion of GDP than any other country on research and development (R&D), and it also has a high share of researchers in total employment (Figure 3.9). However, some 70 per cent of all private-sector R&D in the first half of the 1990s was instigated by just four multinationals (Braunerhjelm, 1998). The country also has one of the highest proportions of patent applications per head of population in the EU, and the highest when adjusted for "patent families".[9] But Sweden's share of medium and high technology exports in manufacturing is lower than both the

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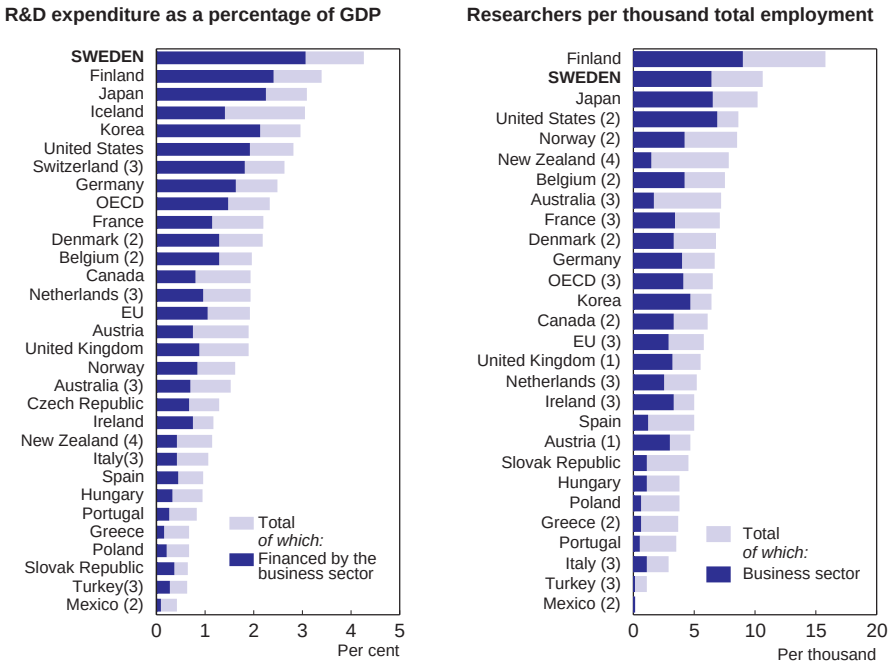
Figure 3.9. **Resources used in R&D**
2001 or latest available year



1. 1998.
2. 1999.
3. 2000.
4. 2002.

Source: OECD, STI Scoreboard.

EU and OECD averages (OECD, 2003c), and does not seem to be correlated with the country's strong investment in R&D.

One logical explanation would be that increasing internationalisation of production could result in Sweden specialising in R&D while manufacturing takes place in lower-cost countries. To the extent that the firms undertaking the R&D were domestically owned, the profits would eventually flow back to Swedish shareholders and be recorded as income earned offshore.[10] However, around 35 per cent of manufacturing R&D in Sweden is carried out by foreign affiliates, slightly higher than their share in production (Figure 3.10). This may present a dilemma for policy: innovation that is both foreign-owned and exploited offshore

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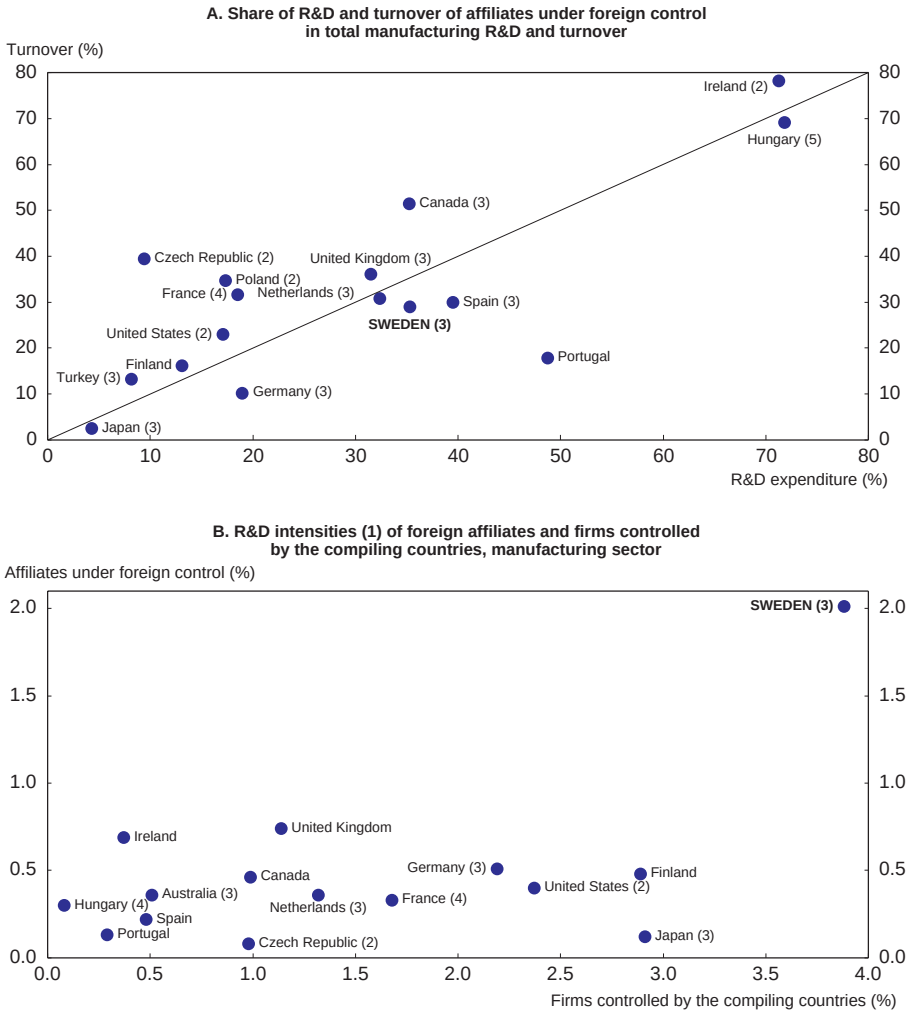
Figure 3.10. **Internationalisation of manufacturing R&D**
2001 or latest year available



1. R&D expenditure as a share of value added in industry.
2. 2000.
3. 1999.
4. 1998.
5. 1997.

Source: OECD, AFA database, May 2003.

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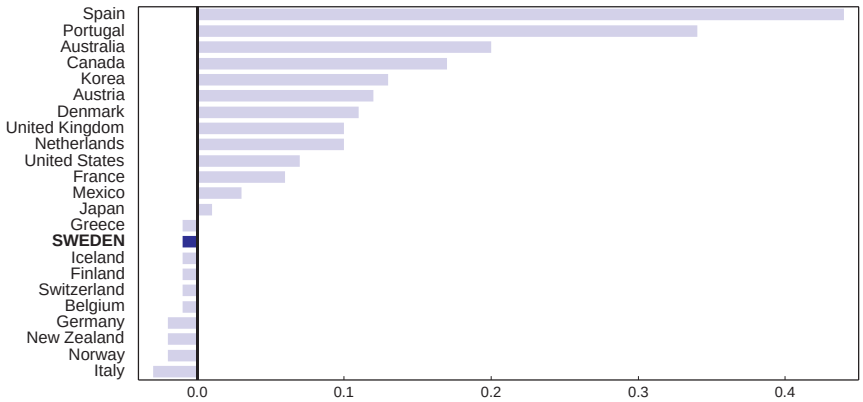
does provide employment in Sweden and generates wage income, while any contribution it would make to raising productivity is realised elsewhere rather than in Sweden, except to the extent that it might provide some spill-over benefits to the rest of the economy. Such effects are hard to evaluate. At the same time, such innovation would not increase Swedish income except in the case where Swedish residents own shares in these foreign affiliates.

This suggests a cautious approach towards policies to promote a faster pace of innovation: if such efforts leaked offshore through greater foreign-owned R&D that was also commercialised overseas, the benefits for Sweden would be rather slim. For the moment, Sweden's public policies towards R&D are more neutral than those of many countries, most notably because it does not offer tax subsidies to R&D (Figure 3.11). On the spending side, most of the expenditures and programmes are geared towards more successfully exploiting the returns from university-oriented research:

- Industrial research institutes have been established as a private-public sector collaboration, with a strong focus on the business sector and international impact, while also maintaining close co-operation with universities.[11]

Figure 3.11. **Tax subsidies for R&D**

Rate of tax subsidy for 1 US dollar of R&D expenditure[1], large firms, 2001



1. The amount of tax subsidy to R&D is calculated as 1 minus the present value of before-tax income necessary to cover the initial cost of R&D investment and to pay corporate income tax, so that it becomes profitable to perform research activities (Warda, 2002).

Source: OECD (2003c).

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- Funds have been allocated for a national incubator programme within which academics and other researchers could develop high-tech start-ups.
- The legal framework has been modified to allow universities and university colleges to set up holding companies to make it easier to commercialise research findings.

Despite the government's initiatives, Swedish university researchers are still much less likely to commercialise their work than their counterparts in the United States. In large part this reflects the institutional framework governing intellectual property rights. Sweden is one of the few countries where the researcher(s) alone retains sole ownership (Table 3.5), which leaves the universi-

Table 3.5. **Intellectual property in public research organisations**

	Universities			Non-university PROs		
	Institution	Inventor	Government	Institution	Inventor	Government
Australia	✦			✦		
Austria	✦			✦		
Belgium	✦			✦		
Canada[1]	✦	✦		✦		
Denmark	✦			✦		
Finland		✦		✦		
France	✦			✦		
Germany	✦			✦		
Iceland		✦		✦		
Ireland	✦			✦		
Italy		✦			✦	
Japan[2]		✦	✧	✦		
Korea	✦			✦		
Mexico	✦			✦		
Netherlands	✦			✦		
Norway	✦			✦		
Poland	✦			✦		
Spain	✦			✦		
Sweden		✦		✦		
Switzerland	✦	✧		✦		
United Kingdom	✦			✦		
United States[3]	✦	✧	✧	✦	✧	

Note: ✦ = Legal basis or most common practice; ✧ = allowed by law/rule but less common.

1. In Canada, ownership of IP at universities generated through institutional funds varies, but IP generated through Crown procurement contracts devolves to the institution performing the contract research.
2. In Japan, the president of a national university or inter-university institution decides upon the right to ownership of inventions made by a staff member of the university/institution on the basis of discussions by the university invention committee.
3. In the United States, universities have the first right to elect title to inventions resulting from federally funded research. The government (*e.g.* federal agency) may claim title if the performer does not. In certain cases, the inventor may retain rights with the agreement of the university/federal partner and the government.

Source: OECD (2003d).

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ties little incentive to actively develop this work. The researcher is likely to find it difficult to obtain leave to exploit the research through independent channels, making it more costly if the venture fails, because once tenure is lost, it may be very difficult to re-establish (Goldfarb and Henrekson, 2003). Indeed, experience from other OECD countries suggests that successful exploitation of university-based research works best when both the institution and the individual inventor can share the royalties (OECD, 2003d). Furthermore those universities in the United States that provide the most streamlined processes for this to happen also appear to be the most successful at generating successful spin-offs (OECD, 1998).

Experience in OECD countries indicates that universities and other public research institutions can also be more successful when there is an appropriate "infrastructure" in place for commercialising activities (see Annex 3.3). This can range from "codes of practice" for managing intellectual property to technology transfer offices. A range of institutional models are currently being tried: most countries have only recently developed such offices and are still learning about the costs and benefits of different approaches. This diversity provides Sweden with an opportunity to look around and identify "best practices" and one common feature to emerge already is the importance of keeping the inventor closely involved in subsequent development phases.

In other parts of the public sector, innovation could also be more actively encouraged by a shift in the incentives facing both the institution and the innovator(s) to patent and commercialise new techniques. This could be done through the same combination of providing shared financial rewards through royalties to the individual and the institution and establishing a legal framework and technology transfer process to facilitate this. A concrete example of this type of approach is available in the British National Health Service (see Annex 3.4).

Innovation is often associated with increasing returns to knowledge accumulation through synergies and the diffusion of both new technologies and more efficient research practices. Sweden, like many other countries, is trying to assist this process by building inter-firm networks, and the state agencies, NUTEK and VINNOVA, are both charged *inter alia* with this task. However, although the importance of networks is reasonably clearly established, fewer insights are available into how public programmes can help (OECD, 2001). There are externalities to setting up a network, because the costs tend to fall mostly on the initiator while the benefits are shared by all participants. However, OECD experience indicates that, once established, these asymmetries disappear and networks can take care of themselves, including sharing out the costs and benefits. Indeed, one role of government is to avoid any negative effects on product market competition that might arise from the "lock-in" tendency of established networks.

Nonetheless, the centre for innovation remains the firm, and possibly the most important factor to boost technical progress across the economy as a whole

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is to ensure that the economic conditions overall encourage all businesses to become more innovative. Product market competition is clearly a key driver in encouraging enterprises to become more innovative (see Chapter 2), especially if the regulatory framework facilitates the entry of new products and services to the marketplace. At the same time, a key feature of innovative firms is their ability to combine and apply knowledge from internal and external sources effectively and to manage the transitions that this approach to business growth implies (OECD, 2002b). In some cases this involves firms completely reinventing themselves – Nokia's transformation from an industrial conglomerate to a specialised telecommunications business is a classic example. It could be noted that non-technology based forms of innovation, involving organisational models, managerial practices and working methods may not only be critical for successful technological change but may also boost multi-factor productivity in themselves. However, labour-market practices that inhibit rapid change within the firm or discourage mobility between firms may prevent such innovation from flourishing.

Concluding remarks and priorities for policies

The foregoing analysis points up a number of areas where government policies might be adjusted in a way that would more effectively harness all the resources available to the economy, thereby helping the country achieve more robust growth and higher employment in the future. This would in turn make it easier to attain ever rising living standards and maintain the welfare state, in the face of a rising dependency ratio. Available inputs stretch from the obvious components of labour and capital to include more dynamic elements such as innovation and entrepreneurship. While Sweden has clearly made progress in all of these areas, more could be done in a number of them.

Labour markets

Although Sweden already has a high employment rate overall, there remain pockets where the rate could be increased; this would not only boost the amount of output that could be produced, but also ease pressures on public finances by expanding the tax base and shifting people from income support into work. Specific measures could include the following:

- Strengthening the powers of the national social insurance board *vis-à-vis* local social insurance offices so as to ensure that the government's new administrative procedures are followed through consistently in all parts of the country.
- Carefully examining ways in which employment among low-skilled immigrants could be raised, including identifying effective programmes for building language and other job-relevant skills, examining the scope

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for applying more rigorous activation strategies, and exploring the potential for appropriately-designed time-limited wage subsidies.

- Speeding up the transition through education by restructuring the university entrance system, and by tailoring the length of study grants to the minimum period required to complete a study programme. The merits of a "study voucher", given directly to students as a mechanism for obtaining a better match between supply and demand for places, would warrant further examination.

While the Swedish welfare state approach makes high tax wedges inevitable, the associated economic losses from making additional work unattractive could be attenuated by reducing the state income tax rates. This could be achieved at low fiscal cost. There would be room to cut labour income tax rates more extensively if the average VAT rate was raised by eliminating the preferential rates applying to certain items or if a larger tax burden was placed on real property.

Job mobility would help the economy to function more smoothly and responsively to shifts in demand and/or costs. This could be boosted by the following:

- Relaxing employment protection legislation to get a better balance between job security and mobility and reduce the risks associated with changing employers. Removing the age bias in the rules would help to make hiring of older workers more attractive.
- Allowing greater wage dispersion to emerge as an outcome of wage bargaining arrangements so as to provide clearer signals about demand for workers and skills.
- Addressing the current distortions in the rental housing market, which can make it particularly difficult for new arrivals in urban areas to find affordable accommodation and thereby discourage geographical mobility.

Capital markets

In most respects, the Swedish capital markets function reasonably well. But adjustments to the tax regime would remove key distortions both to the accumulation of capital and to the way it is allocated to different uses. In particular, the following measures would generate better economic results:

- A more neutral taxation of different types of capital investment and different forms of financing would lead not only to a more efficient allocation of capital, but also remove some of the potential brakes on the rapid shift of capital to uses that generate higher returns.
- Abolition or at least reduction of the various taxes that discourage households from accumulating financial wealth would help boost incomes in the longer term, whether funds are held in Swedish assets or offshore.

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Entrepreneurship

Encouraging entrepreneurship to flourish in Sweden would seem to require both a change in certain parameters that currently make entrepreneurship financially unattractive as well as changing attitudes, which the government is already trying to do. However, efforts to boost entrepreneurial activity directly, through education, incubators and direct financing programmes are unlikely to make much difference while elements of the underlying framework tend to penalise those who are most successful. Areas where policies need to be adjusted to encourage greater entrepreneurial activity are the following:

- Opening up the public sector to greater competition, which would increase the opportunities available to potential entrepreneurs.
- Eliminate those elements of the tax system that discourage individuals from accumulating wealth outside of pension plans (and housing). Most notably this would involve eliminating wealth, gift and inheritance taxes, or at least further raising their thresholds.
- Remove the quantitative restrictions on venture capital investments by institutional investors including private pension funds.
- Address the 3:12 rules affecting closely-held companies, so that they provide greater recognition of the entrepreneurial nature of high earnings.
- The risks associated with failure as an entrepreneur seem to be reinforced by the way previous earnings are used to calculate the major share of unemployment insurance benefits. A careful readjustment of the formula that would produce a more neutral treatment for failed entrepreneurs compared with their earlier employment status might encourage more people to consider trying out a business opportunity.

Innovation

Sweden is keen to boost its rate of innovation and has already taken a number of initiatives. Apart from providing the best conditions for product market competition, specific efforts could concentrate on strengthening the incentives on public institutions to exploit opportunities to commercialise research and development produced by their staff. This could be helped by the following:

- Sharing the ownership of intellectual property rights for publicly-funded research between the institution and the researcher, so that both share the rewards from bringing it to market.
- Developing the supporting organisational structure and expertise to effectively commercialise research.

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Notes

1. Two per cent of this age group were unemployed but also in education.
2. An average marginal effective tax rate of zero indicates that tax has no impact on the marginal investment decision.
3. These institutions include not only pension and other institutional investment funds, but also family-endowed foundations.
4. The logical extension of the system that currently applies for shareholders of unlisted companies would be a full imputation system within the Swedish dual income tax system that provides a lower uniform rate of taxation on capital income received by individuals. Imputation that involved equal capital tax treatment for both foreign and domestic investors and for income from both domestic and foreign shares could prove to be both complex and costly. As capital markets have become more international, several EU countries have moved away from imputation systems. In principle, an alternative "schedular" approach could be designed to produce the same tax outcome as imputation for the domestic investor, and would also have to be applied neutrally to foreign investment and investors. However, the treatment of the foreign investor is the same under both approaches, as foreign investors pay only corporate taxes in Sweden.
5. One study found that there were no firms established after 1975 that had more than 300 employees in 1993 (Rickne and Jacobsson, 1999) while another found that among firms that existed in 1987 and still existed in 1996, only 14 had added more than 250 jobs each (Davidsson and Henrekson, 2002).
6. The Sixth National Pension Fund has made capital investments amounting to around SEK 2 billion by buying out well-established companies considered to show "strong growth potential".
7. However, it could be noted that these exemptions are problematic from a horizontal equity point of view.
8. The basic allowance is SEK 270 per day while the earnings-related benefit is calculated as 80 per cent of reference earnings up to a maximum of SEK 680 (73 euros) per day during the first 100 days of benefit receipt. For self-employed, reference earnings are the taxable income for the previous three years.
9. A patent family provides a better basis for international comparisons, because a patent family groups together all the patents filed at home and abroad for a single invention (OECD, 2003c).
10. An unusual feature of the Swedish ownership structure is the significant gap in some companies between voting rights and dividend rights. This can result in the controlling

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shareholders being Swedish although the bulk of dividend payments would still flow to foreign owners (Henrekson and Jacobsson, 2003).

11. The industrial research institutes have now been converted into limited liability companies with the state as minority shareholder, *via* the IRECO holding company. IRECO is 55 per cent state-owned with a net turnover of SEK 85 million in 2002, mostly from grants directly from the state and *via* the Knowledge Foundation, a capital fund established by the Swedish parliament in 1994.

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Annex 3.1

Income and other direct taxes in Sweden

This Annex sets out the key elements of income and other direct taxes in Sweden, using the rules that applied in 2003.

Income taxes

Sweden operates a dual income tax system where income from employment and income from capital are taxed separately.

Income from employment

Income from employment (*i.e.*"earned" income) is subject to both local and national taxation.

- Local taxes (municipal and county) are proportional and the tax rate varied between 28.90 and 33.72 per cent depending on the municipality, with an average rate of 31.17 per cent.
- National (state) taxes are levied at 20 per cent on taxable incomes between SEK 284 300 and SEK 430 000 and at 25 per cent on income above that bracket.

Spouses are taxed separately and taxable income includes salaries and pensions as well as fringe benefits although deductions are possible for certain work-related expenses. A basic allowance is provided that varies between SEK 11 400 and SEK 25 900 depending on income. Tax credits are granted for 25 per cent of union dues and 40 per cent of unemployment insurance fees.

Employees and the self-employed pay a general pension contribution of 7 per cent of earned income up to a maximum of SEK 23 000. A tax credit equal to 75 per cent of these contributions is granted. Deductions can also be claimed for contributions to private pension arrangements, up to a maximum of SEK 19 300 at the average production worker wage.

Employers pay social security contributions of 32.82 per cent on the total sum of salaries and benefits, while self-employed pay 31.01 per cent on net business income.

Income from capital

Net capital income is taxed at a flat rate of 30 per cent. Capital income includes dividends, interest, capital gains on the sale of stocks, bonds, real estate and personal property, etc. and income from the rental of apartments or houses unless undertaken as a business activity. Only two-thirds of the capital gains resulting from the sale of a private residence is taxable and this can be deferred if the seller purchases another residence within the following year. Income from the sale of other private assets are taxable if the profit exceeds SEK 50 000. Interest payments on qualifying loans can be offset against capital income and a tax credit is given if net capital income is negative.

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Income from business

Income from business activities carried out by individuals is taxed as income from employment, while limited liability companies pay a corporate tax rate of 28 per cent. Special rules apply to "closely held" companies, whereby part of the owners' income is taxed as income from employment and the rest is taxed as capital income (see main text).

Property and wealth taxes

All property and wealth taxes are national (*i.e.* state) taxes.

Real estate (property) tax

The state real estate tax is levied at 1.0 per cent of the taxable value of private homes and 0.5 per cent for apartments. The taxable value is established in an assessment undertaken every six years and must correspond to 75 per cent of the property's market value. Simplified assessments are undertaken in the middle of the six year period to update the taxable value.

Commercial properties are taxed at 1.0 per cent and industrial properties at 0.5 per cent. New dwellings are exempt from tax for the first five years and taxed at half the standard rate for the following five years.

Net wealth tax

Net wealth taxes are levied on household net wealth above SEK 2 million for married couples and SEK 1.5 million for single taxpayers. In general, assets are assessed for taxation at their market value. However, real estate is included at the taxable value for real estate tax and listed shares are assessed at 80 per cent of market value, although shares owned by principal shareholders controlling 25 per cent or more of votes are exempt. Shares in most non-listed companies are also exempt as are assets employed in the owners' trade or business.

Inheritance and gift taxes

Inheritance and gift taxes are progressive and close relatives pay less than other beneficiaries. In calculating the taxable inheritance, spouses can deduct the first SEK 280 000 while children and grandchildren can deduct SEK 70 000. Taxable inheritance is taxed at 10 per cent up to SEK 300 000, 20 per cent between SEK 300 000 and SEK 600 000, and 30 per cent above SEK 600 000. Gifts up to SEK 10 000 may be made to each recipient without incurring gift taxes, which are levied at the same rates as inheritance tax.

Tax on pension fund earnings

Private pension capital funds are taxed at 15 per cent on the estimated yield of the fund capital, which is calculated as the fund capital times the official state lending rate. Other tax privileged investment funds pay a rate of 27 per cent on the estimated yield.

Stamp duty

Stamp duty is levied at 1.5 per cent of the value of real property acquired and at 2.0 per cent of the value of mortgages registered.

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Annex 3.2

Broad policy guidelines for fostering entrepreneurship

The following guidelines for fostering entrepreneurship were developed and presented to the OECD Ministerial Council Meeting in Spring 1998 as part of the Jobs Study follow-up (OECD, 1998):

- Examine the overall institutional framework within which economic activity takes place, to establish whether it provides maximum scope for entrepreneurship to flourish. In particular:
  - Identify and dismantle remaining barriers to competition which limit the incentive on enterprises to innovate and perform more dynamically. Promote competition in all sectors of the economy, including the provision of public services. Provide effective protection of intellectual property.
  - Examine whether current regulations governing financial institutions and/or financial markets inhibit or facilitate the availability and optimal allocation of finance for entrepreneurial activities.
  - Allow scope for flexible employment contracts to be negotiated, with remuneration arrangements and working conditions that are well adapted to the needs of dynamic enterprises. Relax employment protection measures that inhibit restructuring or discourage entrepreneurs from taking on new workers.
  - Examine the costs of complying with government imposed administrative or regulatory requirements and identify where reductions could be made, either by removing the requirement to comply or by reducing the administrative burden involved (including through better co-ordination between different government agencies).
  - Examine the overall effects of the tax system on entrepreneurship, and identify any particular features which act to discourage entrepreneurs or financing of entrepreneurial activity. Ensure that the tax system is transparent and compliance is straightforward.
  - Review and simplify the registration procedures required to create a business. Ensure that firms are able to close quickly should they wish to do so.
  - Ensure that personal bankruptcy legislation provides an appropriate balance between encouraging risk-taking and protecting creditors.
  - Re-examine the effect that social insurance provisions may have on encouraging or discouraging would-be entrepreneurs.
- Ensure that specific programmes designed to foster entrepreneurship operate as part of an integrated and coherent strategy that complements the framework conditions. In particular:

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  - Avoid policies that stem from a too-narrow definition of entrepreneurship (*e.g.* that entrepreneurship is only about start-ups or only about high-technology) and which may divert attention from getting the broader economic policy settings right.
  - Widen the target population for entrepreneurship programmes, where possible, to attract the participation of women, the young and minorities.
  - Undertake regular and comprehensive evaluation of programmes, and ensure that evaluation findings are acted on.
- Improve the effectiveness of entrepreneurship programmes by drawing on the knowledge of sub-national levels of government. In particular:
  - Ensure that resources for programmes to foster entrepreneurship are decentralised where appropriate in order to bettor tailor programmes to the specific needs of an area and its businesses.
  - Provide regular opportunities to exchange information at a national level on the experiences of local authorities in designing and implementing entrepreneurship programmes.
- Seek to identify and implement low-cost and effective programmes with minimal distortionary effects on market incentive. For example:
  - Promote public awareness of entrepreneurship and examine the role the education system could play in developing entrepreneurial skills and attitudes.
  - Increase opportunities for the unemployed to create their own jobs through self-employment schemes.
  - Facilitate networking among firms in order to foster a culture of mutual co-operation and risk-taking.
  - Promote the entrepreneurial non-profit sector by contracting-out where possible the delivery of public services which meet pressing demands in economic and social development.

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Annex 3.3

Turning science into business

A recent OECD Study (OECD, 2003d) drew out the following lessons from OECD countries' experience with management of intellectual property rights in public research organisations (PROs).

Lessons on development of strategies for intellectual property rights

- A PRO's institutional background affects its approach to intellectual property management and commercialisation.
- Intellectual asset management involves more than PRO inventions; it needs to address valuation mechanisms and appropriate exploitation mechanisms.
- Collaborative research requires a particular intellectual asset management strategy.
- Creating revenue is not the main motive.
- Intellectual asset management requires a long-term, broad commitment from PRO management; an intellectual asset management programme will require considerable investment and it may be some time before it breaks even.
- Some non-patented intellectual property may be commercially useful.
- A start-up development strategy needs to be rooted in the PRO's intellectual asset management model.
- The need for professionalism in intellectual asset management and the consequences for hiring professionals.
- The type of Technology Transfer Office models adopted by PROs depends largely on the environment in which they operate but involves trade-offs between economies of scale and proximity.
- The case for limited government support for Technology Transfer Offices at PROs.
- Successful transfer of intellectual property takes time but integrating intellectual asset management into the overall knowledge transfer strategy of a PRO can help increase the type and amount of knowledge that is transferred.

Overall policy lessons

- More strategic approaches to creating and exploiting R&D require reconsidering intellectual property policies.
- Intellectual property policy for PROs can be perceived as contributing to knowledge management in innovation systems as a whole but starts from the availability of an excellent science base.

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- Fostering more explicit intellectual property rights practices and policies at PROs does not necessarily mean amending the legal framework; regulations, recommendations and changing the intellectual property culture in general are also important.
- Reform of intellectual property policies at PROs should involve funding agencies, PROs, scientists and industrial partners as well as other actors in the technology transfer value chain.
- Implementing new approaches to intellectual policy and changing behaviour take considerable effort, require capacity building and time.
- Intellectual property policy should provide incentives to scientists as well as institutions.
- Creating professional Technology Transfer Office infrastructures needs long-term investment in terms of budget and policy attention and professional R&D management.
- The entrepreneurial spirit at PROs – including dealing professionally with intellectual property – should not develop at the expense of PROs' basic task of research and teaching for the benefit of society.
- Intellectual property policy making and management would benefit from close monitoring at PROs.
- Pay attention to the link between intellectual property policies and policy making involving innovation.

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Annex 3.4

Promoting innovation in the British National Health Service

In September 2002, the British government put in place a new set of rules to place innovation at the core of the National Health Service (NHS) business, for the interests of patients and society as a whole. Although innovation was already taking place as a natural process within the NHS, it was recognised that the incentive to innovate and the diffusion of such innovations could be promoted by a better assignment of intellectual property rights. It should be noted that this effort is oriented towards stimulating innovation in everyday working practice, rather than focussing on the outputs of research and development programmes.

The new approach makes it possible for the individuals and institutions in the NHS to profit from their innovations through royalty payments. It allows hospital trusts and other health service institutions to take equity positions in companies set up to commercialise NHS innovations, and has set up state owned companies to hold and/or transfer NHS-based intellectual property rights. It also includes an assurance that the Audit Office will take an open-minded and supportive approach to "well-thought out risk-taking in the exploitation of valuable intellectual property".

The first year of operations has already produced around a dozen commercial ventures. The new framework promises not only to stimulate the search for improved technologies and better treatment regimes throughout the health care sector, but also to provide a stream of public earnings. The NHS recognises that many ideas and best practices are still better disseminated through non-commercial channels and considers that the best solution is a mix of both approaches.

Source: *www.innovations.nhs.uk*.

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4. More sustainable public finances

The demographic changes facing Sweden are likely to lead to significant pressure on public finances, stemming from increased spending on pensions, health and elderly care and slower increases in tax bases. As pointed out in Chapter 1, it is crucial to prepare for these pressures by reducing public debt in the current period of favourable demographics, which should last for about 15 years, if public service standards are to be maintained without leading to higher tax rates or an unsustainable debt position in the longer term. Achieving this reduction will require solid general government budget surpluses in the foreseeable future.

Sweden is in a good position to meet these challenges compared to many other countries with similar demographic prospects. This is partly attributable to the fiscal framework that has been in place since 1997 and which now comprises: a surplus target of 2 per cent of GDP for general government finances over the cycle; nominal ceilings for general government expenditure; and a balanced-budget requirement for local governments. In addition, the government has introduced a target of 80 per cent for the employment rate of 20-64 year-olds and another of halving the number of people receiving social assistance. Both of these will help the authorities to meet the fiscal targets.

Nevertheless, recent developments in public finances suggest that underlying spending pressures have been building up following the consolidation efforts in the second half of the 1990s, which could put at risk the political commitment to the required fiscal strengthening. This might suggest that it is time for a more prudent approach to public finances and a reassessment of the fiscal framework.

This Chapter sets out to identify the main weaknesses in the present fiscal framework and the most obvious ways of improving it in order to make public finances sustainable. First, the recent trends in public finances are described, and then the current framework is briefly reviewed. The specific issues related to central government finances and the expenditure ceilings are then investigated, followed by a section looking at the local government situation. The final section provides some concluding remarks and recommendations.

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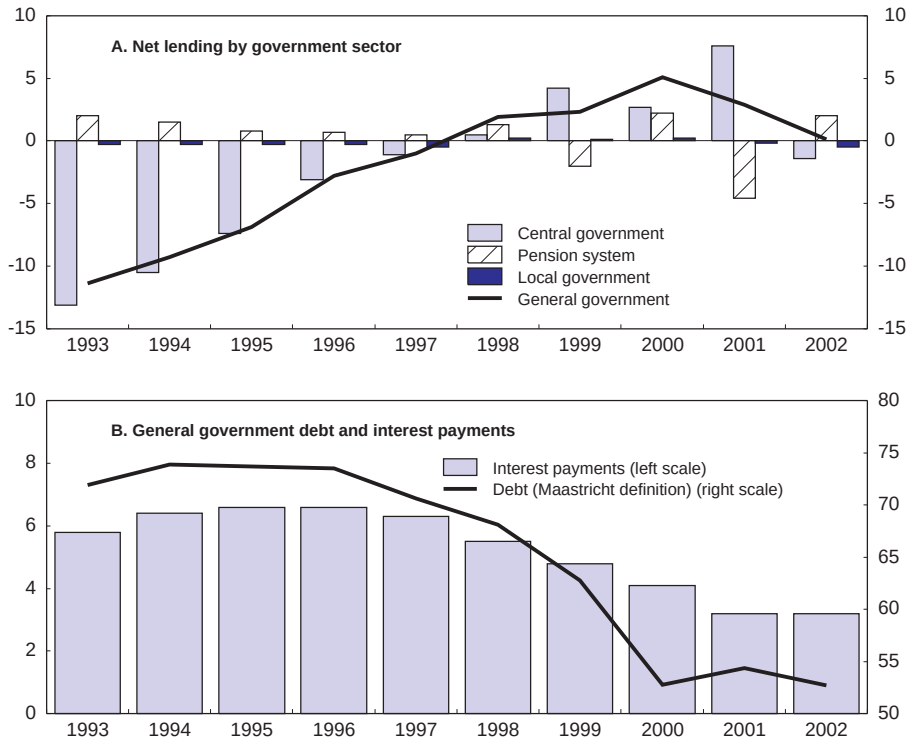
The fiscal position has weakened in recent years

An impressive consolidation of public finances after the crisis in the early 1990s has put Sweden among the minority of OECD countries that are expected to show a general government surplus in 2003, despite the effects of strong automatic stabilisers at work in response to the slowdown in activity. Between 1993 and 1998, central government finances moved from a deficit of 13 per cent of GDP to a surplus of ½ per cent of GDP, while local governments remained more or less in balance and the pension system continued to record surpluses in the range of ½ to 2 per cent of GDP (Figure 4.1). This dramatic turn-around in the central government balance also brought a substantial reduction of public debt and associated interest payments.

Figure 4.1. **Government net lending and debt**

Per cent of GDP



Note: Large transfers to the central government made net lending in the pension system negative in 1999 and 2001.
Source: Statistics Sweden; Ministry of Finance; OECD.

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The measures employed to achieve this consolidation cut to the heart of the welfare state; the public were willing to accept otherwise unthinkable contractions of programmes in order to address the obviously unsustainable public finance position at the time. Efforts included reductions in the generosity of social benefits such as parental insurance and child allowances, cutbacks in public subsidies and diminution of public consumption. These measures allowed central government primary expenditures to be reduced by 10 percentage points of GDP over the consolidation period (Table 4.1). Local government expenditure was also restrained, keeping it in line with a falling tax ratio in the sector and eroding central government grants, while tax reforms helped to raise central government revenue, primarily through base-broadening measures. Public finances improved further in 1999 and 2000 on the back of cyclical improvements.

More recently, there has been a fairly sharp worsening of the fiscal position, mainly as a result of a drop in total tax revenue by 3¼ percentage points of GDP from 2000 to 2002. The international slowdown certainly explains part of the deterioration, especially as revenue from corporate and capital gains taxation

Table 4.1. **Central and local government finances**
Per cent of GDP

	1993	1998	1999	2000	2001	2002
Central government						
Revenue	32.1	35.4	38.5	37.4	39.3	30.8
Taxes and charges	28.2	32.3	33.3	32.3	29.8	28.1
Transfer from the pension system	–	–	2.3	2.1	6.8	–
Other income	3.9	3.2	2.9	2.9	2.7	2.8
Expenditure	45.2	35.0	34.2	34.7	31.7	32.3
Consumption and investment	10.0	9.5	9.5	9.1	9.0	9.4
Transfers to private sector and abroad	23.9	14.9	14.2	13.4	13.1	13.3
Grants to local government	6.2	5.5	5.3	5.0	5.0	5.1
Old age pension charges to pension system	–	–	0.8	0.9	0.9	0.9
Transfer to premium reserve funds	–	–	–	2.6	0.8	0.8
Interest	5.2	5.0	4.4	3.8	2.8	2.8
Net lending	–13.1	0.5	4.2	2.7	7.6	–1.4
Local government						
Revenue	24.6	23.6	23.2	22.4	22.8	23.4
Taxes	15.7	15.5	15.5	15.3	15.9	16.2
Central government grants	6.2	5.5	5.3	5.0	5.0	5.1
Other	2.7	2.5	2.4	2.0	1.9	2.0
Expenditure	24.9	23.4	23.1	22.2	23.0	23.9
Consumption and investment	19.4	19.4	19.3	18.7	19.3	20.1
Other	5.5	4.0	3.9	3.5	3.7	3.8
Net lending	–0.3	0.2	0.1	0.2	–0.2	–0.5

Source: Ministry of Finance.

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plummeted, but a very expansionary fiscal policy stance also contributed. Discretionary fiscal easing amounting to around 3½ per cent of GDP over 2001-02 was partly implemented through supply-oriented reductions of marginal taxes on earned income, intended to compensate for higher social security contributions and to reduce the share of taxpayers paying the state income tax. But it also involved cuts in property taxes and VAT on certain goods, which did not entail obvious structural policy improvements.

In addition, there has been some return to more extensive public sector provisions, including restoration of benefit levels (for instance sickness, unemployment and parental insurance) and new spending programmes to improve health care and education. A cap on certain charges covering child and elderly care has also been introduced. Thus, the public consumption to GDP ratio increased by 1½ percentage points from 2000 to 2002, almost entirely attributable to spending at the local government level, although to some extent this reflects explicit priorities set by Parliament through the granting of activity-specific (tied) state subsidies to counties and municipalities. Higher outlays for sickness benefits and sickness compensation (formerly "disability pension") have pushed up the expenditure ratio further.

The combination of tax cuts and spending increases has put both central and local government in a deficit position, with balances reduced respectively by 4 and ¾ percentage points of GDP from 2000 to 2002. Thus, the recorded general government surplus in 2002 only occurred because of the surplus in the pension system. This is also likely to have been the situation in 2003, when public finances are projected to have remained in a surplus position, primarily because higher local government taxes offset further central government easing.

It should be noted that the pension system is completely separate from the other government sectors, and that the current surplus position corresponds to a build-up of future pension liabilities.[1] The pension system will make central and local government finances by and large resilient to the future pressure on pension expenditure (see Chapter 1). However, while surpluses in the pension system will help in meeting the general government surplus target, they will be needed for future pension payments and therefore will not be available for financing of the future increases in non-pension age-related expenditures or for offsetting lower growth in tax bases. As part of the pension reform, several transfers from the pension system to the central government have taken place in recent years, thus explaining deficits in the pension system in 1999 and 2001 and huge surpluses in central government. A final transfer from the pension system is to be decided before the end of 2004; given its one-off nature, this would be most appropriately used for a reduction of public debt.

Fiscal prudence slipped as revenue boomed

Some fiscal slippage in recent years may have been generated by an overly optimistic assessment of public finances at the political level. Booming

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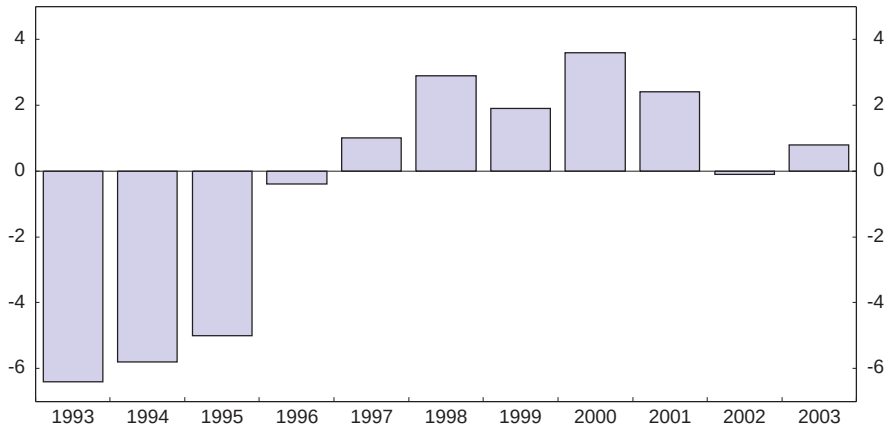
financial markets led to extraordinary increases in revenue from capital gains taxation in 1999 and 2000, which, with hindsight, should have been treated more cautiously. A review of recent budgets and outcomes revealed a continuous upward revision of expenditure in response to ever larger estimates of the surplus for 2000 and 2001, with ratcheting up especially evident for local governments (IMF, 2003).

This overstated picture of budget surpluses, combined with a general perception that previous consolidation measures were only temporary and could be reversed once the economy was back on track, proved damaging to the underlying government surplus position. In fact, during the period 1998-2004, new central government spending programmes and tax reductions have been estimated to be as large as the consolidation measures enacted between 1995 and 1998 (Konjunkturinstitutet, 2003a). This suggests that the improvement of the structural budget balance since 1993 was primarily achieved through a lower level of structural unemployment and reduced indexation of various nominal expenditures. However, recent discretionary policy has brought the OECD's estimate of the structural surplus down to around ¾ per cent of GDP in 2003, somewhat below the government's target of 2 per cent of GDP (Figure 4.2).[2]

Promises to restore and extend various public benefits and services also played a prominent role in the campaign for the general elections to Parliament in 2002 and in the recent referendum on adoption of the euro, and a number of

Figure 4.2. **Cyclically-adjusted general government net lending**

Per cent of potential GDP



Note: Numbers differ from *Economic Outlook 74* because of the new accrual-based national accounts.
Source: OECD.

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welfare improvements were announced in the 121-point plan drawn up by the government and its supporting parties in Parliament after the 2002 election (Box 4.1). While some of the promises have been put on hold due to the weak business cycle position and a more cautious approach after the rejection of the euro, many are likely to be fulfilled once growth has picked up again, thus repeating the pattern from the period 1998-2002. In addition, the planned fourth step of the tax reductions to compensate for higher social security contributions remains to be implemented.

Besides significant spending pressure at the political level, the main challenge to public finances in the medium term is probably to curb the increase in

Box 4.1. Spending reforms and tax expenditure in the 121-point plan

The Social Democrats remained in government after the elections to Parliament in September 2002, with continued support from the Left Party and the Green Party. An agreement signed in October 2002 outlined the basis for co-operation between the parties. While highlighting the parties' commitment to adhere to the expenditure ceilings and to respect the surplus target, as well as to strive for the social targets and pursue a sound and ecologically sustainable economic policy in general, the plan included a number of spending proposals, including:

- Higher benefit levels for parental insurance, pensions, child allowances and housing allowances, and extension of parental leave from 16 to 18 months.
- Increase in public services: hiring an extra 21 000 teachers for pre-school and school; spending programmes in the care sectors amounting to SEK 20 billion over the parliamentary term; more resources to local governments for increased employment, especially in elderly care, and for raising wage levels in public service professions with a high share of female labour.
- Reduction of VAT on construction of rental housing for youth and the elderly.
- Increase international aid to 1 per cent of GNI.

The total costs of the 121-point plan (net of various cost savings also proposed) can be estimated at SEK 60-70 billion (2½ to 2¾ per cent of GDP) in the current Parliamentary term, of which around half is included in the 2003 and 2004 budget bills. A one-month extension of parental leave was already implemented in 2003 along with a rise in the benefit level. Benefit levels have also been raised for those receiving the basic pension, and the upper secondary school study grant has been extended from nine to ten months. Also, municipalities have started to receive grants from a pool of SEK 5 billion set aside for hiring more teachers. In addition, a subsidy for construction of rental housing (designed as a tax expenditure) has been implemented, which, in effect, corresponds to a reduction of the VAT rate from 25 to 6 per cent (see Chapter 2).

Source: OECD estimate based on Sveriges Riksdag (2003).

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the total number of people on sick leave or receiving sickness compensation (disability pension), which rose from 483 000 (in full time equivalents) in 1997 to 677 000 (13 per cent of the working-age population) in 2003. Although the number of people with sickness periods shorter than one year decreased in 2003, this can probably partly be attributed to the weaker business cycle, as employees more prone to taking sick leave might have found themselves among the first to be laid-off, while higher uncertainty about the job situation has led to some reluctance of workers to report sick. However, it remains to be seen whether this recent decline is sustainable as the economy recovers, although the measures implemented in 2003 should contribute to further reductions. In the meantime, net inflows to sickness compensation have continued.

Fiscal rules may have curbed spending pressures

Sweden's fiscal rules have probably prevented what could otherwise have been an even larger deterioration of public finances in recent years. In 1997, the general government surplus target of 2 per cent of GDP over the cycle and nominal expenditure ceilings in the central government were introduced. The ceilings are usually set three years ahead and cover virtually all central government expenditure, including entitlement transfers, local government grants and pension payments (although these are technically outside the central government), but excluding interest payments. In 2000, the framework was complemented with a balanced-budget condition for local governments, which requires that any recorded deficit should be offset by a surplus within two years after the breach.

This framework is undoubtedly more ambitious than in many other countries, and at face value it may be viewed as both transparent and credible. The public should be able to relatively easily monitor observance of the hard rules (nominal ceilings and the balanced-budget requirement), and the nominal ceilings should make for a clearer prioritisation of spending alternatives (see previous S*urvey*). But there are a number of shortcomings that point to possible refinements.

F*irst*, nominal ceilings can be circumvented through accounting practices, for instance moving some payments from one year to the next, although with a multi-year framework this cannot be practiced extensively unless the subsequent ceilings are adjusted as a consequence. This requires only a decision by Parliament, but with the political commitment invested in the ceilings the associated loss of credibility effectively rules out such practices. A potentially more pernicious effect would be the recourse to tax expenditures to avoid the ceiling, although the tax expenditures introduced in the 2002 budget will be converted to normal spending and brought within the ceilings from 2005.

S*econd*, with entitlement transfers included under the ceilings, pro-cyclical effects may result, especially if the budget margins prove insufficient. Conversely,

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in the absence of rules for handling downward expenditure surprises, savings on entitlement transfers can be used for discretionary spending programmes during upturns. Unexpected price changes could have similar effects. This increases the risk of ending up in a pro-cyclical squeeze.

Third, the balance-budget requirement may also foster pro-cyclical behaviour at the local level. Local governments are not subject to any rule on how to deal with expenditure savings or extraordinary revenue during a cyclical upturn. Any additional income can be used freely for spending programmes, but local governments would then need to make expenditure cutbacks or raise taxes in a subsequent prolonged downturn. Given the typical inertia of spending programmes, the reaction would be biased towards higher taxes. A relatively high cyclical sensitivity of the local tax base, which is entirely based on income in contrast to many other countries, may aggravate this problem, although taxing of benefits (which are paid by the central government) has a stabilising effect. Furthermore, no enforcement mechanism is in place to ensure the requirement is actually complied with.

Fourth, while formulating a surplus target over the cycle should avoid pro-cyclical behaviour, observance of the target is not easily monitored, and it could therefore be deemed less credible. For instance, it can be difficult, sometimes even years later, to identify the beginning and end of an economic cycle. While the cyclically-adjusted budget balance may be used as an indicator for any given

Table 4.2. **Central government expenditure**
Billion SEK

	2000	2001	2002	2003[1]
State budget excl. interest expenditure	619	641	658	664
Adjustment to national accounts	64	16	33	77
of which:				
Transfers to the pension system	53	15	17	18
Transfers to local governments	–19	–21	–23	7
Contribution to EU based on value added tax	–13	–15	–10	–5
Transfers financed by EU	–5	–13	–9	–8
VAT paid by central government	20	19	22	23
Capital consumption and imputed pension contributions	19	21	21	23
Interest expenditure and amortisation	–12	–10	–9	–10
Expenditure financed outside the budget	11	12	8	8
Other	11	6	14	20
Primary expenditure in national accounts[2]	684	656	691	742

Note: The change in transfers to local governments in 2003 reflects the central government taking over the payment of local governments VAT with a corresponding reduction in state grants.
1. Ministry of Finance estimate.
2. Based on national accounts before the latest revision in November 2003.
Source: Ministry of Finance.

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year, this too is subject to both measurement problems and to uncertainty about to what extent deviations from 2 per cent of GDP represent non-observance of the target itself.

Fifth, central government nominal expenditure ceilings are stated in budget terms, which differ somewhat from the national accounts, which are the basis for the surplus target. The main discrepancies relate to various state transfers (to the pension system, local governments and the EU) and central government VAT payments (Table 4.2). That blurs the relationship between the ceilings and other targets, and makes it hard to assess whether the ceilings are ambitious enough for meeting the surplus target. In general, the framework could be strengthened by establishing a medium-term reference scenario, as for instance is done in Denmark, that makes it easier to identify the interlinkages between all the targets and highlights the policy action required to reach the targets. That could also make more transparent the link between outlays and the economic costs of raising the revenue needed to fund them.

Expenditure ceilings have come under strain

While the central government expenditure ceilings have been respected in every year since their introduction in 1997, margins have been steadily eroded since 1998 (Table 4.3). Higher discretionary expenditures account for a significant part of the erosion, along with increasing sickness costs. In recent years respecting the ceilings has sometimes been achieved through the application of tax expenditures and special accounting practices. For instance, to avoid breaching them in 2003, the acreage subsidy to farmers and EU contributions amounting to SEK 7¼ billion were brought forward to 2002. There has also been recourse to tax expenditures such as abatement of some social security contributions and general

Table 4.3. **Central government expenditure ceilings**

	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
	Billion SEK									
Expenditure ceiling	722	719	735	744	770	797	827	860	894	931
Budget margin	24.1	2.0	1.5	5.0	4.7	0.4	0.1	0.2	13.8	28.5
	Per cent of GDP									
Expenditure ceiling	38.2	36.5	35.4	33.9	34.0	34.0	34.0	34.1	33.9	33.7
Budget margin	1.3	0.1	0.1	0.2	0.2	0.0	0.0	0.0	0.5	1.0

Note: The ceilings are adjusted for various technical changes, for instance balanced changes in budget accounting. The budget margins for 2003 06 are Ministry of Finance estimates.
Source: Ministry of Finance.

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tax relief for municipalities. Measures have been adopted not only to balance the accounts *ex post*, but also to actively create room for new spending. For instance, as the ceilings have prevented higher grants, the central government has instead redirected tax revenue (the employment subsidy and the so-called "200-kronan" paid by all income-earners to the central government) to local governments in order to provide funds for increased activity in that sector. While the previous S*urvey* recommended such practices be abandoned, these tax expenditures have actually been prolonged for 2004. In 2005 and 2006, however, an equivalent amount is instead supplied *via* higher central government grants to local governments, which are included under the expenditure ceilings.

While the government's measures to battle rapidly rising sickness expenditures (Box 4.2) along with the above-mentioned shifts of expenditures between years and delay of various new expenditures may have been enough to comply with the ceiling in 2003, the margin left for unpleasant surprises was only SEK 0.1 billion this year. A similarly small margin is projected for 2004 in the 2004 budget, which also presented ceilings for 2005 and 2006. Margins in the outer years are significantly higher; however, these projections do not include the 121-point plan, realisation of postponed spending initiatives or reinstatement of sickness benefit levels, while they take into account all the hoped-for effects of the measures to restrain sickness expenditures that were introduced in 2003. Given the experience in recent years, it may not be overly pessimistic to assume that these margins will also be eroded. In that case, the general government balance could be ½-1 per cent of GDP lower than expected, depending on any tax revenue flowing back from the higher spending.

A number of measures could be employed to strengthen the expenditure ceilings and restrain political pressures for new or extended spending programmes. The previous S*urvey* listed a number of possible measures to improve the framework, including the following:

- Introducing a separate ceiling for operating expenses and non-entitlement transfers three years ahead, which would serve as a hard budget constraint, and a separate ceiling for entitlement expenditures that would include a substantial margin for coping with macroeconomic fluctuations. Reallocation of unspent entitlement margins to non-entitlement expenditures would be prohibited.
- Requiring that each new spending proposal be fully financed by concrete and credible proposals for offsetting cuts elsewhere, thereby forcing a clearer prioritisation of the use of public funds.
- Reassessing periodically the priorities within each of the larger spending areas to make reallocations more effective.

It might also be appropriate to look into alternative ways of financing welfare benefits. The government has announced the establishment of a commission of inquiry on social insurance, which will look at ways of financing future expendi-

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Box 4.2. Measures to battle rising sickness expenditures

The increasing number of people on sick leave (in the labour force) or sickness compensation (previously denoted disability pensions) has prompted the government to introduce a number of measures to curb the associated additional budgetary costs. In order to meet the official goals of halving the number of people on sick leave between 2002 and 2008 and reducing the number of people on sickness compensation, the Spring Budget 2003 included measures, which took effect on 1 July 2003, in the following areas (Regeringen, 2003):

- *Strengthening sickness administration*: meetings between the person on sick leave, the doctor, the employment agency and the regional social insurance board are to take place on a more regular basis; more emphasis is to be put on part-time sick leave; sick-leave certificates from doctors are to provide a better assessment of ability to work; fewer people are to be awarded sickness benefits for longer than one year.
- *Adjustment of benefits*: the general replacement ratio for sickness benefits was lowered from 80 to 77.6 per cent, and the benefit levels for unemployed reporting sick were aligned with unemployment benefit levels.
- *Better incentives for employers*: the employers' period for paying sickness benefits was extended by one additional week (employers now pay benefits from the second to the twenty-first day of sickness); protection against sickness was provided for employers with few employees; subsidies for employers who employ persons on long-term sick leave were introduced.

Additional measures were announced in the 2004 budget, including further tightening of administration through more responsibility given to the insured for managing their own situation; more resources devoted to combat fraud in the system; awarding of sickness compensation for no more than three years at a time; and further (non-specified) enhancement of incentives for employers to reduce sick leave (Regeringen, 2003). It is still too early to assess the effect of the new measures. Shifting more of the financing burden to employers, which was recommended in the previous *Survey*, and reducing benefit levels will certainly improve incentives in the system and alleviate the pressure on public finances. However, administration may still need to be strengthened further, as local decisions on awarding sickness benefits and/or sickness compensation are still separated from central financing of the associated costs. In addition, fairly high replacement rates on sickness benefits may still encourage abuse of the system, especially if eligibility criteria are not carefully designed to cover only those who are genuinely sick, and if monitoring and enforcement mechanisms are not tightened. Furthermore, the proposal to award sickness compensation for a limited time period may just result in automatic prolongation, and the insured will have little incentive to take on a less demanding job and earning a correspondingly lower wage after a period of receiving sickness compensation, as benefits will be reduced if (s)he subsequently has to return to the sickness compensation scheme.

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Box 4.2. **Measures to battle rising sickness expenditures** (*cont.*)

While wage subsidies, as proposed by the National Social Insurance Board and the Swedish National Labour Market Administration and as implemented in Denmark, may help to bring people with reduced work ability back into employment, such subsidies need to be carefully targeted and of limited duration in order not to distort the labour market. It would also be better to focus directly on bringing people back to the regular labour market instead of establishing a government company, as proposed by the two organisations, which risks distorting competition in the product market.

It should be noted that in December 2003, the government and its supporting parties in Parliament announced a number of changes of the system to come into force from 1 January 2005. These include for instance restoring the general replacement ratio for sickness benefits to 80 per cent, replacing the third week of employers' payment of sickness benefits with a co-payment amounting to 15 per cent of the benefit costs for the remaining sickness period (beyond the first two weeks), and replacing the current one-day waiting period for sickness benefits with a 20 per cent deduction in sickness benefits (based on weekly salary).

ture on unemployment, sickness and parental benefits, including a possible combination of basic public financing, collective contributions and individual insurance. This kind of financing mixture is currently in place for pensions, and collective agreements on sickness insurance also exist. Nevertheless, further financing reforms should be seriously considered, as they might make it possible to avoid situations where the government would have to cut benefits (temporarily) in order to restore public finances.

Local governments struggle to balance budgets

The strong expansion of the Swedish economy from 1998 to 2000 also led to improvements in local government finances, but the subsequent slowdown pushed them back into deficit. The share of local governments reporting deficits went up from 32 per cent in 2001 to 37 per cent in 2002 (Konjunkturinstitutet, 2003b), and 2003 saw the largest increase in average local government tax rates for 25 years as individual local authorities endeavoured to satisfy the balanced-budget requirement (Figure 4.3). The rapid growth in local governments' income tax base over the last three years has not provided much relief to local finances, as this development to a large extent reflects changes in regulations made by the central government that have been offset *via* reductions in the general state grant to counties and municipalities (Figure 4.4).[3] Despite the tax increases, around two-

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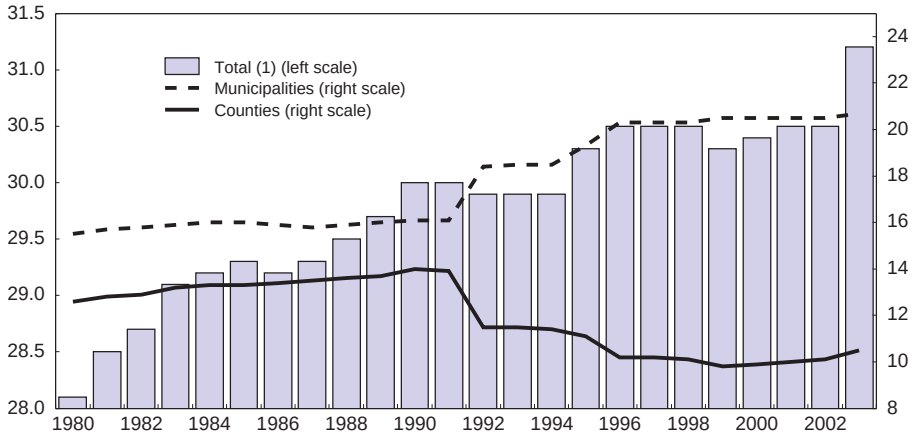
Figure 4.3. **Local government tax rates**

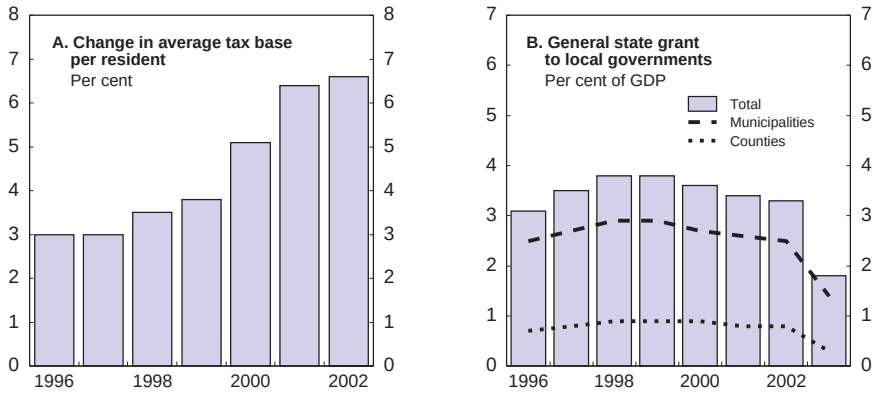
Per cent



1. Excluding church tax.

Source: Statistics Sweden; OECD calculations.

Figure 4.4. **Local government income**



Note: The developments of local governments' tax bases and the general state grant are to some extent affected by changes in regulations made by the central government. For instance, the large reduction of the general state grant in 2003 is basically a result of the central government taking over the payment of local governments' VAT (local governments' charges to the central government are correspondingly lowered).

Source: Statistics Sweden.

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thirds of all municipalities expected to be in deficit in 2003, with further tax hikes by local governments of around 0.3 percentage point on average for 2004 (Svenska Kommunförbundet, 2003).[4] Together with local tax increases in 2000 and 2001, these latest measures will reduce the positive labour market effects from lower central government taxes on earned income.

The scope for local governments to consolidate their finances *via* reallocations of the general state grant has also been reduced as the government has opted for greater use of activity-specific (tied) grants. In 2003, activity-specific grants constituted SEK 41¾ billion out of a total amount of state grants of SEK 85¾ billion (Table 4.4). This has caused tensions between central government priorities and local choices, especially when local governments already devote a relatively large part of resources to the activities in question or in cases where demand for these activities differs markedly between residents in different locations. The government seems to have acknowledged these problems, announcing in the 2004 budget that a number of such grants and tax expenditures will be transferred to the pool for general state grants. Thus, while local politicians might have failed to constrain spending in line with the structural development of revenue during the boom years, a significant part of the expenditure increases at the local level in the last seven years can be attributed to Parliamentary decisions.

Another source of tension within each lower level of government is the comprehensive equalisation scheme, comprising separate systems for counties and municipalities and including both tax revenues and expenditure (see

Table 4.4. **State grants to local governments**
Billion SEK, 2003

	Municipalities	County councils	Total
General state grant	28.3	9.1	37.4
Age related grant	6.7	–	6.7
Activity specific grants	17.3	24.4	41.7
Maximum charges in child care	3.9	–	3.9
School efforts	2.5	–	2.5
Municipal adult education	1.8	–	1.8
Increased accessibility in the care sector	–	1.3	1.3
Medical costs	–	18.6	18.6
Refugees reception	2.5	–	2.5
Other activity specific grants	6.6	4.5	11.1
Total	52.3	33.5	85.8

Note: In addition to the grants, state revenue of SEK 6.4 billion is directed to local governments via "200 kronan" (SEK 1.3 billion) and the employment subsidy (SEK 5.1 billion). These transfers of tax revenue are converted into a higher general state grant from 2005.
Source: Ministry of Finance; SOU (2003); Regeringen (2003).

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Annex 4.1). This system serves the purpose of creating equal conditions for municipalities and counties to supply welfare services; given the decentralised production of these services, some redistribution is necessary. However, the system has been criticised on a number of grounds, including that taking tax revenue from local governments may be in conflict with the constitution, which provides local governments the authority to levy taxes to finance their expenditures. Furthermore, some critics claim that *income equalisation* has gone too far in its attempt to eliminate differences between local governments' tax revenue.[5]

The most important incentive problem arising from the design of *income equalisation* is the resulting bias towards raising tax rates. For a local government levying the average tax rate, any direct gain from increasing its own tax base is, by and large, offset, as 95 per cent of additional revenue will be lost through increased contributions to, or decreased transfers from, the equalisation pool. Furthermore, transfers from the *expenditure equalisation* scheme will be reduced as the shares of unemployed and low-income residents fall (in order for the local government's own tax base to increase). In contrast, if the local government raises its tax rate, total revenue will increase as any consequential decrease in the tax base (from lower labour supply) would be offset by the *income equalisation* scheme. These incentive effects might explain part of the upward drift in local government tax rates since 1999.

While *expenditure equalisation* is only around 85-90 per cent, the scheme is very complex and disregards higher local government wage costs in rapidly growing areas, for instance, Stockholm. Proponents of the current system claim that wage costs should be excluded, because otherwise higher local government wages could be granted with the additional costs offset through equalisation. Opponents focus on the general tendency for factor costs to be driven up in areas that experience significant growth.

A parliamentary committee has recently proposed changes to the system that partly address some of the criticisms levied against it (SOU, 2003). The main proposal is to replace the *income equalisation* scheme and the current general state grant with diversified state grants, which should guarantee every local government a revenue base per inhabitant of 120 per cent of the average tax base per inhabitant. With a guaranteed revenue level of this size, almost all local governments would be net receivers of income-related grants from the government, thereby improving the legitimacy of the system.[6]

However, the proposed changes would only to some extent address the most important problem with the current *income equalisation* scheme, namely, the adverse incentive effects. Furthermore, a guaranteed level of 120 per cent would imply that the state had to provide the municipal sector with more funds than it does today, even if most activity-specific grants were transferred into the general pool of state funds distributed to local governments. Acknowledging that this

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might be unacceptable to the government, the committee proposed a uniform reduction in grants for municipalities, cutting the "guaranteed" revenue to around 116-117 per cent in effective terms. However, because Parliament can raise the size of this uniform reduction at its will, the proposed *income equalisation* system does not protect lower levels of government against real erosion of state grants any more than at present.[7]

The committee proposes the *expenditure equalisation* scheme be continued in its present form, although with the number of sub-models in the municipal part reduced from 16 to 9. A new structural state grant is proposed to provide partial compensation to those local governments that would be made worse off by the changes. This grant is to be financed through uniform reductions in grants to all local governments, thus further reducing the effective "guaranteed" revenue level. Overall, the proposed changes to the equalisation schemes and state grants would imply larger transfers from the Stockholm area to other parts of the country. This has fuelled renewed outcry against the system from politicians in the area, especially as Stockholm County is already in economic difficulty and resorting to large tax increases and spending cuts in order to meet the balanced-budget requirement.

The evolution of local government finances in recent years suggests that several adjustments could lead to better outcomes. F*irst*, more stable revenue bases would avoid fluctuations in revenue leading to a ratcheting up of expenditure in boom periods and higher tax rates in cyclical downturns. Sweden differs from many other countries by having local government taxes entirely levied on personal income, while property taxes are central government revenue. Reassigning the property tax bases to local authorities could provide them with more stable revenue. Another option would be to calculate local governments' tax bases as the average taxable income over a number of years, as was proposed by the STEMU committee as a means of strengthening the role and efficiency of fiscal policy in stabilising short-term fluctuations if Sweden were to join the euro. This proposal is warranted in any event, as was pointed out in the previous S*urvey*.

S*econd*, the central government may want to actively stabilise revenue to local governments by adjusting its grants in a counter-cyclical fashion, as was also recommended by the STEMU committee. That could also help to break the trend for upward creep in spending and tax rates, which reduces incentives to work and thereby undermines the central government's efforts to reach its employment target and ensure sustainable public finances in the future. Alternatively, a rule could be implemented for how to deal with excessive revenue in boom periods, although this might conflict with local governments constitutional rights.[8] Other countries have tried similar ways of stopping trend rises in local government tax rates. Denmark, for example, operates a formal system of negotiations, with

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the central government reaching annual agreements with local governments on aggregate spending and tax levels. Recent agreements on tax levels there have abided by the government's overall tax freeze, but if breached, the central government would offset the aggregate effect by reducing its own tax rates while cutting local government grants correspondingly to keep state finances unaffected (OECD, 2003b).

Third, restructuring of the equalisation schemes to eliminate or substantially reduce the incentives for raising tax rates instead of tax bases could be considered. With the government's renewed focus on growth issues, one way of making the economy more dynamic would be to alleviate the negative effects in faster growing areas. In general, reconsidering the weight given to regional equality objectives at the expense of raising living standards for all parts of the country would probably be even more important in the future, when a steadily smaller part of the population will provide for the rest. Achieving the highest outcome with the available resources will be crucial for maintaining the provisions of the social safety net.

Fourth, exploring the prospects of applying other financing methods than taxes and state grants would be helpful. Local government expenditure is especially sensitive to the ageing of the population, and with wage increases from possible shortages of qualified staff pushing up wages and aggravating expenditure pressure, total spending is likely to grow faster than tax bases. For instance, prepayment of care services through insurance arrangements was suggested in the previous *Survey*, although it was noted that such arrangements would have to include provisions for those unable to contribute adequately for themselves and therefore could be economically indistinguishable from earmarked taxes (given that denying care to anyone would be very unlikely).

Fifth, stricter management of local government resources is necessary in order to prepare for future demographic changes. Until 2015, fewer children and young people should make it possible to reduce spending on education, while the demand for health and elderly care rises only marginally. However, capturing these savings requires cutting spending in the areas with diminishing demand, which may prove difficult. Nevertheless, one county council has recently announced that coverage for a number of hospital treatments will be reduced or terminated altogether, while resources are concentrated on treatment for more serious illnesses.[9] Along the same lines, further measures to increase efficiency in public service delivery could be employed, including extended use of benchmarking, internal competition and contracting out. Although Sweden has already gone quite some way in reforming the operation of education, health and care services, for instance by its system of allocating funding to the pupil's choice of school, the size of the public sector alone makes for substantial potential for reaping efficiency gains through such measures (see previous *Survey*).

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Concluding remarks and priorities for policies

The last 10 years' experience seems to indicate that while the Swedish public is willing to cut deep into the welfare state when public finances are seriously out of balance, Swedes are equally keen on restoring the level of government services once the economy is back in healthier shape. The challenge for the government will be to make it clear to the public that substantial surpluses are needed for a significant period in order to prepare for the future pressure on public finances. Otherwise, heavier burdens will be passed on to future generations, who will have to finance welfare services for a rapidly increasing non-working population. Although it may be difficult to persuade the public that immediate demands should be set aside to better prepare for the future, the importance of doing so should be seen against the risk of having to reduce public service standards and/or cut income transfers to the elderly and infirm at a later point in time.

While Sweden's fiscal framework has undoubtedly helped to curb some of the spending pressure that had accumulated during the consolidation efforts in the middle of the previous decade, it has not managed to prevent a substantial deterioration of public finances in recent years. In reality, the restrictions implied by the framework are only now being seriously tested, with central government expenditure margins estimated to be virtually zero and local governments forced to raise taxes or cut spending in order to counter deficits in the previous two years. To strengthen the fiscal framework and ensure the necessary improvement of public finances, a number of measures could be employed:

- Redesign the system of expenditure ceilings to ensure that budget margins are not used for discretionary spending instead of as a buffer for unexpected macroeconomic fluctuations.
- Establish a comprehensive medium to long-term fiscal framework that integrates all elements of the current fiscal framework as well as the targets for employment rate and social assistance recipients, and which establishes more transparent links between these various objectives. This would include formulating the expenditure ceilings in a way that made them directly comparable to the national accounts-based general government surplus target. That would allow a clear identification of the required effects to come from additional policy measures and provide a benchmark for measuring progress.
- Make sure that implementation of the spending programmes set out in the 121-point plan is fully financed through downsizing of other spending areas, and that the elements in the plan and the financing instruments do not impede the realisation of other government goals, notably the employment target. Along the same line, compensating cost cuts should be found if temporary measures currently in place are extended further.

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- Monitor carefully whether sickness initiatives are effective and are not just leading to a higher inflow to other public income transfer schemes instead. Examine also whether financing structures entail expenditure growth bias, for instance whether the state funding of entitlements for which eligibility is determined by local social insurance and/or unemployment boards gives rise to perverse incentive effects.
- Ensure that the labour supply benefits of state tax cuts are not offset by local governments raising taxes to meet the balanced-budget requirement. Shifting part of local governments' revenue base to property values or making counter-cyclical adjustments for state grants could make for a more stable tax base, which would help to curb expenditure pressure in upturns and reduce the revenue shortfalls in downturns. Furthermore, the equalisation schemes could be redesigned in order to strengthen incentives for local governments to encourage expansion of business and employment, thereby supporting the central government's growth strategy.

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Notes

1. The pay-as-you-go part of the public pension system is financed through the so-called AP funds, which work as a buffer in the pension system. In parallel, pension wealth is build up in the premium pension system (which is fully funded and has no re-distributional element). Contributions to the premium pension system are first placed in the National Debt Office and then redirected to the pension fund chosen by the individual saver. Because the Premium Pension Authority (PPM) formally owns these assets, savings in the premium pension system are attributed to the public sector. During the temporary management by the National Debt Office of the savings in the premium pension system, these savings are included in the central government net lending, thereby reducing central government debt. In the second year after the income year (where savings are recorded), savings for the premium pension are transferred to the PPM, leading to higher recorded saving in the pension system and a correspondingly lower recorded saving in the central government. The phasing-in of the new pension system has involved a transfer of SEK 245 billion (11.2 per cent of GDP) from the AP funds to the central government during 1999-2001 for amortisation of public debt (Regeringen, 2003).
2. Care should be taken when assessing this estimate as recent developments in financial markets and subsequent drops in capital gains taxes might lead to an underestimation of the cyclical component and thus an excessively pessimistic assessment of the government's current position (OECD, 2003a).
3. While local governments in Sweden to a large extent rely on their own taxing powers for financing expenditure, additional funds are provided by the state through grants. A general grant is allotted in equal amounts per inhabitant to all municipalities and counties, supplemented by a smaller grant distributed according to the age-structure of the inhabitants in municipalities (from 2003). Besides functioning as an important financing source, these state grants are used to control the financial resources of the local government sector and for adjustments according to the principle that a transfer of resources is made when the central government decides to change the rules for local government activities (for instance if it assigns new tasks to the local level).
4. While the 2004 State Budget projected balanced budgets in the local government sector in 2004, the associations for municipalities and counties believe that deficits would be the result in this year as well as for the period 2005-07 (Svenska Kommunförbundet, 2003). Differences in the assessment of developments in prices, wages and activity explain only a minor part of the gap.
5. SOU (2003) lists a number of other, often heard, criticisms raised against the current equalisation scheme(s). These include, *i*) that a high general wage level in a local government also implies higher wage costs for the local government administration, but while the *income equalisation* scheme evens out a higher tax base per inhabitant than the

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average, the *expenditure equalisation* scheme does not compensate for higher wage costs, *ii*) while negative marginal effects (arising from the change in contributed/received amounts *via* income equalisation actually exceeding the change in own tax revenue) have generally been dealt with by introducing adjustments to the *income equalisation* scheme, the adjustment method is too complicated, and some municipalities still have to pay more in contributions than they actually gain in revenue from having an above-average tax base per inhabitant, *iii*) the combined effect of the *income equalisation* scheme and age-related state grants provides double compensation to municipalities with large shares of young and elderly inhabitants. In addition to these caveats, it can be argued that differences in tax bases per inhabitant could relate to differences in non-market activities, such as do-it-yourself or black market activities. However, these are by nature difficult to identify, making it practically impossible to take such factors into account.

6. For the few local governments with a tax base per inhabitant exceeding 120 per cent of the average, a contribution should be paid to the state, amounting to the difference multiplied by 0.85 (instead of 0.95 at present) times the county-specific tax rate. By lowering the contribution ratio, the negative marginal effects are reduced and the current adjusted *income equalisation* scheme is made redundant. Nevertheless, the overall change to marginal effects is unclear, as updating of the tax rates used for calculating contributions and subsidies might offset the reduction in marginal effects from lowering the contribution ratio.
7. Technically, an adjustment account (which would be negative for municipalities at the outset) is proposed to keep the effect on state finances neutral. While in principle the "guaranteed" level of revenue per resident is proposed to be fixed at 120 per cent of the average, thereby implying an automatic adjustment of state grants in parallel with tax bases, local governments are not "guaranteed" any level of revenue because Parliament can change the overall level of the adjustment account at any time.
8. From 1991 to 1994, Parliament froze local tax rates at existing levels, and in the period 1997-99 half of any tax increases were clawed back *via* reduced block grants. The measures proved effective as these were virtually the only periods since 1980 with stable or falling average tax rates (exceptions being 1986 and 2002). However, these measures were ultimately ruled unconstitutional and were thus removed.
9. In October 2003, Östergötland County decided on a list of priorities for hospital treatment as a means of saving SEK 300 million in the budget.

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Annex 4.1

Local government equalisation schemes

The current equalisation system was implemented in 1996 and consists of three components: income equalisation, expenditure equalisation and implementation rules. It was designed to be financed within the local government sector in order not to compromise the consolidation of state finances at the time. The system aims to guarantee that all municipalities and counties share the same basis for their activities, regardless of their inhabitants' taxable income and differences in needs and other structural factors, while locally greater ambitions for service provision or lower cost effectiveness would be reflected in a higher tax rate. The system is separated into two similar parts, one for municipalities and one for counties. At each level of lower government, there is an income equalisation scheme and an expenditure equalisation scheme, each of which are self-financing.

The *income equalisation* scheme is designed to ensure every local government has the same revenue base per inhabitant. Local governments with above-average tax bases per inhabitant pay a contribution fee to the state, which comprises the excess amount multiplied by 0.95 of the average local tax rate in 1995 (corrected for any subsequent tax shifts between the county council and municipalities within the county) (Table 4.A1.1). Those with below-average tax bases per inhabitant receive a subsidy from the state amounting to a similar fraction of their shortfall relative to the average tax base. The 0.95 multiplier constitutes the replacement/contribution ratio in the system, and the sum of contributions equals the sum of grants. In 2003, SEK 14 billion were redistributed within the municipal system and SEK 5.5 billion within the county system, a total of 0.8 per cent of GDP; 53 municipalities and 1 county (Stockholm) paid contributions, while the rest (237 municipalities and 20 counties) received transfers.

The *expenditure equalisation* scheme is intended to equalise differences in structural needs and costs, *i.e.* differences that the local governments cannot affect directly. There are 16 submodels within the municipal scheme and 3 within the county scheme. These are based on estimated standard costs per inhabitant, so that local governments with above-average total structural costs receive a subsidy from the government and *vice versa*. The sum of grants by and large equals the sum of contributions. In 2003, the municipal system redistributed SEK 5.2 billion, while SEK 1.6 billion was reallocated within the county system, in total representing 0.3 per cent of GDP. Around half of the municipalities and counties paid contributions, while the other half received transfers.

Implementation rules have served to gradually phase in the system, shielding local governments from substantial changes in subsidies/contributions. The total amount redistributed *via* these rules have been estimated at SEK 1.6 billion each for both municipalities and counties in 2004, with the total of SEK 3.2 billion financed through a reduction of the general state grant to all local governments (SOU, 2003).

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Table 4.A1.1. **An example of income equalisation in 2004**

SEK

	Municipality receiving a transfer	Municipality paying a contribution
	(Övertorneå)	(Håbo)
A. Total tax base according to 2003 taxation (for the income year 2002)	582 976 300	2 766 639 100
B1. Factor for scaling up the tax base to 2003 level	1.050	1.050
B2. Factor for scaling up the tax base to 2004 level	1.035	1.035
C. Adjusted tax base (A x B1 x B2)	633 549 494	3006 645 042
D. Number of residents medio 2003	5 384	18 025
E. Adjusted tax base per resident (C/D)	117 673	166 804
F. Adjusted average tax base per resident (whole country)	149 459	149 459
G. Guaranteed tax base (D × F)	804 687 256	2 693 998 475
H. Basis for income equalisation transfer/contribution (G – C)	171 137 762	–312 646 567
I. Adjusted 1995 average tax rate, per cent	20.13	19.53
J. Income equalisation transfer/contribution per resident (H × I/D/100)	6 399	–3 388
K. Income equalisation transfer/contribution (D × J)	34 450 031	–61 059 875

Note: The average tax rate across municipalities in the whole country was 19.26 per cent in 1995. Subsequent changes in the division of labour between municipalities and county councils within individual counties make the adjusted 1995 average tax rates differ between municipalities in different counties.

Source: Statistics Sweden.

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5. Some aspects of sustainable development

There is a growing concern that long-run sustainable development may be compromised unless measures are taken to achieve balance between economic, environmental and social outcomes both domestically and on a global basis. This Chapter looks at three specific issues of sustainable development that are of particular importance for Sweden: reducing *air pollution* and *water pollution*, and improving *waste management*. In each case, indicators are presented to measure progress and the evolution of potential problems, and an assessment is made of government policies in that area. The section also considers whether institutional arrangements are in place to integrate policy-making across the different elements of sustainable development (Box 5.1).

Air pollution

Main issues

Due to natural conditions, Sweden is particularly susceptible to acidifying air pollutants. In this light, the national authorities have implemented a range of measures to reduce domestic emissions and have supported international initiatives to combat those acidifying air pollutants originating in foreign countries. The main issue for the authorities is to ensure that these measures are cost effective.

Performance

During the 1990s, domestic emissions of sulphur dioxide and nitrogen oxides per unit of GDP fell less rapidly than in the EU area (Table 5.1). Nonetheless, emissions of sulphur dioxide in relation to GDP are considerably lower than the EU average, reflecting developments in the 1980s, such as the expansion of nuclear power (Östblom, 1996). Emissions of nitrogen oxides, by contrast, are slightly above the EU average and substantially higher than the EU 2010 ceiling (Figure 5.1), in large part due to mobile sources that account for three-quarters of domestic emissions, of which non-road mobile sources account for 44 per cent.[1] Despite relatively slow declines in domestic emissions and thanks in large measure to lower transboundary air pollution, the area where acidifying depositions

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Box 5.1. The integration of policies across sustainable development areas*

The main means of policy integration are through the collegiate system of government and the long-established process of widespread consultation in policy formation. A national sustainable development strategy was approved in 2002. The strategy is based on a broad approach and brings together social, economic and environmental priorities. An initial set of indicators was developed to monitor progress towards the main targets and will be reviewed and modified as required at regular intervals (Ministry of the Environment, 2001). A strength of sustainable development decision-making is the attempt to place policies within a coherent framework. The strategic approach may be further enhanced in the future by the Prime Minister's Office taking a leading role in sustainable development policymaking.

While policies and projects require the preparation of environmental impact assessments, cost-benefit analysis is not systematically applied. When such analysis is undertaken, a precautionary approach is adopted, due to the difficulties in estimating all the benefits. In several areas, policymaking is informed by analysis of policy options using computable general equilibrium models. For example, climate change policy options and measures to combat air pollution were evaluated by models of the economy that track emissions and the impact they have on the environment. Policymaking is also assisted by the development of environmentally adjusted national accounts. The approach adopted values the various functions (economic, amenity, etc) of different types of natural capital (forests, wetlands, etc). Such an approach should assist policymakers in prioritising amongst competing environmental and economic concerns.

* The sections in this report dealing with air pollution, water pollution and waste management are inputs into the Organisation's follow up on Sustainable Development as mandated by the Ministerial Council decision in May 2001.

exceeded the capacity of the ecosystem to cope fell by over 60 per cent between the mid 1980s and the late 1990s. While episodes of concentrations of ground-level ozone above health guidelines are rare, mean concentrations appear to have changed little. Finally, concerns about concentrations of particulate matter have been increasing.

Policies

Emission targets for conventional air pollutants are set in international conventions and European Commission (EC) directives, which the Swedish authorities have supported given the importance of transboundary air pollution to

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Table 5.1. **Main indicators: air pollution**

	Change in emissions per unit of GDP, 1990-99[1]			Level of emissions , 1999[2]			Improvement in productive efficiency, 1990-1999	
	Sulphur dioxide	Nitrogen dioxide	VOCs	Sulphur dioxide	Nitrogen dioxide	VOCs	Sulphur dioxide per unit of electricity output	Nitrogen dioxide per vehicle
	Per cent per year			Grams per dollar of GDP				
Australia	3.1	–1.9	–2.9	3.9	5.5	4.1	n.a.	n.a.
Austria	–10.2	–3.5	–6.5	0.2	0.9	1.2	–77.2	–32.5
Belgium	–9.2	–3.6	–4.8	0.8	1.2	1.1	–64.4	–22.6
Canada	–5.3	–2.8	–2.9	3.2	2.6	3.5	n.a.	n.a.
Czech Republic	–19.1	–6.6	–5.7	2.1	3.0	1.9	n.a.	n.a.
Denmark	–14.2	–4.9	–5.2	0.4	1.6	1.0	–79.8	–38.9
Finland	–13.0	–3.8	–4.1	0.7	2.1	1.4	–65.3	–28.7
France	–8.3	–3.8	–5.1	0.5	1.2	1.4	–52.0	–37.8
Germany	–20.1	–7.1	–8.8	0.4	0.9	0.9	–85.2	–43.7
Greece	–1.4	–0.3	–0.2	3.6	2.5	2.6	–4.6	–22.3
Hungary	–6.1	–1.1	–3.8	5.5	2.1	1.4	n.a.	n.a.
Iceland	–1.1	–1.7	–4.9	3.7	3.9	1.4	n.a.	n.a.
Ireland	–8.1	–6.3	–7.9	1.7	1.3	1.0	–35.8	–19.0
Italy	–7.6	–4.3	–4.4	0.8	1.2	1.4	–52.8	–31.3
Japan	–1.7	–1.3	–2.4	0.3	0.5	0.6	n.a.	n.a.
Korea	–8.9	–3.3	–9.7	1.6	1.9	0.2	n.a.	n.a.
Luxembourg	–18.7	–9.1	–7.9	0.2	0.9	0.9	n.a.	n.a.
Mexico	–0.4	–0.9	–9.3	1.6	1.6	1.4	n.a.	n.a.
Netherlands	–10.1	–6.5	–8.8	0.3	1.1	0.7	–50.2	n.a.
New Zealand	–0.4	–0.5	–1.6	0.7	3.0	2.6	n.a.	n.a.
Norway	–9.7	–2.8	–1.8	0.2	2.0	3.0	n.a.	n.a.
Poland	–10.0	–6.6	–4.8	5.1	2.8	2.2	n.a.	n.a.
Portugal	–2.1	–0.9	0.1	2.4	2.3	3.1	–17.7	–18.33
Slovakia	–13.7	–7.4	–5.6	3.3	2.4	2.0	n.a.	n.a.
Spain	–5.8	–2.1	–3.5	2.2	1.7	3.6	–39.4	–14.4
Sweden	**–8.3**	**–4.3**	**–3.9**	**0.3**	**1.3**	**2.1**	**–26.0**	**n.a.**
Switzerland	–6.0	–5.4	–6.3	0.1	0.5	0.9	n.a.	n.a.
Turkey	3.2	1.0	0.0	3.4	2.3	1.6	n.a.	n.a.
United Kingdom	–13.9	–7.9	–6.6	1.0	1.3	1.4	–72.8	–51.0
United States	–5.5	–2.5	–4.7	2.0	2.7	1.9	n.a.	n.a.

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Table 5.1. **Main indicators: air pollution** (*cont.*)

	Change in emissions per unit of GDP, 1990-99[1]			Level of emissions , 1999[2]			Improvement in productive efficiency, 1990-1999	
	Sulphur dioxide	Nitrogen dioxide	VOCs	Sulphur dioxide	Nitrogen dioxide	VOCs	Sulphur dioxide per unit of electricity output	Nitrogen dioxide per vehicle
	Per cent per year			Grams per dollar of GDP				
European Union	–11.1	–5.0	–5.3	0.8	1.2	1.5		
OECD Europe	–10.1	–4.8	–5.0	1.2	1.4	1.5		
OECD	–6.7	–2.9	–4.4	1.5	1.9	1.6		

Note: For the OECD average, data for countries with missing data for either 1990 or 1999, data for the latest year has been substituted. Estimated data for 1999 represents about 5 per cent of the area total. GDP is measured in 1995 prices. Cross-country aggregations use 1995 purchasing power parity exchange rates.

1. Australia: 1995-99 for sulphur dioxide; New Zealand: 1990-98; Mexico: 1994-98 for sulphur dioxide and VOCs and 1990-98 for nitrogen dioxide; Slovakia: 1990-98 for sulphur dioxide and nitrogen dioxide and 1990-97 for VOCs.
2. 1998 for Mexico, New Zealand and sulphur dioxide and nitrogen dioxide in Slovakia; 1997 for VOCs in Slovakia.
3. Between 1990 and 1997 for Portugal.

Source: Cooperative Programme for Monitoring and Evaluating of Long-Range Transmission of Air Pollutants in Europe (EMEP); World Health Organisation; OECD.

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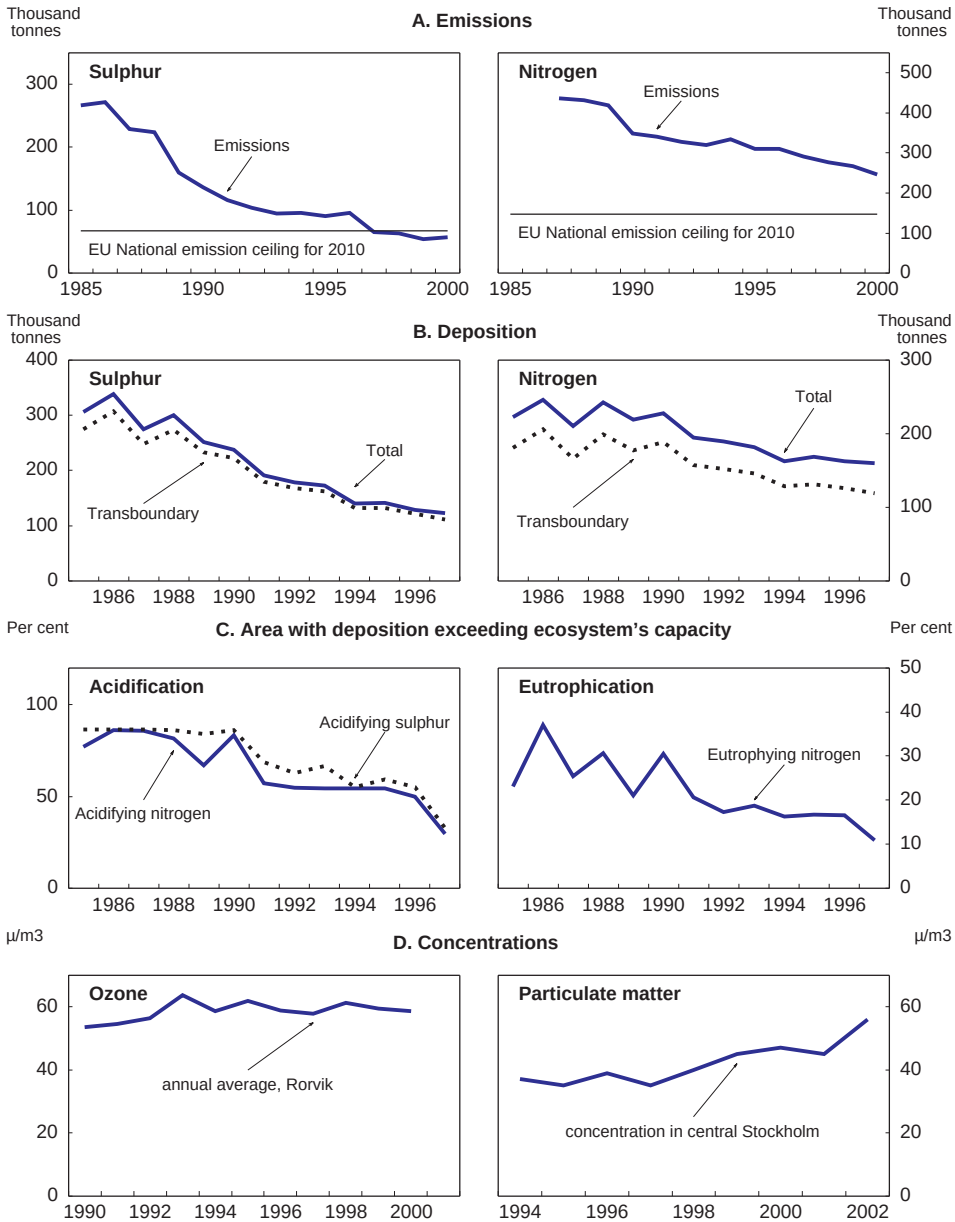
Figure 5.1. **Air pollution**



Source: EMEP; OECD Environmental data, Compendium 2002; European Environment Agency; Johansson (2003).

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Sweden.[2] These targets are complemented by national environmental quality objectives for "natural acidification only" and "clean air" within a single generation. With policy measures already in place, Sweden should meet its international obligations regarding sulphur dioxide emissions relatively easily. Reducing nitrogen oxides emissions presents a greater challenge. Even so, meeting the EU targets for national emission ceilings for sulphur dioxide and nitrogen oxides is estimated to reduce real income by EUR 87 million annually, but the corresponding annual benefits are estimated at between EUR 140 and EUR 260 million (Ministry of Finance, 2001).

Control of domestic emissions of the principal pollutants is based on national emission standards, licensing and economic instruments. National emission standards are already stricter than those applied in the EC Directive on Large Combustion Plants (Cofala and Amann, 2001). The licensing requirements for enterprises include the application of best available technology, though the permitting authority can relax the requirements in order to avoid imposing excessive economic costs. Economic instruments are used to abate sulphur dioxide and nitrogen oxides emissions:

- Since 1991, enterprises have paid a tax on the sulphur content of fuel (the tax rate has remained unchanged at EUR 3.2 per kilogramme), which is refunded at the same rate for each kilogramme of sulphur removed from emissions. The impact of this tax on sulphur emissions has been substantial, accounting for almost one-third of the fall in emissions since its introduction.[3] Nevertheless, estimates of the externality from sulphur emissions indicate that the tax rate may be too low by half to internalise fully the external costs of sulphur emissions (Ahlroth, 2000).
- A different system applies to some sources of nitrogen oxides emissions.[4] A charge of EUR 4.3 per kilogramme of NO_x is levied on emissions from energy generation plants producing at least 25 GWh annually. The revenues from this charge are refunded to generators on the basis of the amount of energy generated, with the amount refunded reduced the higher the emissions. Following the introduction of the nitrogen oxide charge in 1992, emissions covered in the scheme fell by 35 per cent by the end of the decade (Nordic Council of Ministers, 2002).

Additional approaches to addressing the problem of acidification have included encouraging lower emissions from shipping and dealing with its consequences. The Swedish authorities have been innovative in promoting the use of lower sulphur content fuel for shipping and the adoption of improved engine technology for marine vessels.[5] The estimated effects of these measures have been to reduce annual emissions of sulphur dioxide by 50 000 tonnes (or around 6 per cent) and nitrogen oxides by 27 000 tonnes (2.5 per cent), though benefits

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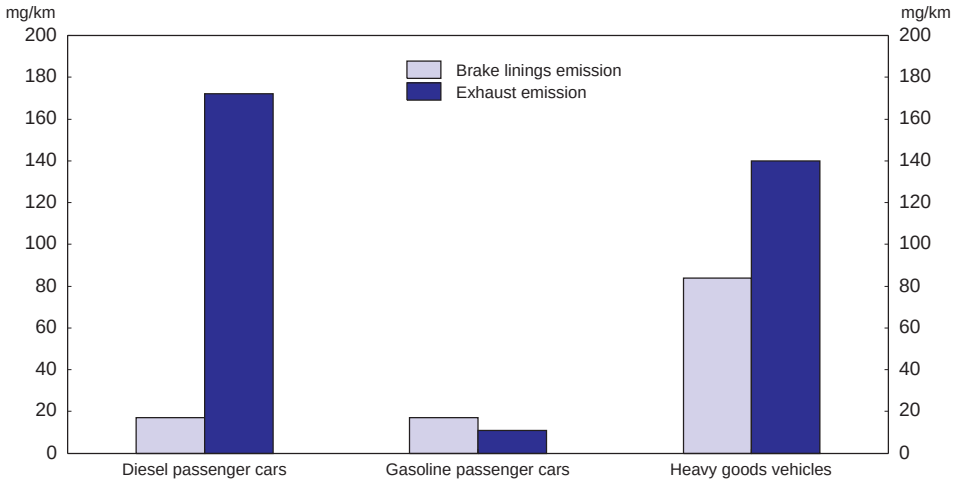
will also accrue to other countries (Swedish Maritime Administration, 2002; ENTEC, 2002). Liming has been used to rectify the consequences of acidification. However, subsequent evaluation revealed that this approach was ineffective in some cases, and government financing was reduced sufficiently for this approach to be adopted only in the most seriously affected areas.

The authorities are moving on two fronts to address pollution from the transport sector, which is the principal source of nitrogen oxides emissions and an important source of particulate matter in urban areas. First, the government is examining the possibility of introducing a distance-based tax. Such a tax based on distance, weight and emission standards has the potential to address the environmental consequences of conventional air pollutants from transport more efficiently. However, the authorities are adopting a cautious approach to distance-based taxation, following its failure in the early 1990s.[6]

The second approach being considered is to alter tax rates for heavy goods vehicles to reflect the different classes of EU-determined emission standards (SOU, 2002). Between 2000 and 2002, the government reduced the annual vehicle tax for newly registered passenger and light commercial vehicles meeting EU emission standards for 2005.[7] Diesel-powered vehicles face a higher annual vehicle tax, but this is offset by lower diesel fuel taxes for a vehicle with average fuel economy once the annual distance travelled exceeds 17 000 kilometres.[8]

Figure 5.2. **Particulate matter emission**

2001



Source: Westerlund, 2001.

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Despite the disincentive to purchase a diesel-powered vehicle for moderate users, such vehicles are estimated to be responsible for almost 80 per cent of particulate matter emissions from passenger cars in Stockholm due to a much higher emission rate (Figure 5.2). In several areas, the authorities have mandated that older commercial vehicles be fitted with particulate filters (Ahlvik, 2001). This in turn has promoted the use of diesel fuel with lower sulphur content, as particulate filters require cleaner fuel to operate efficiently.

While transport is an important source of particulate matter, the residential combustion of wood accounts for over one-third of all particulate matter discharges in Stockholm (Wallin, 1998). Furthermore, while accounting for only around 5 per cent of residential boilers, wood-fired units are responsible for two-thirds of all their emissions. Although the authorities recommend the use of wood boilers that ensure a higher combustion temperature, which helps to reduce particulate emissions, less than half of new wood-fired boilers conformed to these standards in the latter part of the 1990s (Wallin, 1998). Around three-quarters of residential wood boilers in Sweden are older models, which can emit over 60 times the amount of particles emitted by newer models (Nordic Ecolabelling, 2001).

Conclusions

Sweden has in place an effective system to control larger point source emissions of air pollutants. While these measures have been successful in achieving abatement relatively efficiently, the tax rate on sulphur emissions is probably too low. In this light, the authorities should re-estimate the externalities arising from sulphur emissions and adjust the tax rate accordingly. Additional measures are needed to meet international commitments for nitrogen oxides. The approach being adopted for heavy-goods vehicles should also be extended to other vehicles, through aligning vehicle taxes more closely with emission characteristics. In particular, given the large share of particulate matter emissions from diesel vehicles, there would be economic advantages in using taxes to favour diesel vehicles with better emission standards and retro-fitting particulate traps. The taxes on vehicles and fuel are set jointly so that the total tax paid is the same for all types of cars driving around 20 000 to 30 000 km per year. A lower annual tax on diesel cars could be offset by a higher fuel tax on diesel, which would more closely align it with the tax on gasoline. This would result in higher tax payments overall for high-use diesel vehicles. However, as it currently stands, European Directive 99/96/EC effectively limits Sweden's ability to use economic incentives in promoting greater use of particle filters. The authorities should reduce the fuel price advantage enjoyed by diesel over gasoline and pursue the initiative to introduce road pricing. Finally, given the importance of wood fired boilers to particulate matter emissions, the authorities should consider fiscal incentives to encourage new and existing boilers to conform to stricter emission standards.

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Water pollution

Main issues

Drinking water is an essential good, the supply of which depends in large part on the quality of both surface and ground water. The availability of high-quality water has additional advantages as the basis of many economic and social activities. With nitrate contamination of groundwater and eutrophication remaining problems in Sweden, the main issues facing the authorities is implementing effective measures to control diffuse sources of pollution and to meet international obligations to reduce pollution of the Baltic Sea.

Performance

By international comparison, the levels of water pollution in Sweden are relatively low (Table 5.2). During the 1990s, nitrate concentrations in groundwater fluctuated considerably, while concentrations in rivers declined marginally (Figure 5.3). In all, between 10 and 20 per cent of the country is classified as nitrate vulnerable zones, where concentrations are above limits specified in the relevant EC directive or are at risk from nitrate contamination. These areas are largely in the south of Sweden (European Commission, 2002). One of the principal causes of higher pollution loads in the south is the relative importance of agriculture. Pollution loads flowing into the Baltic Sea and contributing to eutrophication, after falling briefly in the mid-1990s, have rebounded in recent years (Figure 5.4).

Policies

The sustainable development strategy targets "zero eutrophication" and "good quality groundwater" as desirable long-term goals, which will require a reduction in water pollution. The EC Nitrates and Water Framework Directives set important targets for Sweden, with the authorities aiming to transpose the latter into national law by 2009. In addition, the states surrounding the Baltic Sea agreed in 1988 to reduce by half the pollution load responsible for eutrophication within ten years. This commitment was reaffirmed in 1998, setting 2005 as a new deadline for compliance as the first deadline had been missed. While Sweden had achieved this target by the late 1990s for phosphorus, progress towards meeting the target for nitrogen is unlikely without additional measures.

As in the case of air pollution, plants that discharge pollutants into water require a licence. Unlike measures to control air pollutants, there is no application of economic instruments on the discharge from licensed point sources. Wastewater treatment facilities, serving almost the entire population, are highly advanced, with the vast majority of plants implementing several stages of water purification.

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Table 5.2. **Performance indicators: water pollution**
Selected rivers

	Biochemical oxygen demand		Nitrates		Total phosphorus	
	mg O2/litre		mg N/litre		mg P/litre	
	Average 1980-85	Average last 3 years	Average 1980-85	Average last 3 years	Average 1980-85	Average last 3 years
Australia	..	..	..	..	..	..
Austria	2.2	2.2	1.2	1.3	0.2	0.1
Belgium	6.6	..	3.8	..	0.7	..
Canada	..	..	0.1	0.1	0.1	0.0
Czech Republic	8.5	4.6	5.1	3.5	..	0.3
Denmark	3.9	2.0	3.5	2.6	0.3	0.1
Finland	..	..	0.2	0.3	0.0	0.0
France	5.2	3.2	2.8	3.1	0.4	0.4
Germany	3.6	2.4	3.5	3.3	0.5	0.2
Hungary	4.4	2.9	2.2	1.6	0.4	0.3
Iceland	..	..	..	..	..	..
Ireland	1.7	1.8	2.2	2.9	0.1	0.1
Italy	..	..	2.0	2.1	0.3	0.2
Japan	2.3	1.4	..	..	..	..
Korea	..	2.8	..	2.6	..	0.1
Luxembourg	3.8	2.7	4.1	4.1	0.6	0.4
Mexico	3.7	15.9	1.9	0.6	..	0.1
Netherlands	2.8	3.1	4.4	3.3	0.5	0.2
New Zealand	..	..	..	..	..	..
Norway	..	..	0.3	0.3	0.0	0.0
Poland	5.0	4.3	1.8	1.9	0.3	0.3
Portugal	..	..	..	..	..	..
Slovak Republic	5.1	3.3	2.0	2.2	0.1	0.2
Spain	4.7	3.6	1.7	3.1	0.6	0.2
Sweden	**..**	**..**	**0.4**	**0.5**	**0.0**	**0.1**
Switzerland	..	..	1.5	1.5	0.1	0.1
Turkey	2.1	2.7	1.3	1.0	0.3	0.2
United Kingdom	3.4	2.9	4.6	5.2	0.8	0.9
United States	1.8	1.6	1.1	..	0.2	0.1

Source: OECD.

However, although wastewater treatment facilities are efficient in reducing phosphorus (93 per cent removal rate), they are less so in removing nitrogen (35 per cent) (Statistics Sweden, 1999). In consequence, in areas where nitrogen is likely to contribute to the pollution load in the Baltic Sea, additional processes are required to remove a larger share of nitrogen.

As in many other countries, introducing measures that effectively address pollution from diffuse sources has proven to be difficult. A major con-

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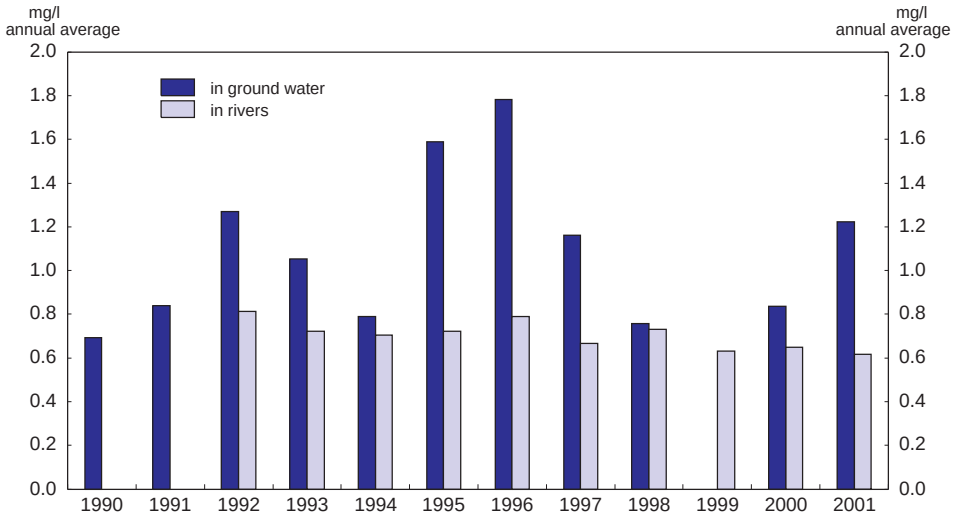
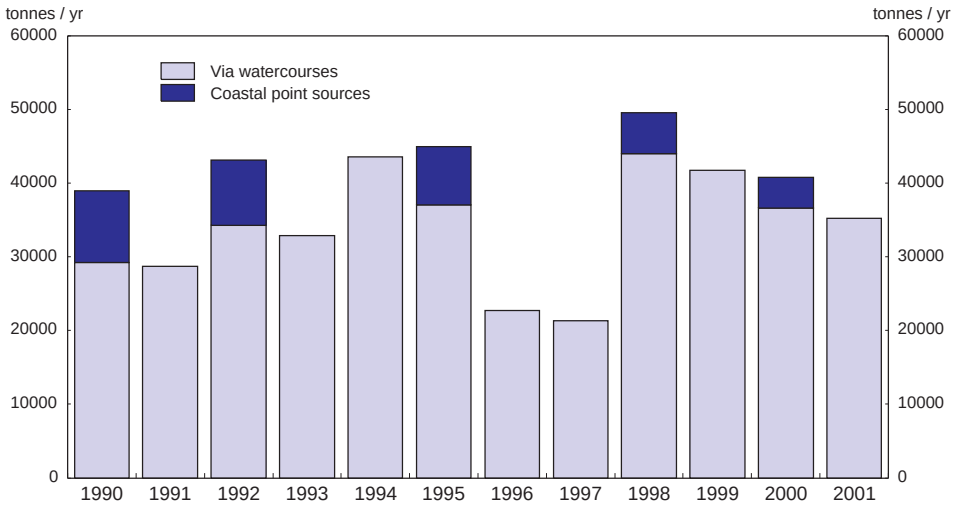
Figure 5.3. **Nitrate concentrations**



Source: Waterbase, European Environment Agency.

Figure 5.4. **Nitrogen supply to the Baltic Sea proper and Öresund**



Source: Swedish EPA.

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straint is agricultural support, which has contributed to increased nitrogen run-off and increased use of pesticides due to the types of crops that benefit, while support of livestock also contributes to ammonia waste and nitrogen run-off (Sjölin and Wadeskog, 2001). Policymakers have attempted to decouple fertiliser and pesticide use from output by imposing taxes on nitrogen content in inorganic fertilisers (EUR 0.2 per kilogramme), cadmium content of phosphorus fertilisers (EUR 3.3 per kilogramme) and the active content of pesticides (EUR 2.2 per kilogramme). While the taxes on phosphorus fertilisers and pesticides have had an effect in reducing use, the nitrogen tax has had only a modest impact (SOU, 2003). This is because the nitrogen tax is inefficient, since it taxes the input instead of the run-off and excludes the nitrogen content of organic fertilisers. These shortcomings could be overcome by targeting farms' nitrogen balances or nitrogen run-off more directly.

With taxes on nitrogen fertiliser use having limited effect, regulations and voluntary measures are also used, which vary regionally depending on the severity of the pollution (Ministry of Agriculture, 2002). Regulations include the permitted density of livestock, the application and storage of manure and a requirement for winter crop cover. Voluntary instruments include compensation for sowing nitrate catch crops, planting vegetation along watercourses and providing wetlands and ponds to reduce nitrate leaching. Estimates of the costs of these subsidies for reducing the pollution load into the Baltic Sea range from EUR 5 per kilogramme of nitrogen reduced for constructed wetlands to EUR 16 per kilogramme of nitrogen reduced for planting catch crops (Finnish Environmental Institute, 2002). These abatement costs are considerably higher than the implicit tax on surplus nitrogen of EUR 0.85 per kilogramme, which suggests that the nitrogen tax rate is set too low.[9]

While the target of a 50 per cent cut in nutrient load to the Baltic Sea could entail overall abatement costs roughly equivalent to estimated benefits, this would be the case only if cost-effective measures were implemented.[10] At present, each individual country bordering the Baltic Sea is obliged to reduce pollution loads by 50 per cent, which almost doubles total abatement costs relative to a more cost-efficient allocation of abatement effort, largely due to the high marginal abatement costs faced by Sweden and Germany. The most cost-effective allocation would require proportionally more abatement effort in the Baltic States and Poland, which would still leave Sweden facing annual costs of over EUR 500 million (Gren *et al.*, 1997). Estimates based on minimising marginal abatement costs suggest that up to 20 per cent of total funds should be directed at abatement in foreign countries (Markowska and ylicz, 1999). However, a proposal to create a Baltic-wide fund in the mid-1990s failed to attract sufficient support. Notwithstanding this outcome, an international trading regime could help achieve the most cost-effective abatement allocation.

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Conclusions

While levels of water pollution are low in comparison with many OECD countries, nitrate contamination and eutrophication remain problems in certain parts of the country. In this context, the authorities should undertake cost-benefit analyses to evaluate whether introducing a wastewater discharge fee and a charge on either the nitrogen balance of farms or nitrogen run-off could be efficient measures in addressing the consequences of nitrate pollution. Given the importance of agriculture to nitrate pollution and eutrophication, the authorities should exploit the possibilities allowed in the reforms to the Common Agricultural Policy (agreed in June 2003) of shifting agricultural support to those measures that would reduce the environmental externalities. Meeting the target for reducing nitrogen pollution in the Baltic Sea could lead to high abatement costs if implemented entirely domestically. Forging agreement with other countries surrounding the Baltic Sea to allow greater abatement efforts in these countries would help ensure that the targets are met in a cost-efficient manner.

Waste management

Main issues

In the field of waste management, the Swedish authorities have set some of the most demanding targets in the world. Policy in this area has largely been determined by the use of environmental impact assessments, with only limited use of cost-effectiveness and cost-benefit analysis (Ministry of Finance, 2001). The main issue for waste management policy is to ensure that its key objectives are attained as cost-effectively as possible.

Performance

The amount of municipal waste generated per capita is lower than the OECD and EU averages but grew fairly rapidly during the 1990s (Table 5.3). The principal waste disposal method, landfilling, declined in importance over the 1990s. However, the most striking feature of that decade was the strong growth in material recovery (Table 5.4), pushing the recycling rate well above the OECD and EU averages. The physical amount of waste incinerated (combined with energy recovery) remained relatively constant during the 1990s, but due to the strong increase in total waste generation its share fell. Finally, composting grew quite rapidly but still accounts for a small overall share of municipal waste treatment.

Policies

Waste management policy embodies a strict waste hierarchy approach, favouring reuse and recycling over incineration, which is in turn preferred to

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Table 5.3. **Performance indicators: waste management**

	Municipal waste					
	Level	Relative to private consumption	Annual growth		Recycling	Composting
	Kg per capita, latest available year	Grammes per US$, 1995 PPPs	Period	Per cent	Per cent of total, latest available year	
Australia	..	..	–	..	..	..
Austria	548	45.5	1990-1999	3.7	24	40
Belgium	534	47.0	1990-1999	3.0	37	15
Canada	..	..	–	..	..	..
Czech Republic	334	54.2	1990-2000	..	..	..
Denmark	626	57.4	1994-1999	3.5	25	14
Finland	464	49.0	1990-1999	–2.8	..	..
France	524	46.4	1993-1999	1.7	10	8
Germany	549	46.0	1991-1998	–1.5	41	..
Greece	371	42.7	1990-1997	3.8	8	1
Hungary	444	85.7	1990-2000	–1.9	0	0
Iceland	718	52.7	1992-2000	2.8	9	2
Ireland	615	49.5	1995-2000	8.2	8	1
Italy	497	40.1	1990-1999	4.0	7	10
Japan	406	34.6	1990-1999	0.2	13	..
Korea	364	50.3	1990-2000	–5.8	41	..
Luxembourg	643	38.7	1991-1999	3.1	0	12
Mexico	315	63.5	1991-2000	4.3	2	0
Netherlands	613	53.3	1991-2000	2.9	23	24
New Zealand	..	..	–	..	..	..
Norway	617	51.2	1992-2000	2.7	22	9
Poland	316	64.0	1990-2000	1.0	0	2
Portugal	437	50.7	1992-1999	4.2	4	5
Slovakia	316	58.2	1992-2000	..	2	5
Spain	464	48.0	1990-1999	4.3	5	18
Sweden	**452**	**47.5**	**1990-1998**	**2.8**	**25**	**8**
Switzerland	655	41.8	1990-2000	1.4	32	14
Turkey	385	101.4	1989-1998	..	0	1
United Kingdom	558	44.8	1990-1999	2.3	9	2
United States	764	35.7	1990-1999	1.3	22	6
OECD average	501	42.3		1.3	17	7
EU average	526	45.6		1.9	18	10

Source: OECD, Eurostat. For Spain: national submissions.

landfill. The authorities have set the target of halving the amount of waste going to landfill by 2005 from 1994 levels. To this end, a tax of EUR 40 per tonne is levied on landfill waste, a rate that likely exceeds its external costs.[11] Furthermore, landfilling of source separated combustable waste was prohibited as from 2002, and the landfilling of organic waste will be prohibited from 2005. With modern technol-

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Table 5.4. **Household waste**

	1980	1985	1990	1995	2001
Total, thousand tonnes	2 700	2 750	2 960	3 746	3 907
Share, per cent					
Landfill	53.7	38.2	30.4	25.4	22.5
Incineration	31.5	47.3	47.3	38.7	38.5
Recycling (*i.e.* materials recovery)	7.4	9.1	15.5	27.9	29.1
Composting	7.4	5.5	6.8	8.0	9.9

Source: RVF.

ogy to reduce methane emissions and prevent leachate discharge, the environmental externalities from landfilling organic waste can be reduced by at least half (ECON, 2001). In less populated regions the loss of amenity value is also likely to be smaller. At present many local authorities that lack incineration or recycling capacity have exemptions from the bans, but in the medium term it is likely that incineration capacity will expand, as this represents the cheapest alternative to landfills. A committee has been appointed to evaluate the introduction of a tax on incineration and measures to increase further the shares of recycling and biological treatment. Reuse is stimulated by deposit refund systems, which are mandatory for aluminium and plastic beverage containers and voluntary for glass containers. For packaging waste and some other products, extended producer responsibility schemes have been introduced in order to meet recycling targets established in national legislation. In some communities, households are obliged to separate waste at source and take recyclable waste to recycling stations.

The local authorities are responsible for household waste not covered by extended producer schemes. The costs of municipal waste management to the household vary significantly across local authorities, with average household fees ranging between EUR 45 and EUR 220, suggesting that large efficiency gains may be available.[12] The impact of the prohibitions on landfilling and the introduction of the landfill tax in 2000 are putting upward pressure on overall costs, with the average costs to the household for municipal waste disposal jumping by over one-third between 1999 and 2001 (RVF, 2000 and 2002). The increasing distances over which municipal waste is transported to incineration and other facilities is also putting upward pressure on waste management costs.

Since the introduction of extended producer responsibility in 1994 for packaging and paper,[13] the organisation REPA has operated recycling systems on behalf of its members, who are responsible for over 90 per cent of packaging material. Companies pay REPA a fee on the type of packaging material to cover recycling costs. These costs are passed onto the consumer in higher product

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Table 5.5. **Recycling rates and targets**
Per cent

	Recycling rate in 2001	Recycling targets set in 2001	Current EU targets	Proposed EU targets
Plastic	16	30	15	20
Metal	48	70	15	50
Glass	84	70	15	60
Paper/cardboard	63[1]/85	40/65	15	55

1. 2000 data.
Source: RVF.

prices, corresponding to between EUR 25-50 per household annually (RVF, 2002). To the extent that packaging fees reflect recycling costs, there are already marked differences in costs by material, ranging from EUR 7 per tonne of steel to EUR 170 per tonne of plastic and aluminium. Furthermore, while the average cost of recycling a tonne of packaging is around EUR 50, the marginal cost of recycling an additional tonne of packaging waste has been estimated to be in excess of EUR 450 (The Environmental Exchange, 2001). The ordinance on producers' responsibility sets some of the most ambitious recovery and recycling targets for packaging waste in Europe (Table 5.5). Furthermore, the target set for plastic packaging requires a near doubling of the recycling rate, which is likely to be particularly expensive to attain.[14]

These recycling costs do not include the costs of the time households incur in cleaning, sorting and transporting waste to collection points. One estimate put the cost to household of the time spent for separation and transport at EUR 180 per tonne of waste (Radetzki, 1999).[15] The mandated sorting of a large number of different types of waste can require individuals to spend 25 hours a year, while transportation of waste to the collection sites could potentially add another 16 hours (Berglund, 2003; Sterner and Bartelings, 1999). However, if the municipality incinerates residual waste, which is now more likely with the ban on landfilling combustible waste, it is allowed to burn separately collected lightweight plastic packaging by adding it back into residual waste, so negating the efforts of the household in sorting the waste (REPA, 2001).

Conclusion

Waste management policy has suffered from a lack of cost-effectiveness and cost-benefit analysis of policy options, resulting in substantial cost increases to households. Targets and tax rates should be set on the basis of such analysis in order to prevent costs rising above the benefits and to ensure that the least costly

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options are employed in achieving stated targets. Given both the size and considerable diversity of the country, the requirement of uniform waste management targets for the local authorities should be reconsidered. In this light, the authorities could introduce mechanisms such as trading between municipalities that would allow national goals to be met cost-efficiently. The authorities should also reconsider the prohibition on landfilling combustible waste. Households have borne a rising direct cost as a result of past initiatives and have likely incurred significant private costs in time spent sorting different types of waste. There appears to be little justification in obliging households to separate waste streams that will in some cases be incinerated. The authorities should also consider introducing pricing systems with a greater emphasis on weight or volume than is presently the case to provide better incentives for both waste reduction and separation. Finally, the targets for recycling of packaging materials are very ambitious for some waste streams. To prevent costs rising excessively, the authorities should determine the external cost for disposal alternatives in order to set a cap on permissible recycling costs.

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Notes

1. Non-road mobile sources include items such as tractors, trains, aircraft and ski-mobiles.
2. For example, Sweden was one of the first countries to ratify the 1999 Gothenburg Protocol.
3. Another factor leading to emission reductions has been increasing energy efficiency.
4. The different approaches reflect the different causes of emissions. While sulphur dioxide emissions largely depend on the sulphur content of fuel, nitrogen oxide emissions are principally dependent on combustion technology.
5. The maritime authorities levy lower fairway and harbour dues for shipping using lower sulphur content fuel and having better engine technology and catalytic converters fitted (OECD, 2001).
6. That scheme suffered from problems in verifying distances travelled. With a view to accession to the European Union the scheme was abandoned and the vehicle tax adjusted in line with practices elsewhere in the EU.
7. The Swedish Environmental Protection Agency reports that the reduction in the annual vehicle tax is available only for vehicles exceeding a certain weight. Non-road mobile sources are not subject to the same emission standards, though the EU is moving to toughen them.
8. The Budget Bill for 2004 will increase slightly the tax on diesel fuel, though it will remain 30 per cent lower than the tax on unleaded gasoline.
9. To estimate the nitrogen fertiliser tax rate in terms of excess nitrogen, "farm-gate method" data on input, output and surplus were used. As around two-fifths of nitrogen in fertilisers is released to the atmosphere as ammonia, the nitrogen surplus was re-estimated to take account of this loss.
10. The estimated benefits from a reduction of the nutrient load entering the Baltic Sea are derived from willingness-to-pay studies (Gren *et al.*, 1997).
11. The external costs associated with the landfill of a tonne of waste in landfills complying with EC requirements are estimated to be in the range of EUR 6 to EUR 24 (COWI Consulting Engineers and Planners, 2000).
12. Independent research by the main association of waste management companies was unable to identify factors leading to such high variations (RVF, 1999).
13. Extended producer responsibility has also been broadened to include motor vehicles, tyres, and waste electronic and electrical equipment.

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14. TNO (2001) suggests that most of the environmental benefits for plastics are realised when the recycling rate is 15 per cent and the remainder is incinerated with energy recovery.
15. While some individuals derive satisfaction from time spent sorting waste, Berglund (2003) showed that others would benefit if the incentive was monetary rather than an imposed obligation.

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